claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

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- Kind: `claude`
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- SHA-256: `42b602e35612011d80ac9bb61c7b59d20b4276d9a35e9c3d5c640ea6f198f648`
- Source modified: `2026-06-08T13:23:59+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_49aab888-e10`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`

Transcript

1. user (2026-06-08T13:21:28.245Z)

You are a completeness critic. Given the study below + the research it drew on, list what is MISSING, any option not considered, any claim still weakly sourced, and any misalignment with our guardrails (Apache-2.0 only, no paid-LLM training data, privacy/local, reuse our labels). Be concise and specific.

STUDY:
{
  "executive_summary": "**One-line thesis:** Train the OWNED rally model in three staged generations off a single, model-agnostic labeled corpus ? (Gen-0) a tiny on-device TAL/TAS head over the WASB features we already have, (Gen-1 / Option A) modular cue fusion into that same head, (Gen-2 / Option B) an end-to-end multimodal VLM fine-tuned on an Apache-2.0 video base ? where every generation reads transcoded views of the SAME human rally CSVs, so labels are never re-collected. The moat is the consented dataset + the eval harness, never the weights.\n\n**What's safe & available NOW:** 128 human rally [start,end]+ending_reason CSVs (96 Adarsh+Avi, 32 Kushagra), per-video WASB trajectory CSVs (Frame,Visibility,X,Y; frozen MIT detector), ~26 GB licensed/owned single-camera GoPro corpus, Gemini silver labels (EVAL/teacher-reference ONLY ? never training signal), and Roora vendor expansion underway. The honest baseline is threshold-tuned F1 ? 0.73 on one match; the learned model loses to the heuristic at n=2. **This is data starvation, not method failure** ? the binding bottleneck is golden DATA + WASB recall, confirmed by our own results.\n\n**Base lock (apply verification corrections):** Gen-2 base = **Qwen3-VL-4B-Instruct (Apache-2.0, explicit Text?Timestamp Alignment, smallest on-device candidate)** as primary, with **Qwen2.5-VL-7B-Instruct (Apache-2.0, best-documented temporal grounding, most mature ms-swift recipes)** as the de-risking fallback. **Gemma 4 is UPGRADED from AMBER to confirmed-clean** ? the primary Google model card (ai.google.dev/gemma/docs/core/model_card_4) verifies BOTH Apache-2.0 AND native video (frames @ 1 fps, 60 s) + audio on E2B/E4B/12B-Unified ? making it the strongest structural match for Option-B frames+audio fusion; lock it as the Option-B end-state base once a timestamp-emission recipe is proven. Gemma 3 stays REJECTED (viral Gemma Terms + PUP flow-down). Qwen2.5-VL-3B (research/non-commercial), Qwen2.5-VL-72B + InternVL3-78B (100M-MAU cap), and DAMO VideoLLaMA3 weights (non-commercial AND trained on OpenAI/Gemini data ? trips our paid-LLM guardrail) are all rejected.\n\n**Gen-0 first owned model = ASFormer (MIT) TAS head over WASB features** ? the only surveyed architecture whose central published result is training from scratch on a tiny dataset (GTEA, 28 videos, ~1.1M params) via convolutional inductive bias; fits the RTX 2070 Super, trains in minutes, fully local. ActionFormer (MIT) runs as the parallel A/B challenger that overtakes once the corpus grows to hundreds of matches. **Caveat made front-and-center:** ASFormer's small-data result was on Kinetics-pretrained I3D 2048-d features; feeding raw ~3-4-dim WASB trajectory is off the validated path and needs a hand-built feature stack (velocity, acceleration, visibility-run-length, net-crossing flags) ? the single biggest empirical unknown, resolvable only by an A/B on the golden set (not runnable in this dev container).\n\n**Platform ranking through our privacy+cost lens:** the fine-tune is a sub-24h, ~$10-50 job on one A100-class GPU, so **privacy terms + operational ergonomics dominate, not GPU $/hr**. Modal wins primary (strongest written no-access contract + per-second scale-to-zero matching retrain-on-each-batch); RunPod Secure Cloud is the cost fallback; Vertex/GCP stays on the bench as the only strong-privacy option with an India region (capacity-gated H100-class A3,

NOT A100). Vertex is explicitly NOT the owned model ? tuning Gemini yields no portable on-device weights.\n\n**Sequencing (owner directive, non-negotiable):** this is a DESIGN blueprint that unblocks PLATFORM P5. NO owned-model implementation starts until P0-P4 scaffolding (async jobs, storage/compute abstractions) is rock-solid AND the golden corpus is healthy. The only items safe to act on now ? and even these need owner sign-off before code ? are the model-agnostic corpus format and growing n?2?n?5 matches.",
  "platform_ranked": [
    {
      "name": "Modal (serverless GPU)",
      "why": "Strongest WRITTEN no-access contract found, verified verbatim against modal.com/docs/guide/security: 'Modal will never access or use your source code, the inputs/outputs to your Functions, or any data you store'; inputs/outputs deleted within a max 7-day TTL; SOC2 Type II; HIPAA via BAA. Per-second billing + scale-to-zero exactly matches the retrain-on-each-golden-batch cadence. Python/recipe-driven model slots cleanly into a one-command agent-runnable harness (ms-swift glue) and the future ComputeBackend abstraction. The privacy contract is the load-bearing reason it wins, not price.",
      "cost": "CORRECTED (verification): A100 80GB ~$2.50/hr (was cited $3.72 ? stale/high), H100 ~$3.95/hr (was $4.29). Full LoRA fine-tune ~$10-50. $30/mo free credits. Non-US region multiplier is 1.5-1.75x (was cited 1.25-2.5x). Re-quote live before budgeting.",
      "privacy": "STRONGEST. Verified no-access + 7-day retention + SOC2 Type II. Meets the hard requirement for consented video out of the box.",
      "verdict": "RECOMMENDED PRIMARY for the fine-tune forge."
    },
    {
      "name": "RunPod (Secure Cloud tier ONLY)",
      "why": "Cheapest vetted-DC self-serve; best value for longer/multi-GPU Option-B runs. Secure Cloud uses vetted partners (SOC2/ISO 27001/PCI), GDPR-compliant, HIPAA-verified; ToS verbatim 'prohibit hosts from inspecting your Pod/worker data'. Drops into the ms-swift-on-rented-GPU plan with full weight ownership.",
      "cost": "CORRECTED (verification): A100 80GB PCIe ~$1.39/hr (SXM $1.49) ? was cited $1.19-1.29 (stale/low); H100 PCIe ~$2.89/hr (SXM $3.29) ? was $2.65. Our fine-tune ~$3-15. NOTE RunPod no longer publishes separate Secure-Cloud dollar figures, so Secure-tier cost is now uncertain ? confirm at booking.",
      "privacy": "STRONG on Secure Cloud (host-no-inspect ToS + GDPR + partner certs). 'AES-256 at rest' and 'has a DPA' could NOT be confirmed from primary docs ? verify. Community/peer-to-peer tier is UNVETTED and MUST NEVER touch consented video.",
      "verdict": "RECOMMENDED FALLBACK / Option-B scale-up. Secure Cloud only."
    },
    {
      "name": "Google Vertex AI / GCP Compute Engine GPUs",
      "why": "Only strong-privacy option with an India region. Written no-train verbatim (verified two sources): 'Customer data is never used by Google. We do not use customer data to train our models... without permission', plus a Vertex Service-Specific-Terms §17 Training Restriction. Mature multi-GPU/VPC/CMEK maps to future Storage/Compute backends. GUARDRAIL TRAP: tuning Gemini yields NO owned/on-device weights ? only tuning an Apache-2.0 open base in Model Garden gives portable weights, and that path is heavyweight.",
      "cost": "Most expensive + most overhead. A100 40GB ~$3.37/hr all-in (compute + mgmt fee); 80GB a2-ultragpu ~$5.07/hr+. CORRECTION: GCP asia-south1 (Mumbai) does NOT offer A100 ? only A3 (H100/H200-class, account-team/invitation gated) + L4/T4/G2/G4. An India-residency run would be on pricier H100-class A3, not A100.",
      "privacy": "STRONG written no-train; ONLY strong option with in-India residency-at-rest ? but that residency is CAPACITY-GATED (asia-south1 high-end GPU is invitation-only), so treat as conditional, not turnkey. Confirm asia-south1 GPU quota before relying on it.",
      "verdict": "KEEP ON BENCH. Adopt only if India residency-at-rest or an audited managed pipeline is mandated. NOT the owned model."
    },
    {
      "name": "Self-hosted RTX 2070 Super 8GB + WSL2",
      "why": "Maximal privacy (data never leaves device ? the local-first moat) and zero marginal cost. The right box for: Gen-0 ASFormer/MS-TCN++ training (~1.1M params, trains in minutes), harness plumbing, overfit-one-batch sanity, and 4-bit 7B inference/eval. It is NOT a training box for any 7B-class video VLM.",
      "cost": "$0 marginal (owned). Dev/CI/inference only.",
      "privacy": "BEST possible ? nothing leaves the machine. No DPA, no egress, no residency question.",

      "verdict": "DEV/CI/INFERENCE ONLY. 8GB cannot QLoRA a 7B video VLM (~20-24GB with video tokens); Turing also lacks bf16/modern kernels. Use for Gen-0 training + all inference, never Gen-2 training."
    },
    {
      "name": "Together AI / Fireworks AI (managed fine-tune)",
      "why": "Lowest operational burden, fastest path to a first checkpoint; LoRA SFT ?16B ~$0.48/M tokens. Fireworks contractually does NOT train on customer content + ZDR-by-default + SOC2 II + HIPAA/BAA. Together asserts data/model ownership + SOC2 II but ? verified ? has NO explicit no-train clause and NO public DPA on its fine-tuning page.",
      "cost": "Together LoRA ?16B $0.48/M tokens (text rate ? VIDEO token counts can balloon, UNVERIFIED for our clip?QA). A 5k-20k-pair run ~$2-30 IF text-priced. Confirm video/multimodal pricing on real frame counts.",
      "privacy": "Fireworks: STRONG (contractual no-train + ZDR). Together: BLOCKED for the consented corpus until a signed DPA + explicit written no-train assurance is obtained.",
      "verdict": "HARD GATE: signed DPA + no-train assurance before any consented video touches them. Fireworks is the stronger managed option; Together is blocked until DPA. Use only if zero-MLOps outweighs weight-portability control."
    },
    {
      "name": "AWS SageMaker (ap-south-1) / Azure ML (Central India)",
      "why": "The other two in-India residency options. Both offer VPC isolation + CMEK + GDPR alignment. AWS ap-south-1 now centers on P5/H100 (P4d/A100 NOT confirmed in Mumbai); Azure Central India has A100/H100 with cheap spot (NC A100 v4 ~$0.82/hr spot).",
      "cost": "Enterprise-heavy + mgmt premium. Azure NC A100 v4 ~$3.67/hr on-demand / ~$0.82 spot; ND H100 v5 ~$6.98-12/hr/GPU. AWS P5 (H100) ~$6.88/hr on-demand / ~$3.83 spot + SageMaker premium.",
      "privacy": "Likely STRONG via signed DPA + VPC/private endpoints, but UNVERIFIED ? neither could be confirmed to carry an explicit no-train clause from public pages. AWS has a default aggregate-metadata opt-out footgun. Read the DPA first.",
      "verdict": "WEAK FIT for a solo founder (operationally heavy). Only relevant if India residency-at-rest is legally mandated AND Vertex capacity is unavailable."
    },
    {
      "name": "Lambda Labs / vast.ai (Secure) / CoreWeave / Paperspace",
      "why": "Cheaper or specialist GPU clouds. Lambda is ML-native (pre-baked CUDA); vast.ai is cheapest marketplace; CoreWeave suits very large H100 fleets; Paperspace is notebook-friendly.",
      "cost": "CORRECTED: Lambda A100 80GB ~$2.79/hr (was $1.99-2.49, stale/low), H100 $3.99/hr. vast.ai specific figures could NOT be confirmed (live dynamic marketplace). CoreWeave A100 ~$2.70/hr, H100 ~$6.16/hr. None in India.",
      "privacy": "Lambda: UNVERIFIED ? obtain/read DPA before the corpus goes on it. vast.ai: Community tier FAILS the requirement; Secure tier acceptable for NON-consented/synthetic only. CoreWeave has the cert stack but no better than Modal/RunPod on no-access specificity.",
      "verdict": "NON-SENSITIVE / synthetic / ablation runs ONLY for vast.ai. Lambda needs DPA review first. None displace Modal-primary / RunPod-fallback for consented video."
    }
  ],
  "model_path_phased": [
    {
      "phase": "Gen-0 (NOW ? unblocks at n?5 matches): small owned TAL/TAS head that beats the heuristic at small n",
      "approach": "Train ASFormer (MIT, ChinaYi/ASFormer) as a TEMPORAL ACTION SEGMENTATION model over a hand-built per-frame feature stack derived from the WASB trajectory we already produce. Per-frame target = rally(1)/non-rally(0) from the rally [start,end] CSVs; recover predicted intervals by merging contiguous rally frames + a min-gap/min-duration prior (reuse the existing net-crossing gate). Treat ending_reason as a SECOND, segment-level 5-class head ONLY after interval detection works ? do not learn it jointly first, and never let it gate the [start,end] milestone (trajectory features alone almost certainly cannot predict winner/forced/unforced/fault/let). SHIP GATE: held-out per-MATCH F1 must beat the threshold-tuned heuristic+LogReg baseline (~0.73) on the golden set ? the nanochat-style pass/fail verdict.",
      "models": "PRIMARY: ASFormer (MIT, ~1.1M params, from-scratch-on-small-data thesis). PARALLEL A/B CHALLENGER: ActionFormer (MIT, regresses rally=one (start,end,class) instance ? cleanest 1:1 map to the CSV schema; expect it to LOSE at n=3-10, OVERTAKE once corpus hits hundreds of matches). LATER drop-in boundary upgrade on ActionFormer: TriDet (MIT). DEFER:

DiffAct (data-hungry), TemporalMaxer (license UNCONFIRMED ? 404 on LICENSE fetch). LICENSE TRAP to record: if MS-TCN++ is ever used, vendor ONLY sj-li/MS-TCN2 (clean MIT) ? yabufarha/ms-tcn carries Commons Clause (non-commercial).",
      "data_needed": "Existing 128 rallies + WASB CSVs are enough to PROTOTYPE; ~3-5 labeled matches needed for STABLE leave-one-match-out cross-val (we're under that at n?2 ? exactly why the learned model currently loses). Grow n?2?n?5 via a small Roora batch to unblock honest tuning. CRITICAL UNKNOWN: ASFormer's small-data result used Kinetics-pretrained I3D 2048-d features; raw ~3-4-dim WASB trajectory is OFF the validated path ? needs a feature-smoothing/windowing front-end (velocity, acceleration, visibility-run-length, net-crossing flags). This is the dominant empirical risk; resolvable only by an A/B on the golden set.",
      "effort": "LOW compute (trains in minutes on the existing RTX 2070 Super; CPU-feasible inference; $0 cloud). Engineering: feature stack + per-frame?interval merge + the eval gate. Fully local, no paid-LLM data, all MIT ? zero licensing/privacy exposure. BLOCKED on owner approval per CLAUDE.md."
    },
    {
      "phase": "Gen-1 / Option A (6-12 months): modular cue fusion into the SAME head",
      "approach": "Concatenate additional fully-local feature channels onto the Gen-0 ASFormer input as each cue matures: WASB trajectory (live) + audio-onset hit-detection (E1 probe AMBER, gated on more golden labels ? ADR-007) + pose/court serve-gating (validated but heavy). Fusion happens in OUR rally layer, not inside the frozen detectors; each cue stays swappable and independently testable. This delivers the Option-A milestone AND generates richer training signal for Gen-2 ? for free, because ASFormer ingests arbitrary-dim per-frame features by concatenation.",
      "models": "Same ASFormer/ActionFormer head, wider input feature vector. No new base model. Open-teacher pseudo-labeling (WASB/our-own-model only, NEVER Gemini, confidence-gated à la PLR/ALQA) over the ~26 GB unlabeled corpus to multiply effective labels; temporal augmentation (moderate speed/time-warp ±20-30%, jitter, reverse, FadeMixUp) as the noisy-student noise.",
      "data_needed": "Same 128 rallies amplified by augmentation + pseudo-labels; route the Roora budget by ACTIVE LEARNING (boundary entropy / WASB-vs-head disagreement) once n?5, mixing in some random/diverse picks to avoid sampling bias. Pseudo-labels inherit WASB's recall ceiling ? confidence-gate + human-spot-check; keep human [start,end] as the boundary authority.",
      "effort": "LOW-MEDIUM. Mostly engineering (feature channels + fusion + pseudo-label loop) + occasional cloud burst for pseudo-labeling. 2-4 week realistic Option-A milestone per cue. Still fully local, guardrail-clean."
    },
    {
      "phase": "Gen-2 / Option B (12-24+ months, AFTER scaffolding P0-P4 + healthy corpus): end-to-end multimodal owned VLM",
      "approach": "Collapse the modular cues into a single fine-tuned end-to-end model: (frames + audio + pose/stance + trajectory tokens) ? rally [start,end] (+ rich semantics). Removes the WASB candidate-window recall ceiling and the academic-weights licensing question. Recipe: quality SFT cold-start on clip?QA pairs THEN GRPO/RFT with a temporal-IoU-vs-golden reward (reuse metrics.match_intervals as the verifiable reward ? no separate reward model). Data-efficiency literature says our small, clean, hand-verified set is the RIGHT asset for RL: VideoChat-R1 reports +31.8 temporal-grounding with 'limited samples' (verified arXiv:2504.06958); VTG-R1 shows a filtered 13K cold-start beats an unfiltered 56K (verified arXiv:2507.18100). So we need amplification + a few clean matches, NOT thousands of new ones.",
      "models": "PRIMARY base: Qwen3-VL-4B-Instruct (Apache-2.0, Text?Timestamp Alignment, smallest on-device). FALLBACK base: Qwen2.5-VL-7B-Instruct (Apache-2.0, most mature ms-swift recipes). OPTION-B END-STATE base: Gemma 4 E4B/12B-Unified (Apache-2.0 + native video+audio ? UPGRADED from AMBER to confirmed-clean per primary Google model card; best structural match for frames+audio fusion; lock once timestamp-emission recipe proven). Also clean: InternVL3-8B (MIT wrapper / Apache Qwen2.5-7B base). BRIDGE PoC alternative for TODAY's data scale: InternVideo2 encoder (MIT) + light grounding head. REJECT: Gemma 3 (viral terms), Qwen2.5-VL-3B (research/NC), Qwen2.5-VL-72B + InternVL3-78B (100M-MAU cap), DAMO VideoLLaMA3 weights (NC + trained on OpenAI/Gemini data ? trips paid-LLM guardrail), TimeChat/VTimeLLM/Momentor/VideoMAE-weights (NC/no-LICENSE ? method-reference only).",
      "data_needed": "Single-digit-thousands of clip?QA pairs (literature 13K-18K; RFT can cold-start from hundreds). Reachable from ~128 rallies × temporal augmentation × QA-task multiplexing (rally-localization + ending_reason) × confident pseudo-labeled clips ? amplification is LOAD-BEARING, not optional; 128 raw rallies alone are not enough. Roora + rally-annotator grow the corpus month-on-month.",
      "effort": "HIGH compute (RFT = multiple rollouts/sample; rented 24-48GB A100/L40S-class GPU mandatory ? does NOT fit the 8GB box even at QLoRA on video). Engine = self-managed ms-swift

(only framework advertising NATIVE video+audio training) on Modal/RunPod. Re-benchmark VRAM on real 1080p/60fps clips (24GB is a floor, not a guarantee). DEFER implementation until P0-P4 ship; benchmark self-hosted-7B $/call vs Gemini before committing budget (COMMERCIALIZATION C7)."
    }
  ],
  "label_reuse_strategy": "**Principle: stop treating the rally CSVs as eval-only files; promote them to ONE model-agnostic labeled corpus that transcodes into every consumer, so a label labeled once feeds heuristics + TAL + VLM + every future generation. This IS the moat.**\n\n**The canonical record (one JSONL row per (video_id, rally_ordinal)):** {video_id (file MD5), rally_ordinal, sport, start, end (seconds), ending_reason, source (human|pseudo|vendor), annotator/provenance, label_type (segment|point), split (train|val|test ? assigned BY MATCH not by clip), trajectory_ref (pointer to the WASB CSV slice for this video)}. Stored once; extends the existing collector export + candidate_segments lineage (ADR-002 bridge already points this way). The provenance/source/label_type/consent fields are the ENFORCEMENT POINT for the legal guardrails (no-paid-LLM, minors-excluded, training-opt-in) ? each record carries who/what produced it and whether training opt-in applies.\n\n**Three deterministic, tested transcoders (one row ? three formats, zero re-labeling):**\n1. **Heuristic/LogReg view** ? candidate-window pos/neg labels. ALREADY EXISTS: backend/eval/training.py::label_candidates runs the same IoU match against backend/eval/metrics.py::match_intervals to label WASB candidate windows. Keep as-is.\n2. **TAL view (ActionFormer/TriDet)** ? ActivityNet JSON: top-level database ? per-video {subset, duration, fps, annotations:[{segment:[start,end] in SECONDS, label:'rally', label_id:0}]}. Our seconds-based [start,end] maps 1:1; only add per-video duration/fps (already in ffprobe_meta) + a train/val/test subset. (Mechanical-support caveat: feeding low-dim WASB features instead of I3D-2048 is off the validated path ? flagged in Gen-0.)\n3. **VLM clip?QA view (Option B)** ? sample a window, ask 'List the [start,end] of every rally in this clip' ? answer = rally intervals normalized clip-relative; ending_reason yields a SECOND free QA task ('Why did rally N end?'). One windowing/sampling policy (clip length/stride, negative no-rally clips) is the one design choice to lock.\n\n**One spine, no metric drift:** reuse metrics.match_intervals as the SINGLE scorer for heuristic-labeling, TAL eval, AND the Gen-2 RFT temporal-IoU reward ? this prevents metric inflation from data leakage and is already proven in code.\n\n**Leakage guardrail (baked into the format):** split is enforced by MATCH, never by clip ? kills the leakage failure mode the platform doc calls out. Gemini silver labels stay EVAL/teacher-reference ONLY, never in any training transcoder output.\n\n**Amplification levers (all guardrail-clean, never touch paid-LLM outputs):** (1) open-teacher pseudo-labeling (WASB/our-own-model, confidence-gated); (2) temporal augmentation (multiplies effective segments, boundary-preserving by transforming timestamps with the signal); (3) active learning to spend the Roora budget on highest-uncertainty matches; (4) timestamp/point supervision for cheap scale-out to new sports + ending_reason (~? the labeling time of full segments, near-full-supervision accuracy ? directionally from the Breakfast action-segmentation study, not measured on rally detection); (5) for Gen-2, GRPO/RFT over SFT for label-efficiency. ending_reason becomes a near-free second supervised task from the SAME intervals.\n\n**WASB trajectory** attaches as an auxiliary per-frame feature stream keyed by (video_id, frame) via trajectory_ref ? usable as TAL input features AND as a Gen-2 'trajectory-token' modality, without bloating the rally record.",
  "licensing_privacy_guardrails": "**Hard guardrails (non-negotiable, from CLAUDE.md + COMMERCIALIZATION C14 + PLATFORM §5A/§7) ? all respected in this study:**\n\n**1. Apache-2.0/MIT bases ONLY, verified per-checkpoint (licenses change per size/revision):**\n- APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0 ? verified verbatim on HF), Qwen3-VL-4B/8B (Apache-2.0 ? verified), Gemma 4 (Apache-2.0 ? UPGRADED to confirmed via PRIMARY Google source: opensource.googleblog.com + google/gemma-4-E4B HF card + ai.google.dev/gemma/docs/core/model_card_4), InternVL3 ?38B (MIT wrapper / Apache Qwen2.5 ?32B base), InternVideo2-Stage2_6B (MIT) / InternVideo2.5_Chat_8B (Apache-2.0). Architectures: ASFormer/ActionFormer/TriDet/DiffAct = MIT (verified); MS-TCN++ ONLY via sj-li/MS-TCN2 (clean MIT).\n- REJECTED: Gemma 3 (custom Gemma Terms + PUP viral flow-down ? verified ai.google.dev/gemma/terms), Qwen2.5-VL-3B (Qwen Research License, non-commercial ? verified), Qwen2.5-VL-72B + InternVL3-78B (100M-MAU cap + 'Built with Qwen' attribution ? verified), Llama Vision (700M-MAU + naming mandate), DAMO VideoLLaMA3 released weights (non-commercial AND trained on OpenAI/Gemini data), TimeChat/VTimeLLM(CC-BY-NC-ND)/Momentor(no-LICENSE) (method-reference only), VideoMAE pretrained weights (CC-BY-NC), yabufarha/ms-tcn (Commons Clause), TemporalMaxer (LICENSE UNCONFIRMED ? verify before vendoring).\n- Gemma 4 SIZE-NAME CORRECTION: official sizes are E2B / E4B / 26B-A4B (MoE) / 31B (dense) + a 12B-Unified ? there is NO plain dense '4B'; the on-device small model is E4B; native audio is limited to E2B/E4B/12B-Unified. Pull the right checkpoint.\n\n**2. No paid-LLM training data (the single biggest legal landmine):** Gemini/OpenAI/Anthropic outputs are inference/eval/teacher-REFERENCE only ? NEVER a

training signal (competing-model clauses + breach of terms). Training teachers = open Apache-2.0/MIT models only (Qwen-VL/InternVL/Gemma 4). Gemini silver labels are eval-only and stay out of every training transcoder. This is why VideoLLaMA3's instruction data is doubly disqualified (it was generated by OpenAI/Gemini).\n\n**3. Exclude minors by default + separate training opt-in:** training requires a distinct, explicit opt-in consent (separate from processing consent). Minors' video excluded from the training pool by default (DPDP under-16 + verifiable parental consent; GDPR Art. 9). Per-user adapters trained ONLY on that user's own consented video; deletion = drop the adapter. Enforced at the corpus-membership layer (consent/provenance fields) BEFORE any transcoding.\n\n**4. Generic data schema:** coach?athlete is an optional overlay, not baked in; the canonical rally record stays reusable for solo users, academies, power users.\n\n**5. Owned footage + human labels are clean to train on:** we own the video; timestamps/ending_reasons are non-copyrightable facts; WASB code is MIT. (Strong reading, not formal legal advice ? confirm with counsel before scale; per-clip consent ledger + minors exclusion + separate training opt-in are the operational guardrails.)\n\n**6. Privacy lever = local inference:** the on-device model (Gen-0 ASFormer ~1.1M params; Gen-2 4B quantized) keeps user video on the device ? no cross-border/DPDP/GDPR transfer issue at all. For any cloud TRAINING: signed DPA + explicit no-train assurance before consented video touches a backend; Modal (no-access) and Vertex (no-train-without-permission) are the two with clear written assurances; Together is BLOCKED until DPA. Route ONLY owned/consented footage, never user uploads.",
  "cost_estimates": "**Headline: the fine-tune is cheap; the deciding factor is NOT GPU $/hr but privacy terms + ops ergonomics + (above all) golden-label spend. Spend the budget on labels, not GPUs.** All $/hr figures are 2026 snapshots within a ±15-25% volatility band ? re-quote live before committing.\n\n**Gen-0 (small TAL/TAS head):** ~$0 marginal on the existing RTX 2070 Super (8GB). ASFormer/MS-TCN++ (~0.8-1.1M params, ~4.5GB training VRAM) train locally in minutes; inference is CPU-feasible. If ever rented, a full re-train is single-digit dollars. ZERO licensing/privacy exposure. This is the cheapest path AND attacks the real bottleneck ? so it is where near-term effort goes.\n\n**Gen-2 (LoRA/RFT on a 7B-class video VLM):** does NOT fit the 8GB box ? rented 24-48GB GPU mandatory (even QLoRA on video lands ~20-24GB; 24GB is a FLOOR, not a guarantee, on real 1080p/60fps clips ? must re-benchmark). Per-run ranges:\n- Self-managed LoRA on rented GPU: ~$5-40/run (?4-30 GPU-hrs). CORRECTED rates: Modal A100 80GB ~$2.50/hr (not $3.72), H100 ~$3.95/hr; RunPod Secure A100 ~$1.39-1.49/hr (not $1.19-1.29), H100 ~$2.89-3.29/hr; Lambda A100 ~$2.79/hr. RunPod is cheapest self-serve.\n- Managed token-priced (Fireworks/Together): LoRA SFT ?16B ~$0.48/M tokens ? ~$2-30 for a 5k-20k-pair run IF text-priced. VIDEO/multimodal token counts can BALLOON cost ? UNVERIFIED; confirm on real frame counts before budgeting.\n- Full multimodal fine-tune (not LoRA): ~10-15× heavier ? ~$70-300/epoch on 4×A100.\n- RFT/GRPO: label-efficient but compute-heavy (multiple rollouts/sample) ? plan cloud burst.\n\n**Vertex/GCP (India-residency option):** most expensive + overhead. A100 40GB ~$3.37/hr all-in; 80GB ~$5.07/hr+. CORRECTION: a real India-residency run is on H100-class A3 (pricier), NOT A100, and is capacity-gated.\n\n**Inference economics (COMMERCIALIZATION C7):** self-hosted 7B VLM (?16GB VRAM, ~14GB FP16 / ~3.5GB INT4) beats Gemini per-call ONLY once (a) sustained QA volume is high, (b) the consented dataset is defensibly large, (c) per-call API cost pain is real. Below those thresholds, paid Gemini (~$0.05 input for a 10-min clip) is cheaper than a standing GPU + second stack. BENCHMARK self-hosted-7B $/call vs Gemini on the target workload before committing budget ? hence paid-primary now, owned-primary/paid-fallback later.\n\n**Highest-ROI spend is NOT compute ? it is golden labels (Roora/rally-annotator).** Getting n?2?n?5 matches unblocks honest heuristic tuning immediately and is a small Roora batch.",
  "risks": [
    "DOMINANT EMPIRICAL UNKNOWN: ASFormer's from-scratch-on-small-data result used Kinetics-pretrained I3D 2048-d features; feeding raw ~3-4-dim WASB trajectory is OFF the validated path and may underperform until a hand-built feature stack (velocity/acceleration/visibility-run-length/net-crossing) is added. A from-scratch learned model still may not clear the ~0.73 heuristic bar at n=3-10. The actual ship/no-ship gate is an A/B on the golden set, which cannot be run in this dev container (no GPU/torch/cv2/numpy).",
    "DATA STARVATION, not method failure, is the binding bottleneck: the learned model loses to the heuristic at n=2 because n<~5 is too few for stable cross-val. The fix is golden DATA + WASB recall, confirmed by our own docs. No model choice solves this.",
    "WASB recall ceiling propagates: pseudo-labels can only be as complete as the WASB teacher sees; confidence-gate + human-spot-check, keep human [start,end] as the boundary authority. Option B is partly motivated by removing this ceiling ? but only after the corpus is healthy.",
    "Gemma 4 native-video premise was internally inconsistent across the source findings (one section VERIFIED, one AMBER). Verification RESOLVES this in favor of clean+native-video per the primary Google model card ? but timestamp-EMISSION format for grounding is LESS documented than Qwen's, so Gemma 4 needs more grounding-output teaching; confirm a working timestamp recipe

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before locking it as the Option-B base.",
    "Amplification is LOAD-BEARING for Gen-2: 128 raw rallies alone cannot reach the single-digit-thousands clip?QA band; if augmentation + QA-multiplexing + pseudo-labels underdeliver, the Option-B fine-tune is under-fed. The 5k-20k band is itself an estimate; amateur single-camera 60fps footage may need MORE labels than broadcast-trained literature implies (domain transfer is the dominant unknown).",
    "VIDEO/multimodal fine-tuning VRAM + token cost is UNVERIFIED ? most cited data is text-LLM-centric. Real 1080p/4K-60fps clip sequences can push VRAM and token counts (hence cost) far beyond the $10-50/one-A100 LoRA estimate; re-benchmark on real clip-QA before trusting Gen-2 budget.",
    "Licenses change per-checkpoint/per-revision ? the 'Gemma' shorthand is a live foot-gun (only Gemma 4 is Apache; Gemma 3 stays rejected). Re-verify the EXACT license file on the specific HF revision before any code lands; verify Gemma 4 size names (E2B/E4B/26B/31B, no plain 4B) so the right checkpoint is pulled.",
    "GPU prices are volatile (several cited figures were stale by ±8-50%); RunPod no longer publishes separate Secure-Cloud rates; vast.ai is a live marketplace. Re-quote live before budgeting. Together AI has no public no-train clause/DPA ? BLOCKED for the corpus until signed.",
    "India residency-at-rest is capacity-gated on GCP (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100) ? treat as conditional, not turnkey, if a legal residency requirement emerges.",
    "Sequencing risk: starting owned-model implementation before P0-P4 scaffolding is solid would strand the model on a single box and violate the owner directive. This study is DESIGN-only; even the safe-now items (corpus format, n?5) need owner sign-off before code."
  ],
  "open_questions": [
    "Empirical A/B on the golden set: does ASFormer (or ActionFormer) over a hand-built WASB feature stack actually beat the ~0.73 heuristic+LogReg baseline at n?5 matches? Cannot be answered in this dev container ? needs the GPU/WSL machine-side suite.",
    "What feature stack does the low-dim WASB trajectory need (velocity, acceleration, visibility-run-length, net-crossing flags, smoothing/windowing) to give ASFormer's dilated convs enough signal? Unproven in the literature for ~3-4-dim input.",
    "What is the real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on OUR 1080p/60fps clip?QA pairs (actual frame counts), vs the cited text-LLM estimates?",
    "Does Gemma 4 have a working, documented timestamp-EMISSION recipe for moment localization (its grounding output format is less documented than Qwen's), and which exact checkpoint (E4B vs 12B-Unified) is the right Option-B base?",
    "What is the measured self-hosted-7B $/call + latency vs Gemini paid on the target workload (COMMERCIALIZATION C7) ? i.e. is sustained QA volume high enough to justify the owned-model serving stack?",
    "Will Together AI / Lambda / Azure sign a DPA with an explicit no-train assurance for consented video, and is in-region (India) configurable? (Gates whether they're usable for the corpus at all.)",
    "Can GCP asia-south1 H100-class A3 quota actually be obtained (account-team gated) if India residency-at-rest becomes a legal requirement?",
    "How many clip?QA pairs does augmentation + QA-task-multiplexing + confident pseudo-labels realistically yield from 128 rallies, and is that enough to clear the RFT cold-start bar for Option B?",
    "What is the right clip windowing/sampling policy (clip length, stride, negative no-rally-clip ratio) for the VLM clip?QA transcoder ? the one design choice that affects label balance?"
  ],
  "sources": [
    "https://modal.com/docs/guide/security (Modal no-access contract + 7-day retention + SOC2 II ? verified)",
    "https://modal.com/pricing (CORRECTED Modal A100 ~$2.50/hr, H100 ~$3.95/hr, region 1.5-1.75x)",
    "https://www.runpod.io/pricing (CORRECTED A100 ~$1.39-1.49/hr, H100 ~$2.89-3.29/hr)",
    "https://docs.runpod.io/references/security-and-compliance (Secure Cloud host-no-inspect ToS + GDPR + partner certs)",
    "https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards",
    "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-data-residency-guarantees-for-data-stored-at-rest (Vertex no-train verbatim; gen-AI residency excludes asia-south1)",
    "https://docs.cloud.google.com/compute/docs/regions-zones/gpu-regions-zones (asia-south1 = A3 H100/H200, NOT A100; capacity-gated)",
```

"https://cloud.google.com/vertex-ai/pricing",
"https://lambda.ai/pricing (CORRECTED A100 ~$2.79/hr, H100 ~$3.99/hr)",
"https://www.together.ai/fine-tuning (no explicit no-train clause / no public DPA ? BLOCKED until signed)",
"https://docs.together.ai/docs/fine-tuning-pricing (LoRA ?16B $0.48/M tokens, text rate)",
"https://fireworks.ai/pricing + https://docs.fireworks.ai/guides/security_compliance/data_security (contractual no-train + ZDR)",
"https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct (Apache-2.0 + temporal grounding via mRoPE/absolute-time)",
"https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE (Qwen Research, non-commercial ? REJECT)",
"https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE (100M-MAU cap + attribution ? REJECT)",
"https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct + .../Qwen3-VL-8B-Instruct (Apache-2.0 + Text?Timestamp Alignment)",
"https://ai.google.dev/gemma/docs/core/model_card_4 (Gemma 4 Apache-2.0 + native video @1fps/60s + audio E2B/E4B/12B ? UPGRADED to confirmed)",
"https://opensource.googleblog.com/2026/03/gemma-4-expanding-the-gemmaverse-with-apache-20.html + https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/",
"https://ai.google.dev/gemma/terms (Gemma 1/2/3 viral terms + PUP flow-down ? Gemma 3 REJECT)",
"https://huggingface.co/OpenGVLab/InternVL3-8B + .../InternVL3-78B (MIT wrapper; 78B inherits Qwen2.5-72B MAU cap ? REJECT 78B)",
"https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B (MIT) + .../InternVideo2_5_Chat_8B (Apache-2.0)",
"https://huggingface.co/MCG-NJU/videomae-base (CC-BY-NC weights ? REJECT)",
"https://github.com/DAMO-NLP-SG/VideoLLaMA3 (Apache code, NC weights trained on OpenAI/Gemini data ? paid-LLM guardrail trip)",
"https://github.com/ChinaYi/ASFormer + https://ar5iv.labs.arxiv.org/html/2110.08568 (MIT; from-scratch-on-28-videos, ~1.1M params; I3D-2048 input nuance)",
"https://github.com/happyharrycn/actionformer_release (MIT; benchmark-scale, no few-shot claim)",
"https://github.com/dingfengshi/TriDet/blob/master/LICENSE (MIT)",
"https://github.com/yabufarha/ms-tcn/blob/master/LICENSE (Commons Clause ? non-commercial) vs https://github.com/sj-li/MS-TCN2/blob/master/LICENSE (clean MIT)",
"https://github.com/Finspire13/DiffAct/blob/main/LICENSE (MIT)",
"https://github.com/huangb23/VTimeLLM (CC-BY-NC-ND) + https://github.com/DCDmllm/Momentor (no LICENSE) + https://github.com/RenShuhuai-Andy/TimeChat (NC research preview; BSD code layer)",
"https://arxiv.org/abs/2504.06958 (VideoChat-R1: RFT +31.8 temporal grounding with limited samples)",
"https://arxiv.org/abs/2507.18100 (VTG-R1: filtered 13K cold-start beats unfiltered 56K)",
"https://openaccess.thecvf.com/content/ICCV2023/papers/Xia_Learning_from_Noisy_Pseudo_Labels...ICCV_2023_paper.pdf (SS-TAL confidence-gated pseudo-labels)",
"https://arxiv.org/pdf/2103.06669 (timestamp supervision near-full-accuracy 64.1% vs 68.0% on Breakfast)",
"https://github.com/modelscope/ms-swift (only framework advertising native video+audio training)",
"Project docs: docs/COMMERCIALIZATION.md (C4/C7/C14), docs/PLATFORM_ARCHITECTURE.md (§5A/§6 P0-P5/§7), docs/ROADMAP.md (ADR-004 WASB substrate), backend/eval/metrics.py::match_intervals + backend/eval/training.py::label_candidates (the label-reuse spine, already in code)"
]
}

RESEARCH:

[
{
"dimension": "Training compute platforms for an owned video-QA (rally temporal segmentation) model",
"summary": "A LoRA/QLoRA fine-tune of an Apache-2.0 7B VLM on our 5k-20k clip-QA pairs is a small job: under 24h on one A100 80GB, $10-50 total. So the deciding factor is privacy terms and operational ergonomics, not GPU dollar-per-hour. Modal has the strongest written no-access contract; Vertex AI is the only strong-privacy option with an India region. The RTX 2070 Super 8GB is dev/inference-only, too small for real 7B-VLM training.",

```
    "options": [
      {
        "name": "Self-hosted RTX 2070 Super 8GB + WSL2",
        "what": "Local single-GPU fine-tuning/inference via ms-swift/Unsloth/QLoRA. Zero
marginal cost, zero data egress.",
        "pros": [
          "Maximal privacy: data never leaves the machine - the local-first moat",
          "Zero cost; ideal free dev/CI smoke-test box and 4-bit 7B inference",
          "No lock-in, no DPA, no billing surprises"
        ],
        "cons": [
          "8GB too small for a real 7B-VLM QLoRA run (needs ~16-24GB with vision/video tokens;
4-bit 7B text inference alone ~7.1GB)",
          "Turing lacks bf16/modern FlashAttention kernels",
          "Single GPU; unsuitable for Option B end-to-end multimodal training"
        ],
        "cost_note": "$0 marginal (owned). Dev/CI/inference only.",
        "license": "N/A (own hardware)",
        "privacy_note": "BEST possible - data never leaves device.",
        "fit_for_us": "medium"
      },
      {
        "name": "Modal (serverless GPU) - RECOMMENDED PRIMARY",
        "what": "Serverless Python-native compute, per-second billing, scale-to-zero, A100/H100.
Has a Training product.",
        "pros": [
          "Strongest privacy contract: Modal 'will never access your source code, the
inputs/outputs to your Functions, or any data you store'; inputs/outputs deleted within 7 days;
SOC2 Type II; HIPAA via BAA",
          "Per-second billing + scale-to-zero = no idle cost, perfect for retrain-on-each-
golden-batch cadence",
          "Python/recipe-driven - fits a one-command agent-runnable harness and the future
ComputeBackend abstraction",
          "Asia-Pacific regions (Singapore, Seoul); $30/mo free credits"
        ],
        "cons": [
          "No India asia-south1 region (Singapore closest) - fine for batch training but not
India residency-at-rest",
          "A100 ~$3.72/hr, H100 ~$4.29/hr - pricier per-hour than RunPod/Vast",
          "Regional multipliers 1.25x-2.5x for non-US"
        ],
        "cost_note": "A100 80GB ~$3.72/hr, H100 ~$4.29/hr per-second. Full LoRA fine-tune
~$10-$50. $30/mo free credits.",
        "license": "Commercial cloud; you keep all weights/data. Qwen2.5-VL-7B Apache-2.0 base
clean.",
        "privacy_note": "STRONGEST written no-access/short-retention terms found. Meets the hard
requirement.",
        "fit_for_us": "strong"
      },
      {
        "name": "RunPod (Secure Cloud tier) - RECOMMENDED FALLBACK",
        "what": "Rented GPU pods + serverless. Secure Cloud (vetted DCs) vs Community Cloud
(unvetted hosts).",
        "pros": [
          "Cheap A100 80GB ~$1.19-$1.29/hr, H100 ~$2.65/hr (spot ~$1.19) - best value for
longer/multi-GPU runs",
          "Secure Cloud: SOC2 Type II, ISO 27001, GDPR, HIPAA; AES-256 at rest; ToS bars hosts
inspecting pod data",
          "Per-tenant container isolation; instant data destruction on delete; serverless +
checkpointing; has a DPA"
        ],
        "cons": [
          "Community Cloud tier unvetted peer-to-peer - MUST restrict corpus to Secure Cloud
only",
          "No India/Asia datacenter - no India residency",
```

          "Less one-command-reproducible than Modal; USD billing"
        ],
        "cost_note": "A100 80GB ~$1.19-$1.29/hr (Secure), H100 ~$2.65/hr on-demand / ~$1.19 spot. Our fine-tune ~$3-$15.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "STRONG on Secure Cloud. MUST avoid Community Cloud for sensitive data.",
        "fit_for_us": "strong"
      },
      {
        "name": "Google Vertex AI / GCP Compute Engine GPUs - KEEP ON BENCH",
        "what": "Managed Vertex custom-training jobs or raw Compute Engine GPU VMs. The platform the user named.",
        "pros": [
          "Written: 'Customer data is never used by Google. We do not use customer data to train our models, or models used by others, without permission'",
          "India region asia-south1 (Mumbai) for Compute Engine GPUs - genuine India residency-at-rest, unique among strong options",
          "Mature multi-GPU/checkpointing/VPC/CMEK; strong ecosystem maps to future Storage/Compute backends"
        ],
        "cons": [
          "Most expensive + most overhead: Vertex mgmt fee (A100 ~$3.37/hr all-in, H100 ~$11.27/hr); easy to overspend on always-on resources",
          "Heavyweight IAM/quota/capacity friction; slowest path to a reproducible harness",
          "gen-AI residency region list excluding asia-south1 applies to MANAGED gen-AI APIs, not custom training on your own weights (asia-south1 GPUs ARE usable)"
        ],
        "cost_note": "A100 ~$3.37/hr all-in, H100 ~$11.27/hr; L4/T4 cheap. Highest per-hour + overhead.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "STRONG written no-train; ONLY strong option with an India (Mumbai) region for residency.",
        "fit_for_us": "medium"
      },
      {
        "name": "Lambda Labs (Lambda Cloud)",
        "what": "Pure ML-focused GPU cloud, simple on-demand pricing, free egress.",
        "pros": [
          "ML-native UX, pre-baked CUDA/PyTorch; A100 80GB ~$1.99-$2.49/hr, H100 ~$3.29-$4.29/hr",
          "Free data egress (vs AWS $0.09/GB, GCP $0.12/GB)",
          "Good multi-GPU clusters for Option B"
        ],
        "cons": [
          "No spot/preemptible tier",
          "Privacy terms less explicit publicly - needs DPA review before the corpus goes on it",
          "US-only, no India/Asia region; historically tight capacity"
        ],
        "cost_note": "A100 80GB ~$1.99-$2.49/hr, H100 ~$3.29-$4.29/hr, free egress. Our fine-tune ~$5-$20.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "UNVERIFIED - obtain/read DPA before using for consented video.",
        "fit_for_us": "medium"
      },
      {
        "name": "Together AI (managed fine-tuning API)",
        "what": "Managed LoRA/full fine-tuning of open models priced per training-token; hosted serving.",
        "pros": [
          "Lowest operational burden: managed LoRA ~$0.48/M tokens (<=16B); fastest path to a first checkpoint",
          "Supports Qwen and 'any open-source model from HF Hub'; resume-from-checkpoint",
          "SOC2 Type II, HIPAA-aligned, Asia/ME deployment; 'data and models remain fully under

your ownership'"
        ],
        "cons": [
          "Could NOT confirm a verbatim no-train clause or public DPA - BLOCKED for the
consented corpus until signed DPA obtained",
          "Token pricing opaque for VIDEO/multimodal; Qwen2.5-VL video FT support is the open
question",
          "Managed = ship data to their cloud; weakest fit with 'data never leaves device'"
        ],
        "cost_note": "LoRA ~$0.48/M tokens (<=16B); ~$8-$12/M typical. Video token cost
unverified.",
        "license": "Commercial managed; you own resulting weights (per claim) - verify
downloadability.",
        "privacy_note": "UNCONFIRMED no-train clause + no public DPA -> BLOCKED for the corpus
until signed DPA + no-train assurance.",
        "fit_for_us": "weak"
      },
      {
        "name": "vast.ai (marketplace) - Secure tier only",
        "what": "Peer-to-peer GPU marketplace, cheapest rates. Community (unvetted) vs Secure
Cloud (ISO-27001+ vetted DCs).",
        "pros": [
          "Cheapest anywhere: A100 80GB ~$0.67/hr, H100 ~$1.87/hr, spot to ~$0.34/hr - great for
non-sensitive experiments",
          "Secure Cloud: vetted DCs (ISO 27001 min), SOC2 Type II platform-level, per-tenant
containers, instant data destruction"
        ],
        "cons": [
          "Community/peer-to-peer tier unvetted - DIRECT conflict with our hard privacy
requirement for consented video",
          "Price volatility; reliability varies host-to-host; no India region; rough
operability"
        ],
        "cost_note": "A100 80GB ~$0.67/hr, H100 ~$1.87/hr (spot ~$0.34/hr). Cheapest by 2-5x -
Secure tier only, ideally non-sensitive.",
        "license": "Marketplace; you keep weights/data.",
        "privacy_note": "Community tier FAILS the requirement. Secure Cloud acceptable for non-
consented/synthetic data.",
        "fit_for_us": "weak"
      },
      {
        "name": "AWS SageMaker (EC2 P4d/P5) - Mumbai ap-south-1",
        "what": "Enterprise managed training or raw EC2 GPU. ap-south-1 (Mumbai) fully
supported.",
        "pros": [
          "India residency: ap-south-1 full region with SageMaker + P4d (A100)",
          "VPC isolation by default, CMEK/KMS encryption, GDPR-aligned; can opt out of
aggregated-metadata sharing",
          "P4/P5 price cuts up to 45% (2025); spot + savings plans; mature distributed training"
        ],
        "cons": [
          "High complexity + cost for a solo founder; SageMaker mgmt premium; spot
interruptions",
          "Default aggregate library-usage metadata collection (opt-out is a footgun)",
          "Steep learning curve; heavy SDK lock-in; among the most expensive per-hour"
        ],
        "cost_note": "p4d (8xA100) ~$3.07-$3.43/hr/GPU-equiv; P5 (H100) ~$6.88/hr on-demand /
~$3.83 spot; plus SageMaker premium.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "STRONG (VPC+KMS+GDPR, India region) IF you opt out of metadata sharing
and use VPC. Operationally heavy.",
        "fit_for_us": "weak"
      },
      {
        "name": "Azure ML - Central India",

        "what": "Enterprise managed ML on Azure NC/ND A100/H100 VMs; Central India region.",
        "pros": [
          "India residency: Central India with A100/H100",
          "Cheap spot: NC A100 v4 ~$0.82/hr spot vs ~$3.67/hr on-demand",
          "Enterprise compliance (GDPR/ISO/SOC), VPC/private endpoints, CMEK; good multi-GPU"
        ],
        "cons": [
          "Could NOT confirm an Azure ML 'we don't train on your data' statement - needs
Microsoft DPA review",
          "On-demand H100 very expensive (~$6.98-$12/hr/GPU); only spot attractive",
          "Heavy enterprise tooling; high overhead; pipeline lock-in"
        ],
        "cost_note": "NC A100 v4 ~$3.67/hr on-demand / ~$0.82/hr spot; ND H100 v5
~$6.98-$12/hr/GPU. India region available.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "Likely STRONG via Microsoft DPA + private endpoints, UNVERIFIED here -
read the DPA first.",
        "fit_for_us": "weak"
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        "name": "CoreWeave / Paperspace (DigitalOcean Gradient)",
        "what": "CoreWeave = GPU-specialist for large reserved fleets; Paperspace = simpler
notebook/VM GPU cloud.",
        "pros": [
          "CoreWeave: SOC2/HIPAA/GDPR/ISO 27001; strong for very large multi-GPU H100 fleets
(Option B if big)",
          "Paperspace: no minimum commitment, notebook-friendly (A100 ~$3.09/hr, H100 ~$5.95/hr
promo)"
        ],
        "cons": [
          "CoreWeave enterprise/reserved-oriented (H100 ~$6.16/hr/GPU, A100 ~$2.70/hr) -
overkill/pricey for small jobs",
          "Paperspace mid-to-high pricing, no spot; tight capacity",
          "Neither has an India region; neither beats Modal/RunPod on privacy specificity"
        ],
        "cost_note": "CoreWeave A100 ~$2.70/hr, H100 ~$6.16/hr/GPU. Paperspace A100 ~$3.09/hr,
H100 ~$5.95/hr. No India region.",
        "license": "Commercial cloud; you keep weights/data.",
        "privacy_note": "CoreWeave has the cert stack; Paperspace less differentiated.
Acceptable, not better than Modal/RunPod.",
        "fit_for_us": "weak"
      }
    ],
    "key_sources": [
      "https://modal.com/docs/guide/security",
      "https://modal.com/pricing",
      "https://modal.com/blog/region-selection-launch",
      "https://www.runpod.io/pricing",
      "https://docs.runpod.io/references/security-and-compliance",
      "https://www.runpod.io/press/runpod-meets-hipaa-and-gdpr-standards",
      "https://cloud.google.com/blog/products/ai-machine-learning/google-cloud-generative-ai-
data-residency-guarantees-for-data-stored-at-rest",
      "https://cloud.google.com/vertex-ai/pricing",
      "https://lambda.ai/pricing",
      "https://www.together.ai/fine-tuning",
      "https://docs.together.ai/docs/fine-tuning-pricing",
      "https://vast.ai/pricing",
      "https://docs.vast.ai/guides/reference/faq/security",
      "https://aws.amazon.com/sagemaker/pricing/",
      "https://docs.aws.amazon.com/sagemaker/latest/dg/data-protection.html",
      "https://azure.microsoft.com/en-us/pricing/details/machine-learning/",
      "https://learn.microsoft.com/en-us/azure/virtual-machines/sizes/gpu-
accelerated/nca100v4-series",
      "https://www.coreweave.com/pricing",
      "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",

"https://willitrunai.com/can-run/qwen-2.5-7b-on-rtx-2070-super-8gb",
"https://kaitchup.substack.com/p/qwen25-qlora-lora-and-full-fine-tuning"
],
"recommendation": "Decouple from raw GPU price: our fine-tune (Apache-2.0 Qwen2.5-VL-7B + LoRA/QLoRA on 5k-20k clip-QA pairs) is a sub-24h, $10-$50 job on one A100 80GB, so privacy terms + operational ergonomics dominate. Plan: (1) DEV/CI on the RTX 2070 Super (8GB) - free; harness plumbing, overfit-one-batch sanity, 4-bit 7B inference, NOT real training. (2) PRIMARY = Modal - strongest written privacy contract ('Modal will never access your source code, inputs/outputs, or stored data'; 7-day max retention; SOC2 Type II), per-second billing with scale-to-zero matching the retrain-on-each-golden-batch cadence, Python/recipe-driven model that slots into a one-command agent-runnable harness and the future ComputeBackend abstraction; run a thin glue over ms-swift on Modal A100s. (3) FALLBACK / Option-B scale-up = RunPod Secure Cloud (A100 ~$1.19-$1.29/hr, SOC2+GDPR+HIPAA, contractual no-inspect) - Secure Cloud only, never Community. (4) KEEP Vertex AI / GCP on the bench: ONLY strong-privacy choice with an India (asia-south1 Mumbai) region + written no-train-on-customer-data - adopt if/when India residency or an audited managed pipeline is mandated. (5) Use vast.ai/Lambda ONLY for non-sensitive synthetic/ablation runs - never the consented corpus on vast.ai community tier. (6) HARD GATE before any consented video reaches Together / Lambda / Azure: signed DPA + explicit written 'we do not train on or retain your data' assurance; until then BLOCKED for the corpus. Sequencing: unblocks PLATFORM P5 design only - do not implement until P0-P4 (async jobs, storage/compute abstractions) ship, per owner directive.",
"confidence": "medium",
"caveats": [
"Pricing is volatile and sourced largely from aggregator blogs plus some primary pages; treat all dollar-per-hour figures as +/-15-25% and re-check live pricing pages before committing budget.",
"Biggest UNVERIFIED gap: VIDEO/multimodal fine-tuning support and cost. Most data found is LLM-text-centric. Qwen2.5-VL VIDEO fine-tuning (frames as image tokens) can balloon token counts and VRAM far beyond text-7B numbers - the $10-$50/one-A100 estimate is for a modest LoRA run and must be re-benchmarked on real clip-QA data with actual frame counts before trusting it for Option B.",
"Together AI, Lambda Labs, and Azure ML privacy terms could NOT be confirmed to include an explicit no-train-on-your-data clause from public pages; flagged as needing a signed DPA before the corpus touches them.",
"Vertex/GCP nuance: the published gen-AI residency region list (excludes asia-south1) governs Google's MANAGED gen-AI APIs, not custom training on your own weights via Compute Engine GPUs - asia-south1 GPUs exist and the no-train principle applies, but confirm A100/H100 quota/capacity in asia-south1 (capacity-constrained).",
"RTX 2070 Super: 4-bit 7B text inference fits ~7.1GB and QLoRA training of a 7B can fit ~7GB in ideal text-only conditions, but a 7B-VLM with video frames realistically needs 16-24GB; treat the 2070S as dev/inference-only. Turing also lacks bf16/modern kernels.",
"India-region: only Vertex/GCP, AWS (ap-south-1), Azure (Central India) offer in-India residency among providers studied; Modal/RunPod/Lambda/Vast/CoreWeave/Paperspace do not. For batch TRAINING this is usually irrelevant, but if a legal residency-at-rest requirement emerges only the three hyperscalers qualify.",
"Compliance certs (SOC2/GDPR/HIPAA) attest to security controls, NOT a 'we won't train on your data' promise. The two clear written no-train/no-access assurances found are Modal (no-access) and Google/Vertex (no-train-without-permission); everything else needs contract review."
]
},
{
"dimension": "Model architecture for rally temporal localization/segmentation (action = rally; labels = [start,end] intervals) over per-frame features",
"summary": "For OUR situation ? per-frame features already in hand (WASB trajectory ~3-4 dims + optional audio/pose), only ~3-10 labeled matches, RTX 2070S/8GB, commercial intent, on-device-capable ? the right model class is a LIGHT feature-input temporal model, NOT a heavy end-to-end video network. Two viable families, both consuming pre-extracted per-frame feature sequences (so no RGB pipeline required and they ride directly on WASB output):\n\n(1) TEMPORAL ACTION LOCALIZATION (TAL) ? ActionFormer / TriDet / TemporalMaxer. These output (start, end, class) instances directly via a multi-scale feature pyramid + classification/regression heads. This is the most NATURAL fit for rally [start,end] + ending_reason: each rally is one detected instance; ending_reason becomes the instance class (multi-class head) or a second head. ActionFormer and TriDet are MIT and accept ARBITRARY input feature dimension via a learned input projection (confirmed: the projection maps any D-dim feature to the model width), so the WASB (X,Y,Visibility) trajectory can be fed directly ? I3D is NOT required, just the default choice

on benchmarks.\n\n(2) TEMPORAL ACTION SEGMENTATION (TAS) ? MS-TCN++ / ASFormer / DiffAct. These do per-frame dense labeling (frame -> {rally, non-rally}); rally segments are recovered by merging contiguous 'rally' frames. ASFormer is the standout for OUR data-hunger: it is explicitly designed to train FROM SCRATCH on tiny datasets (GTEA = 28 videos, 11 classes) with only ~1.1M params, by injecting local convolutional inductive bias to compensate for the Transformer's missing inductive bias on small data. This is the closest published analogue to our n=3-10 regime.\n\nKEY LICENSE FINDING (a real trap): the ORIGINAL ms-tcn repo (yabufarha/ms-tcn) is MIT + Commons Clause = NON-commercial; do NOT vendor it. The MS-TCN++ repo (sj-li/MS-TCN2), ASFormer (ChinaYi/ASFormer), DiffAct (Finspire13/DiffAct), ActionFormer (happyharrycn/actionformer_release), and TriDet (dingfengshi/TriDet) are all clean MIT and commercially usable. TemporalMaxer's LICENSE file could not be confirmed (404 on raw fetch) ? treat as 'verify before vendoring'.\n\nBADMINTON PRECEDENT: published systems for the adjacent task (per-frame hit detection from shuttle trajectory + pose) use exactly the light-sequence-model recipe ? GRU-based recurrent nets over court/pose/shuttle features predicting per-frame hit/no-hit (the MonoTrack lineage), and rule/threshold pipelines over TrackNet+YOLO. None use heavy end-to-end video models for the localization step. This validates: a SMALL sequence model over per-frame features is the field-standard approach for racket-sports temporal structure at our data scale.\n\nNONE of these are the 'Option B' end-to-end multimodal VLM (Qwen-VL/Gemma) from the grounding brief ? they are the bridge between today's heuristic+LogisticRegression and that future VLM. They are the right FIRST owned model precisely because they learn from the labels and features we ALREADY have, with no paid-LLM data and no large pretraining corpus.",
    "options": [
      {
        "name": "ASFormer (TAS) ? RECOMMENDED first owned model",
        "what": "Transformer for action segmentation that adds local convolutional inductive bias (dilated temporal convs) to a vanilla Transformer so it trains from scratch on small data. Per-frame dense classifier: every frame -> {rally, non-rally(, ending_reason)}; rally [start,end] recovered by merging contiguous rally frames + boundary cleanup. Input = per-frame feature sequence of arbitrary dim (we feed WASB X,Y,Visibility +/- audio-onset +/- pose). ~1.1M params.",
        "pros": [
          "Explicitly engineered for SMALL training sets ? trains from scratch on GTEA (28 videos); the only surveyed model whose paper makes the small-data claim its central thesis, matching our n=3-10 regime",
          "Tiny (~1.1M params) -> trivially fits RTX 2070S 8GB; trains in minutes; on-device/edge inference is easy",
          "Consumes per-frame features directly ? rides on existing WASB trajectory output; no RGB/I3D extraction pipeline needed; trivially fuses audio-onset + pose by concatenating feature channels (directly enables Option A modular fusion)",
          "Clean MIT (ChinaYi/ASFormer) ? commercially usable, fine-tunable, wrappable",
          "Per-frame formulation is forgiving of label noise and variable rally length; ending_reason maps to extra per-frame classes or a per-segment head",
          "Beats our current ceiling plausibly: it LEARNS temporal structure rather than threshold-tuning, and degrades gracefully where the LogisticRegression+windowing baseline plateaued (~F1 0.73)"
        ],
        "cons": [
          "Per-frame segmentation needs a post-merge step to produce clean [start,end] intervals (minor engineering; over/under-segmentation is the classic TAS failure mode)",
          "No native 'instance' concept ? two rallies separated by a 1-frame gap can merge; needs a small gap/duration prior (we already have the net-crossing gate to reuse)",
          "Standard recipe assumes I3D features; using raw low-dim WASB features is off-the-beaten-path and may need a feature-smoothing / windowing front-end to give the convs enough signal",
          "Boundary precision is frame-granular, not sub-frame regressed (TAL models regress boundaries more precisely)"
        ],
        "cost_note": "Free/OSS. Training cost ~minutes on the existing RTX 2070S; no cloud GPU required at this data scale. Inference is CPU-feasible.",
        "license": "MIT (ChinaYi/ASFormer) ? verified, commercially clean",
        "privacy_note": "Fully local; ~1.1M-param model runs on-device, user video/features never leave the box. No paid-LLM data anywhere in the loop.",
        "fit_for_us": "strong"
      },
      {

        "name": "ActionFormer (TAL) ? RECOMMENDED parallel/second model",
        "what": "Single-stage Transformer for temporal action localization. Multi-scale feature pyramid + shared classification/regression heads emit (start, end, class) action instances directly at every time step. Input = per-frame feature sequence; a learned input-projection layer accepts ARBITRARY feature dimension (confirmed), so WASB trajectory feeds in without I3D.",
        "pros": [
          "Most NATURAL label mapping: rally = one detected instance with regressed [start,end]; ending_reason = the instance class (multi-class head). Matches our CSV schema (rally_number,start,end,ending_reason) almost 1:1",
          "Sub-frame boundary regression -> more precise [start,end] than frame-merged TAS, directly attacking our F1 ceiling on boundary localization",
          "Input projection accepts any feature dim ? WASB (X,Y,Visibility) + audio + pose concatenated; no RGB pipeline",
          "Clean MIT (happyharrycn/actionformer_release); strong, well-maintained, widely forked codebase; TriDet/TemporalMaxer build on it so swapping later is cheap",
          "Standard, agent-runnable training recipe (config-driven) fits the 'one command + report card' harness goal"
        ],
        "cons": [
          "More data-hungry than ASFormer ? designed/validated on THUMOS/ActivityNet (hundreds-to-thousands of videos); benchmark papers do NOT claim few-shot, so at n=3-10 it risks overfitting and may UNDERPERFORM ASFormer initially (same failure we saw when a learned model lost to the baseline at n=2)",
          "Bigger than ASFormer (multi-scale pyramid + attention) though still small vs a VLM; needs more careful regularization/augmentation on tiny data",
          "Default features are I3D; low-dim trajectory-only input is unproven and may need feature engineering to give attention enough to attend to",
          "NMS / instance de-duplication tuning adds knobs"
        ],
        "cost_note": "Free/OSS. Trains on RTX 2070S; somewhat heavier than ASFormer but still no cloud GPU needed at our scale.",
        "license": "MIT (happyharrycn/actionformer_release) ? verified, commercially clean",
        "privacy_note": "Fully local, small model, on-device inference; no paid-LLM data.",
        "fit_for_us": "strong"
      },
      {
        "name": "TriDet (TAL)",
        "what": "One-stage TAL built ON TOP of ActionFormer's codebase. Adds a Trident-head modeling the action boundary as a relative probability distribution + Scalable-Granularity Perception (SGP) layer replacing some self-attention. Same (start,end,class) instance output and same arbitrary-feature input as ActionFormer.",
        "pros": [
          "Better boundary modeling than ActionFormer (the Trident-head is purpose-built for precise [start,end] ? directly relevant to our boundary-localization ceiling)",
          "Drop-in once ActionFormer is wired (shares ActionFormer's data/feature plumbing), so it's a cheap upgrade path, not a separate integration",
          "Clean MIT (dingfengshi/TriDet); SGP reduces reliance on full self-attention -> a bit lighter/faster",
          "Releases VideoMAEv2 feature configs ? useful if we later add an RGB cue"
        ],
        "cons": [
          "Same small-data risk as ActionFormer (benchmark-scale validation; no few-shot claim) ? not the right thing to START with at n=3-10",
          "More moving parts than ActionFormer; marginal gains may be invisible until the corpus grows",
          "Off-benchmark low-dim trajectory input equally unproven here"
        ],
        "cost_note": "Free/OSS; same hardware envelope as ActionFormer.",
        "license": "MIT (dingfengshi/TriDet) ? verified",
        "privacy_note": "Fully local; no paid-LLM data.",
        "fit_for_us": "medium"
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      {
        "name": "MS-TCN++ (TAS)",

```
        "what": "Multi-stage temporal convolutional network: stage 1 makes an initial per-frame prediction, refinement stages iteratively smooth it. Pure dilated temporal convs (no attention). Per-frame {rally/non-rally} -> merge to intervals. ~0.8M params.",
        "pros": [
          "Smallest/simplest (~0.8M params, conv-only) -> fastest to train, easiest on 8GB, strongest edge/on-device story",
          "Conv inductive bias handles small data reasonably (precursor to ASFormer's design); multi-stage refinement directly fights over-segmentation",
          "Clean MIT on the CORRECT repo (sj-li/MS-TCN2)",
          "Battle-tested, minimal dependencies, easy to read/modify -> good fit for the 'minimal/readable harness' goal"
        ],
        "cons": [
          "LICENSE TRAP: the original yabufarha/ms-tcn repo is MIT + Commons Clause (NON-commercial). MUST vendor sj-li/MS-TCN2 (clean MIT), not the original ? easy to get wrong",
          "Generally a notch below ASFormer in accuracy on the same benchmarks (no attention; ASFormer was designed to beat it)",
          "Same per-frame->interval merge + over/under-segmentation concerns as all TAS",
          "Frame-granular boundaries, no sub-frame regression"
        ],
        "cost_note": "Free/OSS; cheapest to train/run of the whole set.",
        "license": "MIT ? but ONLY via sj-li/MS-TCN2. AVOID yabufarha/ms-tcn (Commons Clause, non-commercial).",
        "privacy_note": "Fully local, tiny model, strong on-device fit; no paid-LLM data.",
        "fit_for_us": "medium"
      },
      {
        "name": "Plain 1D-CNN / GRU / small Transformer over features (DIY sequence model)",
        "what": "A from-scratch per-frame or windowed sequence classifier (1D-CNN, BiGRU, or a small Transformer) over WASB+audio+pose features, emitting per-frame rally prob -> threshold+merge -> [start,end]. This is the direct learned successor to today's windowing+LogisticRegression and matches the published badminton hit-detection recipe (BiGRU over court/pose/shuttle).",
        "pros": [
          "Maximum control + minimum dependencies; trivially small, trivially on-device, trains in seconds",
          "Matches the field-standard racket-sports recipe (MonoTrack-style GRU hit detector over trajectory/pose) -> lowest-risk, known-to-work pattern at our data scale",
          "No license questions at all (our own code, our own weights)",
          "Easiest to make deterministic/reproducible for the agent-runnable harness; cleanest baseline to A/B against ASFormer"
        ],
        "cons": [
          "We essentially already have this (distilled LogisticRegression) and it UNDERPERFORMED the heuristic at n=2 ? a bare GRU may not clear the ~0.73 bar without the multi-stage refinement / inductive-bias tricks ASFormer/MS-TCN++ bake in",
          "Reinventing temporal-modeling tricks (multi-scale, refinement, boundary regression) that ASFormer/ActionFormer already implement and validate",
          "No community-tuned recipe -> more of our own tuning time; easy to under-build"
        ],
        "cost_note": "Free; negligible compute. Mostly costs our engineering time.",
        "license": "N/A (own code) ? no third-party weights/data",
        "privacy_note": "Fully local; no paid-LLM data.",
        "fit_for_us": "medium"
      },
      {
        "name": "TemporalMaxer (TAL)",
        "what": "Parameter-free max-pooling-only TAL on the ActionFormer codebase: drops self-attention for local max-pooling. Outputs (start,end,class) instances. ~2.8x fewer GMACs and ~3x faster inference than ActionFormer.",
        "pros": [
          "Cheapest/fastest TAL at inference (best edge/on-device latency of the TAL family); fewer params than ActionFormer",
          "Shares ActionFormer's data/feature plumbing -> cheap to try alongside it",
          "Claims to beat ActionFormer on several TAL benchmarks despite simplicity"
```

        ],
        "cons": [
          "LICENSE UNCONFIRMED ? could not fetch a LICENSE file (raw 404); must verify it is MIT/Apache before any commercial use",
          "Same benchmark-scale validation as ActionFormer; no few-shot claim -> not a starting choice at n=3-10",
          "Max-pooling-only may lose the fine temporal context that matters for rally-boundary precision on noisy amateur footage"
        ],
        "cost_note": "Free/OSS; lowest inference cost of the TAL set.",
        "license": "UNVERIFIED (likely permissive given ActionFormer base, but confirm the LICENSE file before relying on it)",
        "privacy_note": "Fully local; no paid-LLM data.",
        "fit_for_us": "weak"
      },
      {
        "name": "DiffAct (TAS, diffusion)",
        "what": "Diffusion-based action segmentation: iteratively denoises per-frame action predictions conditioned on video features. ICCV 2023. Per-frame {rally/non-rally} -> merge.",
        "pros": [
          "SOTA-class accuracy on standard TAS benchmarks; principled boundary refinement via denoising",
          "Clean MIT (Finspire13/DiffAct)",
          "Same per-frame feature input as MS-TCN++/ASFormer -> would consume WASB features"
        ],
        "cons": [
          "Most complex + most data-hungry of the TAS set; diffusion training is finicky and the LAST thing you want at n=3-10",
          "Iterative denoising = slower inference -> weakest on-device story among TAS",
          "No few-shot / small-data claim; overkill until the golden corpus is large",
          "Highest harness/operability burden vs the 'minimal/readable, deterministic' goal"
        ],
        "cost_note": "Free/OSS but heaviest to train and tune; multi-step inference.",
        "license": "MIT (Finspire13/DiffAct) ? verified",
        "privacy_note": "Fully local; no paid-LLM data (but heavier model).",
        "fit_for_us": "weak"
      }
    ],
    "key_sources": [
      "https://arxiv.org/abs/2202.07925",
      "https://github.com/happyharrycn/actionformer_release",
      "https://github.com/dingfengshi/TriDet",
      "https://github.com/dingfengshi/TriDet/blob/master/LICENSE",
      "https://arxiv.org/abs/2303.09055",
      "https://github.com/TuanTNG/TemporalMaxer",
      "https://github.com/sj-li/MS-TCN2",
      "https://github.com/sj-li/MS-TCN2/blob/master/LICENSE",
      "https://github.com/yabufarha/ms-tcn/blob/master/LICENSE",
      "https://github.com/ChinaYi/ASFormer",
      "https://ar5iv.labs.arxiv.org/html/2110.08568",
      "https://arxiv.org/abs/2110.08568",
      "https://github.com/Finspire13/DiffAct",
      "https://github.com/Finspire13/DiffAct/blob/main/LICENSE",
      "https://arxiv.org/pdf/2006.09220",
      "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/",
      "https://ar5iv.labs.arxiv.org/html/2204.01899"
    ],
    "recommendation": "FIRST owned model = ASFormer (MIT, ChinaYi/ASFormer), as a TEMPORAL ACTION SEGMENTATION model over our existing per-frame features. Rationale: it is the ONLY surveyed architecture whose central published result is training FROM SCRATCH on a tiny dataset (GTEA, 28 videos) with ~1.1M params, by adding convolutional inductive bias to compensate for small-data overfitting ? exactly the failure mode we hit when a learned model lost to the heuristic at n=2. It consumes per-frame feature sequences directly, so it rides on the WASB trajectory we already produce (X,Y,Visibility) and trivially fuses audio-onset and pose by concatenating channels ? which simultaneously delivers the Option A modular-fusion milestone. It

is tiny (fits the RTX 2070S, trains in minutes, runs on-device), clean MIT, fully local (no paid-LLM data, preserving the privacy lever and all guardrails), and learns temporal structure rather than threshold-tuning, giving it a credible path past the ~F1 0.73 ceiling.\n\nConcrete plan:\n1. Map labels: per-frame target = rally(1)/non-rally(0) from the rally [start,end] CSVs; recover predicted intervals by merging contiguous rally frames + a min-gap/min-duration prior (reuse the existing net-crossing gate). Treat ending_reason as a SECOND, segment-level head (5 classes) once interval detection works ? do NOT try to learn it jointly first.\n2. Features: feed WASB (X,Y,Visibility) per frame; add audio-onset and pose channels as they come online (this IS Option A fusion). Smooth/window features so the dilated convs have signal on amateur 60fps footage.\n3. Split DISCIPLINE: split train/eval by MATCH, never by clip (prevents the leakage the brief flags). Gemini silver labels stay eval/teacher-only, never in training.\n4. Gate: ship only if held-out per-match F1 beats the heuristic+LogisticRegression baseline (~0.73) on the golden set ? the nanochat-style pass/fail verdict.\n\nPARALLEL second model = ActionFormer (MIT, happyharrycn/actionformer_release) as the TAL formulation: rally = one regressed (start,end,class=ending_reason) instance ? the cleanest 1:1 map to our CSV schema and the better long-term boundary-precision play. Run it as the A/B challenger; expect it to LOSE to ASFormer at n=3-10 (it has no few-shot claim and is benchmark-scale) but to OVERTAKE once the Roora-fed golden corpus grows into the hundreds of matches. TriDet is the cheap drop-in boundary-precision upgrade on top of ActionFormer for that later stage.\n\nEXPLICITLY DEFER: DiffAct and TemporalMaxer (too data-hungry / license-unconfirmed for a first model). And note these models are the BRIDGE to the brief's Option B end-to-end Qwen-VL/Gemma VLM ? not a replacement for it; they unblock P5 design using only data and features we already own.\n\nLICENSE GUARDRAIL TO RECORD IN COMMERCIALIZATION.md: if MS-TCN++ is ever chosen, vendor ONLY sj-li/MS-TCN2 (clean MIT) ? the original yabufarha/ms-tcn carries a Commons Clause (non-commercial) and MUST NOT be used.",
    "confidence": "high",
    "caveats": [
      "TemporalMaxer's license is UNCONFIRMED ? the raw LICENSE fetch 404'd and search did not surface a license. It is built on ActionFormer (MIT) so it is likely permissive, but verify the actual LICENSE file before any commercial vendoring. I marked it 'weak' partly for this reason.",
      "LICENSE TRAP confirmed by hand: yabufarha/ms-tcn = MIT + Commons Clause (NON-commercial); sj-li/MS-TCN2 (the MS-TCN++ repo) = clean MIT. These are DIFFERENT repos for what is colloquially called 'MS-TCN' ? vendor the right one.",
      "The few-shot evidence for ASFormer is GTEA = 28 videos, which are MULTI-CLASS densely-labeled cooking videos, not 3-10 single-action sports matches. The small-data claim is well-supported but our regime is even smaller and differently structured; expect to need feature smoothing, augmentation (time-warp/jitter on trajectories), and possibly leave-one-match-out CV. A from-scratch learned model still may not beat the ~0.73 heuristic until the golden corpus grows ? the brief itself notes the bottleneck is golden DATA + WASB recall, not the method. This recommendation lowers the floor and unblocks design; it does not guarantee a win at n=3-10.",
      "All TAL/TAS benchmark recipes default to I3D (2048-d) features. Feeding raw ~3-4-dim WASB trajectory is OFF the validated path. ActionFormer's input-projection and ASFormer's input layer both accept arbitrary feature dim (so it is mechanically supported), but accuracy on such low-dim input is unproven in the literature ? a learned feature front-end or hand-built feature stack (velocity, acceleration, visibility-run-length, net-crossing flags) will likely be needed.",
      "ending_reason (winner/forced_error/unforced_error/service_fault/let) is a SEMANTIC per-rally label that trajectory features alone almost certainly cannot predict well; treat it as a separate, later head and do not let it gate the core [start,end] detection milestone.",
      "I did not benchmark these models against each other on OUR data (cannot run GPU/torch here per the environment note). The ranking is from architecture properties, published small-data behavior, and license ? empirical A/B on the golden set is still required and is the actual ship/no-ship gate.",
      "These models are NOT the brief's Option B end-to-end multimodal VLM (Qwen-VL/Gemma). They are the intermediate owned model that beats the heuristic baseline using current assets; the VLM remains the 12-24mo target after scaffolding + corpus health, per the roadmap."
    ]
  },
  {
    "dimension": "OPTION B ? Owned end-to-end multimodal model: open, video-capable Apache-2.0/MIT bases fine-tunable for rally temporal grounding (frames+audio+pose ? rally [start,end]).",
    "summary": "For an OWNED, on-device-capable rally-temporal-segmentation model we must split candidates into two classes. (1) BASES we can legally own and ship commercially ? the only

viable path ? are generic video-VLMs with clean Apache-2.0/MIT licenses, NOT the temporal-grounding-specialized research models. (2) The temporal-grounding specialists named in the brief (TimeChat, VTimeLLM, Momentor) and the classic video encoders (VideoMAE) are LICENSE-POISONED for a product: they are research-only / CC-BY-NC / LLaMA-derived-with-no-LICENSE, so they are usable ONLY as METHOD references (how to tokenize timestamps, how to format moment-localization QA), never as the weights we fine-tune and serve. Top approved bases: Qwen2.5-VL-7B (Apache-2.0, native timestamp grounding via mRoPE+absolute-time-alignment ? best-documented temporal-grounding VLM in the legal set), Qwen3-VL-4B/8B (Apache-2.0, explicit Text?Timestamp Alignment, 4B fits the on-device/privacy lever), Gemma 4 4B/12B (Apache-2.0 ? VERIFIED, a genuine reversal from Gemma 1/2/3's restrictive terms, native frames+audio+video without separate encoders, ideal for the Option-B multimodal fusion), and InternVL3-8B (MIT, built on Qwen2.5-7B Apache base). CRITICAL GUARDRAIL CONFIRMATIONS: Gemma 3 stays REJECTED (Gemma Terms of Use, viral) but Gemma 4 is Apache-2.0 and clean; Qwen2.5-VL-3B (Qwen Research, non-commercial) and Qwen2.5-VL-72B (100M-MAU cap) are REJECTED per-size ? only the 7B and 32B Qwen2.5-VL sizes are Apache-2.0. COMPUTE REALITY for our situation: the lone 8GB RTX 2070 Super CANNOT train any 7B-class video VLM (video sequences push even QLoRA to ~20-24GB); it is an inference/eval box only. Training MUST be a rented GPU (one 24-48GB A100/L40S-hour-class job). DATA REALITY: 128 human-labeled rallies (96+32) is FAR below the 5k-20k clip?QA band needed for end-to-end fine-tuning ? so Option B is correctly a 12-24mo horizon; the near-term move is LoRA on a 4B Apache base as a PoC once Roora/rally-annotator grow the corpus, with the modular Option-A fusion shipping first.",
    "options": [
      {
        "name": "Qwen2.5-VL-7B-Instruct",
        "what": "7B generic video-VLM; native video temporal grounding (mRoPE + absolute-time alignment lets it pinpoint relevant video segments / emit timestamps). The single best-documented temporal-grounding-capable model inside the Apache-2.0 fence. Built on the Qwen2.5-7B LLM with an optimized ViT (SwiGLU+RMSNorm).",
        "pros": [
          "EXACT license = Apache-2.0 at the 7B size (verified on the HF model card) ? commercial-clean, derivative-wrappable, plugs into the provider registry with no MAU cap",
          "Native timestamp/moment-localization ability out of the box ? directly the rally [start,end] task class; minimizes the grounding behavior we must teach",
          "Massive ms-swift / LoRA / QLoRA tooling + community fine-tuning recipes (the brief's chosen engine targets exactly this family)",
          "Mature, stable checkpoint (not a research preview) ? safe to standardize the harness on"
        ],
        "cons": [
          "7B cannot train on our 8GB RTX 2070 Super; LoRA needs ~20GB and even QLoRA on VIDEO sequences lands ~24GB ? training is rented-GPU only",
          "7B is heavy for the on-device privacy lever; local INFERENCE at 1080p/60fps video is feasible but not snappy on 8GB (expect quantized + low-fps sampling)",
          "Newer Qwen3-VL supersedes it on temporal alignment; choosing 7B trades cutting-edge accuracy for a battle-tested baseline",
          "Still needs 5k-20k clip?QA pairs to beat the F1?0.73 heuristic ceiling ? base license being clean does not solve the data scarcity"
        ],
        "cost_note": "Inference: runs locally on the 2070 Super quantized (8GB) for eval; comfortable on a 16GB+ rented card. Training: one LoRA run ~4-12h on a 24GB rented GPU (RunPod/Lambda L40S/A100 class), low-tens-of-USD per run.",
        "license": "Apache-2.0 (verified on huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct). NOTE: 3B=Qwen Research (non-commercial, REJECT), 72B=Qwen License (100M-MAU cap, REJECT). Only 7B and 32B are Apache-2.0.",
        "privacy_note": "Local inference satisfies the privacy lever (video never leaves device) but 8GB is the binding constraint; per-user LoRA adapters are tiny and deletable (drop the adapter = delete the user's training).",
        "fit_for_us": "strong"
      },
      {
        "name": "Qwen3-VL-4B-Instruct (and 8B)",
        "what": "Newer Qwen VLM (released 2025-10) with explicit 'Text?Timestamp Alignment' (evolves beyond T-RoPE) for precise timestamp-grounded event localization; second-level video indexing, 256K-1M context. 4B is the small, on-device-friendly variant.",
        "pros": [
          "EXACT license = Apache-2.0 at BOTH 4B and 8B (verified on HF) ? commercial-clean",

"4B is the best fit for the on-device/privacy lever: small enough to LoRA on a modest rented 24GB card and to run quantized locally for inference",
"Temporal grounding is a headline feature (explicit textual-timestamp alignment) ? purpose-built for rally [start,end] emission",
"Newest of the approved family ? strongest temporal modeling per the technical report; future-proofs the base lock"
],
"cons": [
"Still cannot train on the 8GB local box (4B QLoRA on video is ~12-16GB+); training is rented-GPU only, though cheaper than 7B",
"Younger checkpoint ? fewer hardened community fine-tuning recipes than Qwen2.5-VL-7B (more harness debugging on us)",
"'Thinking' variants add latency; the Instruct variant is the right pick for deterministic [start,end] output",
"Same data-scarcity ceiling: 128 rallies is far below the training band"
],
"cost_note": "Cheapest train of the strong options: 4B LoRA fits a single 24GB rented GPU comfortably, likely <$10/run. Local quantized inference plausible on the 2070 Super.",
"license": "Apache-2.0 (verified: Qwen3-VL-4B-Instruct and 8B-Instruct on HF). Larger/MoE Qwen3-VL sizes must be re-checked per-size before any lock.",
"privacy_note": "Best on-device candidate for the privacy lever given the 4B size; per-user adapters deletable.",
"fit_for_us": "strong"
},
{
"name": "Gemma 4 (4B / 12B, multimodal)",
"what": "Google's 2026 open multimodal family (2B/4B/12B/26B/31B). Processes text+image+audio+VIDEO natively without separate encoder networks; 12B runs on a 16GB laptop. The native frames+audio path maps directly onto our Option-B fusion goal (frames+audio+pose ? rally).",
"pros": [
"EXACT license = Apache-2.0 ? VERIFIED, and a genuine reversal: Gemma 1/2/3 were under the restrictive 'Gemma Terms of Use', Gemma 4 (from the 2026-04 launch) is true Apache-2.0 with full commercial rights. This directly clears the guardrail that REJECTS Gemma 3",
"Native audio+video in ONE model is uniquely aligned with Option B (collapse WASB-trajectory + audio-hit + pose into a single owned model) ? fewer modality bridges to build",
"4B variant is genuinely on-device (16GB laptop / quantized smaller) ? strong privacy lever",
"Backed by Google's tooling + ecosystem; broad fine-tuning support expected"
],
"cons": [
"Temporal-grounding/timestamp-emission is NOT as explicitly documented as Qwen's ? we'd likely have to teach moment-localization output format more from scratch",
"Very new (mid-2026) ? fine-tuning recipes for the temporal-grounding task are immature; more harness risk",
"Must KEEP verifying per-checkpoint: the brief's rejection was specifically Gemma 3; do not let 'Gemma' shorthand leak a Gemma-3 weight into the pool",
"Audio-native is a future-facing bet; our near-term audio cue (E1) is AMBER, so the multimodal advantage is partly latent"
],
"cost_note": "4B trains cheaply on a rented 24GB card; 12B runs locally for inference on a 16GB machine (our 8GB box would need the 2B/4B quantized).",
"license": "Apache-2.0 (VERIFIED across multiple 2026 sources; replaces Gemma Terms of Use used by Gemma 1/2/3). Per-checkpoint re-verify before lock ? only Gemma 4 is clean; Gemma 3 remains REJECTED.",
"privacy_note": "4B on-device satisfies the privacy lever; native local audio+video keeps the whole multimodal pipeline on-device ? strongest privacy story of all options.",
"fit_for_us": "strong"
},
{
"name": "InternVL3-8B",
"what": "OpenGVLab VLM, MIT-licensed wrapper built on a Qwen2.5-7B Apache-2.0 LLM + InternViT vision encoder; supports interleaved image/video/text and extended video training samples.",
"pros": [

"EXACT license = MIT on the InternVL3 wrapper, and the underlying Qwen2.5-7B base is Apache-2.0 ? dual clean, commercial-safe",
"Strong general video understanding; competitive VLM benchmarks",
"MIT is the most permissive option ? zero ambiguity for derivative-wrapping",
"Same Qwen2.5 lineage means ms-swift recipes largely transfer"
],
"cons": [
"Temporal-grounding/timestamp localization is LESS first-class than Qwen2.5-VL/Qwen3-VL (general video QA, not purpose-built moment localization) ? more to teach for rally [start,end]",
"Per-SIZE base differs: 9B uses InternLM3 and 38B/78B use Qwen2.5-32B/72B ? the 72B-based 78B inherits the 100M-MAU cap; ONLY the 1B/2B/8B/14B/38B (Qwen2.5 ?32B) sizes are clean. Verify per size before lock",
"8B has the same 8GB-box training impossibility as Qwen2.5-VL-7B",
"Adds a vision-encoder dependency (InternViT) to the stack"
],
"cost_note": "Comparable to Qwen2.5-VL-7B: rented-GPU LoRA, local quantized inference only.",
"license": "MIT (InternVL3 wrapper) + Apache-2.0 (Qwen2.5 base) for the ?8B/14B/38B sizes. REJECT InternVL3-78B (Qwen2.5-72B base = 100M-MAU cap). Verify per size.",
"privacy_note": "Local inference feasible quantized; same adapter-deletion model as Qwen.",
"fit_for_us": "medium"
},
{
"name": "VideoLLaMA2 / VideoLLaMA3 (DAMO)",
"what": "DAMO video-LLMs; v3 (Jan 2025) is built on Qwen2.5 (7B and 1.5B/2B), code Apache-2.0, with demonstrated temporal-grounding capability. v2 added explicit audio understanding (relevant to Option B fusion).",
"pros": [
"CODE is Apache-2.0; v3 base LLM (Qwen2.5) is Apache-2.0; demonstrated temporal-grounding notebook ? closest 'specialist behavior on a clean base' in the set",
"v2's audio-understanding branch is a useful reference for our frames+audio fusion target",
"2B (Qwen2.5-1.5B) variant is small enough for the on-device lever"
],
"cons": [
"BLOCKER: the released VideoLLaMA3 SERVICE/checkpoint is declared a 'research preview intended for non-commercial use ONLY', subject to extra Qwen + data-source terms ? so we CANNOT ship DAMO's fine-tuned weights. We could only re-train from the Apache Qwen2.5 base ourselves using their (Apache) code, i.e. it collapses to 'use Qwen2.5-VL/Qwen3-VL + their training recipe'",
"Therefore not a true 'owned base' ? it's a method/recipe reference plus an Apache code dependency",
"Mixed signal: 'Apache code' vs 'non-commercial weights' is exactly the trap to avoid asserting as clean"
],
"cost_note": "Same rented-GPU training profile; no advantage over directly using Qwen3-VL/Gemma 4 as the base.",
"license": "Code Apache-2.0; RELEASED WEIGHTS = research/non-commercial preview (NOT shippable). Use as recipe reference only.",
"privacy_note": "Moot for product use given the non-commercial weights; if re-trained from Qwen2.5, privacy story = same as Qwen options.",
"fit_for_us": "weak"
},
{
"name": "TimeChat / VTimeLLM / Momentor (temporal-grounding specialists)",
"what": "Academic video-LLMs purpose-built for temporal grounding / moment localization. TimeChat (CVPR'24, time-sensitive), VTimeLLM (CVPR'24, 'grasp video moments'), Momentor (fine-grained temporal reasoning, trained on Moment-10M).",
"pros": [
"State-of-the-art METHODS for exactly our task: timestamp tokenization, temporal position encoding (Momentor's TPM), moment-localization instruction formats ? high-value as design references for our QA-pair schema and the grounding head",
"Papers/datasets (e.g. Moment-10M format) inform how to convert rally CSVs ? clip?QA

pairs"
        ],
        "cons": [
          "LICENSE-POISONED as bases: TimeChat = research-preview non-commercial; VTimeLLM = CC-BY-NC-ND 4.0 (NonCommercial + NoDerivs ? cannot even fine-tune/redistribute); Momentor = built on LLaMA with NO LICENSE file in-repo (legally unusable + LLaMA's own restrictions). NONE are Apache-2.0/MIT",
          "Cannot be the owned, commercial, fine-tunable base under our hard guardrail ? full stop",
          "Older LLM backbones (Vicuna/LLaMA-class) lag the approved Qwen3-VL/Gemma 4 bases on raw capability anyway"
        ],
        "cost_note": "N/A as a base (cannot ship). Zero-cost as reading/method references.",
        "license": "TimeChat = research-only/non-commercial; VTimeLLM = CC-BY-NC-ND 4.0; Momentor = no LICENSE + LLaMA-derived. ALL REJECTED as bases; method-reference only.",
        "privacy_note": "N/A ? not deployable.",
        "fit_for_us": "weak"
      },
      {
        "name": "Video encoders: InternVideo2 / VideoMAE",
        "what": "Pure spatiotemporal video encoders (not generative VLMs) usable as a backbone for a custom temporal-localization head (e.g. encoder ? CG-DETR-style grounding head, the Option-A/modular or a from-scratch e2e route that avoids a full VLM).",
        "pros": [
          "InternVideo2-Stage2_6B = MIT (VERIFIED) and InternVideo2.5_Chat_8B = Apache-2.0 ? clean encoder weights we CAN use commercially",
          "An encoder+grounding-head is far CHEAPER to train than a full VLM and could run on the 8GB box at smaller sizes ? a realistic near-term owned-model PoC at our data scale",
          "InternVideo2 is a proven temporal-grounding backbone (evaluated on QVHighlights/Charades-STA via CG-DETR)",
          "Sidesteps the 'generative VLM is huge' problem for the privacy/on-device lever"
        ],
        "cons": [
          "VideoMAE weights = CC-BY-NC 4.0 ? REJECTED for commercial use (research-only); only its Apache-2.0 code components are reusable, not the pretrained checkpoints",
          "Encoder-only path is NOT the generative video-QA Option-B end-state ? it answers 'where is the rally' but not the rich rally semantics (ending_reason etc.); it's a complementary/bridge architecture, not the VLM the brief ultimately wants",
          "Building+maintaining a custom grounding head is extra engineering vs. fine-tuning an off-the-shelf VLM",
          "Need to confirm InternVideo2's largest checkpoints per-file (some sub-weights vary)"
        ],
        "cost_note": "Cheapest training of any option; a small InternVideo2 encoder + light head is plausibly trainable on the 8GB box or a single small rented card ? the only option that fits TODAY's 128-rally data scale.",
        "license": "InternVideo2-Stage2_6B = MIT; InternVideo2.5_Chat_8B = Apache-2.0 (both VERIFIED, usable). VideoMAE checkpoints = CC-BY-NC 4.0 (REJECTED); VideoMAE code Apache-2.0 only.",
        "privacy_note": "Small encoders are the most on-device-friendly option ? strongest fit for the local privacy lever at small data scale.",
        "fit_for_us": "medium"
      }
    ],
    "key_sources": [
      "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",
      "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",
      "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",
      "https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",
      "https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",
      "https://github.com/qwenlm/qwen3-vl",
      "https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/",
      "https://www.infoq.com/news/2026/04/google-gemm4/",
      "https://venturebeat.com/technology/googles-new-open-source-gemma-4-12b-analyzes-audio-video-and-runs-entirely-locally-on-a-typical-16gb-enterprise-laptop",
      "https://huggingface.co/OpenGVLab/InternVL3-8B",

      "https://huggingface.co/OpenGVLab/InternVideo2-Stage2_6B",
      "https://huggingface.co/OpenGVLab/InternVideo2_5_Chat_8B",
      "https://github.com/DAMO-NLP-SG/VideoLLaMA3",
      "https://github.com/DAMO-NLP-SG/VideoLLaMA2/blob/main/LICENSE",
      "https://github.com/RenShuhuai-Andy/TimeChat",
      "https://github.com/huangb23/VTimeLLM",
      "https://github.com/DCDmllm/Momentor",
      "https://huggingface.co/MCG-NJU/videomae-base",
      "https://arxiv.org/html/2403.15377v1",
      "https://arxiv.org/html/2402.11435v1",
      "https://www.labellerr.com/blog/fine-tune-qwen-2-5-vl/",
      "https://kaitchup.substack.com/p/qwen3-vl-fine-tuning-on-your-computer"
    ],
    "recommendation": "LOCK the base as a two-stage decision, not one model. (1) NEAR-TERM PoC base = Qwen3-VL-4B-Instruct (Apache-2.0, explicit Text?Timestamp Alignment, smallest strong on-device candidate) as the primary, with Qwen2.5-VL-7B-Instruct (Apache-2.0, best-documented temporal grounding, most mature ms-swift recipes) as the de-risking fallback. Both are clean for commercial ownership and the privacy lever. (2) WATCH/EVALUATE Gemma 4 4B (Apache-2.0, VERIFIED reversal from the rejected Gemma 3) as the Option-B end-state base ? its native frames+audio+video in one model is the best structural match for collapsing WASB-trajectory + audio + pose into a single owned model; lock it for Option B once a fine-tuning recipe for timestamp emission is proven. DO NOT use TimeChat/VTimeLLM/Momentor/VideoMAE as bases (all non-commercial or no-LICENSE) ? mine them only for the timestamp-tokenization and clip?QA-format methods. DO NOT touch Qwen2.5-VL-3B (Qwen Research), Qwen2.5-VL-72B / InternVL3-78B (100M-MAU cap), DAMO's released VideoLLaMA3 weights (research-preview), or any Gemma ?3. COMPUTE: treat the 8GB RTX 2070 Super as an inference/eval-only box; budget a single rented 24-48GB GPU (RunPod/Lambda L40S or A100 class) for every fine-tune ? even QLoRA on video exceeds 8GB. SEQUENCING (matches the brief's directive): ship modular Option-A fusion first; do NOT start any owned-model fine-tune until (a) the async job model / ComputeBackend P0 lands and (b) the golden corpus crosses ~thousands of clip?QA pairs (Roora + rally-annotator). At today's 128 rallies the only owned-model experiment that is data-realistic NOW is a small InternVideo2-encoder (MIT) + light grounding head as a bridge PoC ? the full VLM fine-tune is correctly a 12-24mo Option-B horizon.",
    "confidence": "high",
    "caveats": [
      "Licenses change per-checkpoint and per-version. Re-verify the EXACT license file on the specific HF revision before any code lands ? especially: Qwen2.5-VL sizes (only 7B/32B Apache; 3B/72B rejected), InternVL3 sizes (78B inherits Qwen2.5-72B MAU cap), and ANY Gemma weight (only Gemma 4 is Apache; Gemma 3 stays REJECTED). The 'Gemma' shorthand is a live foot-gun.",
      "Gemma 4's Apache-2.0 status is confirmed across multiple 2026 secondary sources and the official Google blog, but I did not open the raw LICENSE file on each Gemma 4 HF checkpoint ? do that before locking it as the Option-B base.",
      "Compute/VRAM numbers are from community fine-tuning write-ups (Labellerr, Kaitchup, Unsloth), not a controlled benchmark on OUR footage. Real 1080p/4K-60fps video sequences with long context can push VRAM higher than the text/OCR LoRA figures cited; treat 24GB as a floor for a 7B video LoRA and validate empirically.",
      "The 5k-20k clip?QA training band is the brief's own estimate; the true minimum to beat the F1?0.73 heuristic with an end-to-end VLM is unproven for this niche task and could be higher (or, with strong base priors + good QA formatting, somewhat lower). The honest n=2 result (learned model underperforms baseline) is the binding evidence that data ? not base choice ? is the current bottleneck.",
      "Momentor having no LICENSE file is from a single fetch of the repo root; a LICENSE may exist in a subdir or have been added since ? but absence-of-license + LLaMA derivation is enough to disqualify it as a base regardless.",
      "I treated 'on-device' against the stated 8GB RTX 2070 Super. If the owner provisions a 16-24GB local card later, the local-training calculus changes materially (4B QLoRA becomes locally trainable)."
    ]
  },
  {
    "dimension": "LABEL & DATA REUSE / DATA ENGINEERING ? maximizing the value of our human rally CSVs + ending_reason + WASB trajectories across model generations (heuristics ? TAL ? owned video-VLM \"Option B\")",
    "summary": "Our labels are far more valuable than their raw count (96 + 32 rallies ? 2 matches) suggests, IF we stop treating them as one-format eval files and start treating them as

```
a model-agnostic \"labeled-corpus\" that we transcode into every consumer. (a) FORMATS: the
rally [start,end]+ending_reason+sport rows already ARE the ActivityNet/ActionFormer schema
modulo a transcode ? a `segment:[start,end]` (seconds) + `label` + per-video
`duration/fps/subset` JSON; the SAME rows transcode into (i) candidate-window labels for our
heuristic/LogReg distillation (already done via `training.label_candidates`), (ii) ActivityNet-
style JSON for any TAL net (ActionFormer/TriDet), and (iii) clip?QA text pairs (\"When does each
rally happen?\" ? \"[start,end]?\") for the Option-B video-VLM. ending_reason becomes a second
QA task for free. WASB trajectories are an auxiliary feature stream keyed by the same (video_id,
frame). (b) AMPLIFICATION ? five legitimate levers that never touch paid-LLM outputs: (1)
teacher-student / pseudo-labeling where the TEACHER is OUR OWN model or an open MIT model
(WASB+heuristics, or a TAL net trained on the few labels), self-training on our ~26 GB unlabeled
corpus ? SS-TAL is an active field and PLR/ALQA-style confidence-gated pseudo-labels are the
standard recipe; (2) temporal data augmentation (speed/time-warp ±20?30%, temporal jitter,
reverse, segment cut-mix/mixup) that multiplies effective segment count for any TAL/VLM trainer
while preserving boundaries; (3) active learning ? spend the Roora budget where the model is
least certain (boundary entropy / disagreement vs WASB), measurably beating random labeling
under a fixed budget; (4) timestamp/point supervision for cheaper future labels (~? the time of
full segments, SkipTag ~?, reaching near-full-supervision accuracy) ? relevant for ending_reason
and for scaling new sports; (5) for Option B specifically, RL/RFT (GRPO) is dramatically more
LABEL-efficient than SFT for temporal grounding (VideoChat-R1: +31.8 temporal-grounding with
\"limited samples\", and quality-filtered 13K beat unfiltered 56K in VTG-R1) ? meaning our small
hand-verified set is the right asset class. (c) HOW MANY MATCHES: heuristic threshold tuning
needs ~3?5 matches for stable cross-val (we're under that at n?2, which is exactly why the
learned model underperforms baseline); a from-scratch TAL net (ActionFormer-class) wants
~50?200+ labeled videos to beat heuristics; an Option-B VLM fine-tune is the sweet spot ?
single-digit-thousands of clip?QA pairs (literature uses 13K?18K), which our ~128 rallies ×
augmentation × QA-task-multiplexing can approach without thousands of new matches, and RFT can
cold-start from hundreds. (d) The reusable format: one canonical JSONL \"rally record\" per
(video_id, rally_ordinal) carrying [start,end], ending_reason, sport, source/provenance, split
tag, + a pointer to the WASB trajectory slice; deterministic transcoders emit the heuristic-
label view, the ActivityNet-TAL view, and the VLM clip?QA view, so EVERY label labeled once
feeds all three model classes and all future generations.",
    "options": [
      {
        "name": "(a) Format transcoders ? one corpus, three consumer formats",
        "what": "Treat the existing rally CSV rows
(rally_number,start_time,end_time,ending_reason,sport) as the source of truth and write
deterministic transcoders to each model class's expected format, instead of re-labeling per
consumer. View 1 (heuristic/LogReg distillation): we ALREADY do this ?
`backend/eval/training.py::label_candidates` runs the same IoU `match_intervals` to turn rally
intervals into positive/negative labels on WASB candidate windows; keep it. View 2 (TAL nets):
emit ActivityNet/ActionFormer JSON ? top-level `database`, per-video `{subset, duration, fps,
annotations:[{segment:[start,end] in SECONDS, label:'rally', label_id:0}]}` (verified against
ActionFormer's thumos14 loader). Our seconds-based [start,end] maps 1:1; we only add per-video
duration/fps (already in ffprobe_meta) and a train/val/test `subset`. View 3 (Option-B VLM):
emit clip?QA text pairs ? sample a window, ask 'List the [start,end] of every rally in this
clip' ? answer is the rally intervals normalized to clip-relative seconds; ending_reason yields
a SECOND free task ('Why did rally N end?' ? winner/forced_error/...). One row, three formats,
zero re-labeling.",
        "pros": [
          "Zero new labeling: every consumer reads the same ~128 rallies; labels labeled once
feed heuristics + TAL + VLM and every future model generation",
          "Our schema is already 90% of the ActivityNet TAL schema ? `segment:[start,end]` in
seconds is exactly what ActionFormer/TriDet ingest; transcode is a thin map, not a re-
annotation",
          "ending_reason is a near-free second supervised task (rally-outcome QA) that
multiplies QA-pair yield from the SAME intervals",
          "Reuses the existing deterministic scorer (`metrics.match_intervals`) as both the
heuristic-labeler and the TAL/VLM eval grader ? one spine, no metric drift",
          "WASB trajectory CSV (Frame,Visibility,X,Y) attaches as an auxiliary per-frame feature
stream keyed by the same video_id+frame ? usable as TAL input features OR as a 'trajectory-
token' modality for Option B"
        ],
        "cons": [
          "VLM clip?QA generation needs a windowing/sampling policy (clip length, stride,
```

negative clips with no rally) ? a design choice that affects label balance",
          "ActivityNet-JSON path assumes you extract per-clip FEATURES (I3D/VideoMAE) for classic TAL nets; that is extra preprocessing, though it's open-model and one-time",
          "Clip-relative vs absolute timestamps must be handled carefully in the VLM answer format (Qwen2.5-VL aligns MRoPE to absolute time ? pick one convention and lock it)"
        ],
        "cost_note": "Engineering-only; no labeling spend. A few hundred lines of transcoder + tests; reuses existing metrics/annotations modules.",
        "license": "N/A (our own data + open formats). ActivityNet JSON schema and ActionFormer loader are open/permissive references.",
        "privacy_note": "Transcoders run locally on licensed/owned corpus only; no user video, no egress. Minors excluded at the corpus-membership layer before transcoding.",
        "fit_for_us": "strong"
      },
      {
        "name": "(b1) Self-training / teacher-student / pseudo-labeling ? open-teacher ONLY",
        "what": "Amplify the few labels by pseudo-labeling our ~26 GB unlabeled corpus with a teacher that is OUR OWN model or an open MIT/Apache model (NEVER Gemini). Concretely: train a first-pass rally model (heuristics+WASB, or a small TAL net) on the ~128 labeled rallies, run it over unlabeled matches, keep only high-confidence pseudo-rallies (confidence-gated, à la ALQA/PLR semi-supervised TAL), retrain on labeled+pseudo, iterate (noisy-student style with temporal augmentation as the 'noise'). This is the canonical SS-TAL recipe; it directly attacks the n?2 data starvation that currently makes the learned model lose to the baseline.",
        "pros": [
          "Stays inside guardrails: teacher is WASB/our-model/open-MIT ? paid-LLM outputs are never a training signal",
          "SS-TAL with confidence-gated pseudo-labels (ALQA, PLR) is an established, published recipe for exactly our 'few labeled + many unlabeled videos' regime",
          "Noisy-student framing (student ? teacher + augmentation noise) is a proven data-efficiency multiplier (Xie et al. CVPR'20 lineage)",
          "Uses the corpus we already own; no per-rally vendor spend to grow training signal",
          "Pseudo-labels can bootstrap Option-B VLM SFT data (clip?QA pairs from confident pseudo-rallies) without human labeling"
        ],
        "cons": [
          "Pseudo-label noise: low-quality pseudo-rallies degrade boundaries ? MUST confidence-gate and ideally human-spot-check a sample (the PLR/ALQA papers exist precisely because naive pseudo-labels hurt)",
          "Teacher blind spots propagate (WASB recall ceiling becomes the pseudo-label recall ceiling) ? confirmed bottleneck in our own results",
          "Boundary precision from pseudo-labels is weaker than human [start,end]; treat pseudo as recall-boosters, keep human labels as the boundary authority",
          "Iteration adds GPU cost on the single RTX 2070 Super (8 GB) ? feasible but slow; cloud burst may be needed"
        ],
        "cost_note": "Mostly compute on owned GPU + occasional cloud burst. No new human labeling required to START; small human audit budget to validate pseudo-label quality.",
        "license": "Teachers restricted to Apache-2.0/MIT/BSD (WASB=MIT, our own LogReg, open TAL nets). Explicitly excludes Gemini/OpenAI/Anthropic outputs as training data.",
        "privacy_note": "All pseudo-labeling on owned/licensed corpus, local. Pseudo-labeled clips inherit the same minors-excluded, training-opt-in constraints.",
        "fit_for_us": "strong"
      },
      {
        "name": "(b2) Temporal data augmentation ? multiply effective segments",
        "what": "Apply temporal augmentations that preserve rally semantics while changing the signal: speed/time-warp (±20?30%, rescale [start,end] accordingly), temporal jitter of boundaries, VideoReverse (rally played backward still a rally), and segment-level CutMix/MixUp/FadeMixUp to synthesize new rally/non-rally compositions. For VLM clip?QA, also vary clip windows/stride so the same rally appears in many contexts. Scale-aware sampling (randomize action timescale) is reported superior to spatial-only aug for video.",
        "pros": [
          "Multiplies effective labeled-segment count with ZERO new annotation ? directly addresses tiny-n",
          "Speed/time-warp + scale-aware sampling are validated to improve temporally-

```
localizable features (RandAugment-T, FadeMixUp lineage)",
          "Boundary-preserving by construction (you transform the timestamps with the signal),
so it doesn't corrupt our hard-won [start,end] authority",
          "Cheap and deterministic; runs in the data loader for both TAL nets and VLM SFT",
          "Window/stride variation for clip?QA is a free QA-pair multiplier for Option B"
        ],
        "cons": [
          "Augmentation realism risk: extreme speed changes can break shuttle-physics cues WASB
relies on ? keep ranges moderate",
          "CutMix/MixUp across rallies can create unnatural boundaries; FadeMixUp (soft) is
safer than hard cut-paste for localization",
          "Doesn't add NEW information (same matches, same players, same court) ? complements
but cannot replace corpus diversity from new matches/venues",
          "Gains are incremental, not a substitute for the ~3?5 real matches the heuristic tuner
needs for stable cross-val"
        ],
        "cost_note": "Compute-only, in-loader; negligible marginal cost.",
        "license": "N/A ? transforms our own data.",
        "privacy_note": "Local, owned corpus; no egress.",
        "fit_for_us": "strong"
      },
      {
        "name": "(b3) Active learning ? label the most-informative match/segment next",
        "what": "Spend the Roora vendor budget where it buys the most: rank unlabeled
matches/clips by model uncertainty (boundary entropy, prediction disagreement between WASB-
heuristic and a TAL net, or high alignment-cost vs labeled prototypes) and send THOSE to Roora
first, rather than random/sequential matches. Active-learning-for-TAL papers (uncertainty/TPE
scoring, boundary-centric acquisition) show this beats random labeling under a fixed budget.",
        "pros": [
          "Maximizes IoU-lift-per-rupee on the constrained vendor budget ? exactly the solo-
founder cost constraint",
          "Published SS-TAL+active-learning results show uncertainty-based selection beats
random under fixed label budget",
          "Pairs naturally with (b1): pseudo-label everything, then human-label only the
segments the model is LEAST sure about (the highest-value oracle queries)",
          "Boundary-centric acquisition targets our actual weak point (dStart/dEnd boundary
error), not just presence/absence",
          "Disagreement signal is free ? we already run WASB-heuristic AND can run a TAL net;
route their disagreements to humans"
        ],
        "cons": [
          "Needs a deployable uncertainty signal first; our LogReg gives probabilities but a
richer TAL net gives better entropy estimates ? small upfront build",
          "Cold-start problem: with n?2 the model's uncertainty is itself unreliable; first 3?5
matches should still be diverse/random to bootstrap, THEN switch to active selection",
          "Risk of sampling bias (always querying hard cases) can skew the train distribution ?
mix in some random/diverse picks",
          "Operationally couples the eval/training loop to the vendor pipeline (must re-rank
between Roora batches)"
        ],
        "cost_note": "Saves vendor money net-net; small engineering cost for the scoring/ranking
step. Highest ROI once n?~5.",
        "license": "N/A.",
        "privacy_note": "Selection runs locally; only the SELECTED owned/licensed clips go to
the vendor through the existing single DPA-gated provider boundary.",
        "fit_for_us": "strong"
      },
      {
        "name": "(b4) Timestamp / point supervision ? cheaper labels for scale-out",
        "what": "For FUTURE labeling (new sports, ending_reason, scaling beyond Roora), prefer
point/timestamp supervision (one click per rally + outcome) over dense [start,end] where
boundary precision isn't the bottleneck. User study on Breakfast: timestamp supervision ? ? the
annotation time of full supervision, SkipTag ? ?, and timestamp-supervised models reach near-
full-supervision accuracy (e.g. 64.1% vs 68.0%). Point-level TAL is a mature sub-field.",
        "pros": [
```

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          "Cuts annotation time substantially (timestamp ? ?, SkipTag ? ? of full) ? more
matches per rupee",
          "Near-full-supervision accuracy is achievable from point labels with the right losses
(HR-Pro, point-TAL methods)",
          "ending_reason is naturally point-like (one outcome per rally) ? fits timestamp
supervision perfectly",
          "Lowers the barrier to expand to tennis/pickleball/table-tennis (collector's 5-sport
shape) without dense re-labeling each sport",
          "Our existing dense [start,end] golden set becomes the high-quality 'boundary
authority' that calibrates the cheaper point-labeled bulk"
        ],
        "cons": [
          "Point labels alone give weaker boundaries ? keep a dense-labeled golden CORE for eval
and boundary calibration",
          "Requires point-supervision-aware training losses (not just feeding points to a full-
supervision trainer)",
          "Mixing dense + point labels needs careful handling in the corpus format (label_type
field) and the trainer",
          "Our current dense CSVs are already collected ? this is a forward-looking lever for
the NEXT thousands of labels, not the existing 128"
        ],
        "cost_note": "Reduces future labeling cost ~33?66%. Existing dense labels unaffected.",
        "license": "N/A.",
        "privacy_note": "Same owned-corpus + vendor-DPA path; minors excluded.",
        "fit_for_us": "medium"
      },
      {
        "name": "(b5) Option-B specific: RL/RFT (GRPO) over SFT for label efficiency",
        "what": "When training the owned video-VLM (Qwen2.5-VL-7B / Qwen3-VL-8B / InternVL2.5,
all Apache-2.0/MIT), prefer reinforcement fine-tuning (GRPO/RFT) with a temporal-IoU reward over
pure SFT for the temporal-grounding objective. RFT is markedly more LABEL-efficient:
VideoChat-R1 reports +31.8 on temporal grounding with 'limited samples' and GRPO > SFT in their
ablation; VTG-R1 shows a quality-filtered 13K cold-start set beating an unfiltered 56K set. Our
reward = temporal IoU of predicted [start,end] vs our golden rally labels (the metric we already
compute).",
        "pros": [
          "RFT is data-efficient ? matches our exact situation: small, high-quality, hand-
verified label set is the RIGHT asset for RL, not a liability",
          "Reward = temporal IoU vs golden labels = our EXISTING deterministic scorer
(`metrics`) repurposed as the RL reward ? no new eval machinery",
          "Quality-over-quantity is empirically validated (filtered 13K > unfiltered 56K), so
our small clean set + augmentation can punch above its weight",
          "GRPO needs no separate reward model (uses verifiable IoU reward) ? operationally
simple, fits the nanochat-style one-command harness goal",
          "Bases are guardrail-clean (Qwen2.5-VL-7B/32B Apache-2.0; InternVL2.5 MIT; Gemma 4 if
Apache-2.0 confirmed)"
        ],
        "cons": [
          "RFT has higher COMPUTE cost than SFT (multiple rollouts per sample) ? heavy for one 8
GB RTX 2070 Super; likely needs cloud burst (RunPod/Lambda/Modal)",
          "Still wants a quality SFT cold-start first (cold-start data matters per VTG-R1) ? so
you need BOTH a clean SFT set AND the RL loop",
          "Reward hacking risk if IoU reward is naive (model emits over-long intervals) ? needs
precision/recall-balanced reward shaping",
          "VLM temporal grounding is still maturing; absolute accuracy on amateur single-camera
footage is unproven vs the broadcast-trained literature"
        ],
        "cost_note": "Compute-heavy (RL rollouts); plan cloud-GPU burst. But LABEL cost is low ?
the whole point. Benchmark self-hosted 7B vs Gemini per-call before committing (per
COMMERCIALIZATION C7).",
        "license": "Bases Apache-2.0/MIT only (Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B, InternVL2.5;
Gemma 4 pending license verify; Gemma 3 REJECTED). Reward = our own IoU, no paid-LLM signal.",
        "privacy_note": "Training data = owned/licensed corpus, training-opt-in only, minors
excluded. Final model runs locally (privacy lever).",
        "fit_for_us": "medium"
```

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      },
      {
        "name": "(c) How many labeled matches per model class ? realistic bands",
        "what": "Heuristic threshold tuning (current WASB+windowing+gate): ~3?5 labeled matches for STABLE leave-one-video-out cross-val; at n?2 the variance is too high, which is exactly why our learned model loses to the baseline (this is a data-starvation artifact, not a method failure). Small distilled classifier (our LogReg on trajectory features): also needs n?~5 diverse matches before it reliably beats threshold tuning. From-scratch TAL net (ActionFormer/TriDet class): wants ~50?200+ labeled videos to be competitive (these nets are tuned on THUMOS14/ActivityNet-scale sets) ? NOT realistic for us soon, so deprioritize training one from scratch. Option-B VLM fine-tune: the sweet spot ? single-digit-thousands of clip?QA pairs (literature: 13K?18K), reachable from ~128 rallies × temporal augmentation × QA-task multiplexing (rally-localization + ending_reason) × pseudo-labeled clips; RFT can cold-start from HUNDREDS of high-quality pairs.",
        "pros": [
          "Sets honest, non-aspirational targets per model class ? prevents over-investing in a from-scratch TAL net we can't feed",
          "Explains the current 'learned < baseline at n=2' result as expected data starvation, not a dead end ? the fix is data, confirmed by our own docs",
          "Identifies the VLM path as the most label-EFFICIENT owned-model route given our asset shape (few, clean labels + pretrained base)",
          "Gives a concrete corpus-growth milestone: get to ~5 matches to unblock honest heuristic tuning NOW; build toward low-thousands QA pairs for Option B",
          "Aligns with the staged roadmap (heuristics ? Option A modular fusion ? Option B e2e)"
        ],
        "cons": [
          "The ~50?200 for from-scratch TAL is a literature-scale heuristic (THUMOS/ActivityNet), not measured on our footage ? treat as order-of-magnitude",
          "'Single-digit-thousands of QA pairs' for the VLM assumes augmentation + multiplexing reach it; raw human rallies alone (128) are not enough without amplification",
          "Numbers are domain-transfer-sensitive: amateur single-camera footage may need MORE labels than broadcast-trained literature implies",
          "Pose/audio-fused Option-A may shift these bands (more signal per label) ? revisit after E1/pose probes"
        ],
        "cost_note": "Planning guidance; the actionable spend is getting from n?2 to n?5 matches (small Roora batch) to unblock the heuristic loop immediately.",
        "license": "N/A.",
        "privacy_note": "N/A.",
        "fit_for_us": "strong"
      },
      {
        "name": "(d) Model-agnostic 'labeled-corpus' format ? the reusable spine",
        "what": "Define ONE canonical JSONL record per (video_id, rally_ordinal): {video_id (file MD5), rally_ordinal, sport, start, end (seconds), ending_reason, source (human|pseudo|vendor), annotator/provenance, label_type (segment|point), split (train|val|test, assigned BY MATCH not by clip), trajectory_ref (pointer to the WASB CSV slice for this video)}. Store it once (extends the existing collector export + candidate_segments lineage). Then deterministic, tested transcoders emit: (1) the heuristic-label view (candidate-window pos/neg via `match_intervals`), (2) the ActivityNet/ActionFormer TAL-JSON view (`database`?per-video?`annotations:[{segment:[s,e],label,label_id}]`), (3) the VLM clip?QA view (rally-localization + ending_reason tasks). Split is enforced by MATCH to prevent leakage (CODE_MAP/PLATFORM 'split by match, not clip'). This is the asset the moat actually lives in.",
        "pros": [
          "Label once, consume everywhere and forever ? survives model-generation churn (heuristics?TAL?VLM?whatever's next) because consumers read transcoder output, not raw labels",
          "Provenance + source + label_type fields make pseudo-labels, vendor labels, and human labels first-class and auditable (critical for the no-paid-LLM + minors-excluded guardrails ? each record carries who/what produced it and whether training-opt-in applies)",
          "split-by-match is baked into the format ? kills the data-leakage failure mode the platform doc calls out (train|eval split by match not clip)",
          "Reuses existing infra: candidate_segments table is already the detector-agnostic extension point; collector export already shares (video_id, ordinal, start, end) shape across 5 sports",
          "trajectory_ref keeps WASB as an attachable modality without bloating the rally record
```

? feeds TAL features AND Option-B multimodal tokens",
          "Same record powers eval (golden set) and training ? one corpus, graded by one scorer"
        ],
        "cons": [
          "Requires upfront schema discipline + migration of current CSVs into the JSONL (one-time; the collector bridge/ADR-002 already points this way)",
          "Provenance/consent fields add bookkeeping the current flat CSV doesn't have ? but they're non-negotiable given the guardrails",
          "Two timestamp conventions already exist in-repo (generic start,end vs golden rally_number,start_time,end_time) ? the canonical record must absorb both and the transcoders own the conversion",
          "trajectory_ref coupling means trajectory CSVs must be retained/versioned alongside labels (storage, but cheap)"
        ],
        "cost_note": "One-time engineering: schema + migration + 3 transcoders + tests. No labeling spend. Pays back on every future model.",
        "license": "Our own schema; emits open formats (ActivityNet JSON). No third-party license entanglement.",
        "privacy_note": "Provenance/consent/minors-excluded/training-opt-in are FIELDS in the record, enforced at corpus build; the format is the enforcement point for the legal guardrails.",
        "fit_for_us": "strong"
      }
    ],
    "key_sources": [
      "https://arxiv.org/html/2407.07673v1",
      "https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/",
      "https://openaccess.thecvf.com/content/ICCV2023/papers/Xia_Learning_from_Noisy_Pseudo_Labels_for_Semi-Supervised_Temporal_Action_Localization_ICCV_2023_paper.pdf",
      "https://github.com/happyharrycn/actionformer_release/blob/main/README.md",
      "https://arxiv.org/abs/2208.14856",
      "https://joonyoung-cv.github.io/assets/paper/18_eccv_what_do.pdf",
      "https://arxiv.org/html/2604.15173v1",
      "https://arxiv.org/pdf/2103.06669",
      "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640276.pdf",
      "https://arxiv.org/abs/2504.06958",
      "https://arxiv.org/html/2507.18100v1",
      "https://arxiv.org/pdf/2502.13923",
      "https://openaccess.thecvf.com/content_CVPR_2020/papers/Xie_Self-Training_With_Noisy_Student_Improves_ImageNet_Classification_CVPR_2020_paper.pdf",
      "https://arxiv.org/abs/2211.04888",
      "https://ar5iv.labs.arxiv.org/html/2008.05721",
      "https://arxiv.org/pdf/2308.12608"
    ],
    "recommendation": "Stop treating the rally CSVs as eval-only files; promote them to a single model-agnostic \"labeled-corpus\" JSONL (d) ? one record per (video_id, rally_ordinal) carrying [start,end], ending_reason, sport, provenance/source, label_type, split-by-match, and a WASB-trajectory pointer ? with deterministic transcoders to three consumer formats (heuristic-window labels, ActivityNet/ActionFormer TAL JSON, VLM clip?QA pairs). This is the highest-leverage, lowest-cost move and it IS the moat. Then, in priority order: (1) IMMEDIATE ? grow from n?2 to n?5 matches via a small Roora batch; below ~5 matches the heuristic/LogReg tuner is data-starved and that, not the method, is why the learned model currently loses to the baseline. (2) NEAR-TERM, free ? temporal augmentation (moderate speed/time-warp, jitter, reverse, FadeMixUp) + open-teacher pseudo-labeling (WASB/our-model only, confidence-gated à la PLR/ALQA) over the owned ~26 GB corpus to multiply effective labels without violating the no-paid-LLM guardrail. (3) ONGOING ? route the Roora budget by active learning (uncertainty/disagreement/boundary entropy) once n?5, and adopt timestamp/point supervision for cheap scale-out to new sports and ending_reason. (4) DEFER but design-for-now ? Option B: fine-tune a guardrail-clean base (Qwen2.5-VL-7B Apache-2.0 / InternVL2.5 MIT) with a quality SFT cold-start THEN GRPO/RFT using temporal-IoU-vs-golden as the verifiable reward ? the data-efficiency literature (VideoChat-R1 +31.8 with limited samples; VTG-R1 filtered-13K > unfiltered-56K) says our small, clean, hand-verified set is exactly the right asset for RL, so we do NOT need thousands of new matches, we need amplification + a few clean ones. Do NOT invest in training a from-scratch TAL net (ActionFormer-class wants ~50?200+ labeled videos we won't have soon). Reuse `metrics.match_intervals` as the single spine for heuristic-labeling, TAL eval, and the RL

reward to avoid metric drift and data leakage.",
    "confidence": "medium",
    "caveats": [
      "The 'how many matches' bands are calibrated from public TAL/VLM literature (THUMOS14/ActivityNet scale, VTG datasets of 13K?18K) and our own n=2 results, NOT measured on our amateur single-camera 60fps footage ? domain transfer from broadcast-trained literature is the dominant unknown and likely pushes our requirements UP (our own research doc flags this exact broadcast?amateur gap).",
      "'Single-digit-thousands of clip?QA pairs' for Option B assumes augmentation + QA-task multiplexing + confident pseudo-labels reach it; ~128 raw human rallies alone are NOT enough ? amplification is load-bearing, not optional.",
      "Pseudo-labeling inherits WASB's recall ceiling (our confirmed bottleneck): pseudo-rallies can only be as complete as the teacher, so they boost recall on what WASB sees and must be confidence-gated + human-spot-checked; treat human [start,end] as the boundary authority.",
      "RFT/GRPO is LABEL-efficient but COMPUTE-heavy (multiple rollouts/sample) ? infeasible to run fully on one 8 GB RTX 2070 Super; budget cloud-GPU burst (RunPod/Lambda/Modal) and benchmark self-hosted-7B vs Gemini per-call before committing (COMMERCIALIZATION C7).",
      "Timestamp-supervision savings (~? full time; SkipTag ~?; 64.1% vs 68.0% acc) come from the Breakfast action-SEGMENTATION user study, not racket-sports rally detection ? directionally relevant but not a measured number for our task.",
      "Per the owner sequencing directive, this is a DESIGN blueprint: no Option-B implementation starts until P0?P4 scaffolding is solid and the golden-set corpus is healthy; the corpus-format (d) and n?5 (c) moves are the only items safe to act on now, and even those should get owner sign-off before code lands (CLAUDE.md: do not start implementation without explicit approval).",
      "Base-model license must be re-verified per-checkpoint at build time (Qwen2.5-VL Apache-2.0 only at 7B/32B; 3B/72B custom; Gemma 4 Apache-2.0 status pending confirmation; Gemma 3 REJECTED) ? licenses change."
    ]
  },
  {
    "dimension": "COST + LICENSING + PRIVACY/COMPLIANCE for an owned, on-device-capable rally temporal-segmentation model (and its Option-B end-to-end multimodal successor) for a Bangalore-based solo-founder racket-sports highlight indexer.",
    "summary": "Three findings dominate. (1) COST: the two training paths are an order of magnitude apart. A small TAL/segmentation head (ActionFormer-class) trains on ~4.5 GB VRAM, so it fits the existing RTX 2070 Super (8 GB) locally at ~$0 marginal cost, or a few dollars of rented GPU per full re-train; this matches today's bottleneck (golden data, not method). LoRA fine-tuning a 2-7B video VLM needs ~16-24 GB VRAM ? it does NOT fit the 2070, so it is cloud-only: realistic ranges are roughly $5-40 for a single LoRA run on rented GPU (RunPod A100 80GB at ~$1.19-1.39/hr, ~4-30 GPU-hrs depending on clip count/epochs) or $2-30 on managed token-priced backends (Together/Fireworks LoRA SFT ?16B at ~$0.48-0.50/M tokens). Full multimodal fine-tuning is far heavier (~15 GPU-hrs/epoch on 4xA100 reported for a 7B VLM ? roughly $70-300/epoch rented). (2) LICENSING: Apache-2.0-clean bases exist but the size matters enormously for Qwen ? verified per-checkpoint: Qwen2.5-VL-7B = Apache-2.0 (clean), but 3B = Qwen Research License (NON-commercial) and 72B = custom Qwen License (commercial-OK but 100M-MAU gate + 'Built with Qwen' attribution + competing-clause concerns). Qwen3-VL 4B/8B/30B/235B are Apache-2.0. InternVL2.5 is MIT but some variants embed a Qwen-3B research-licensed component ? verify the specific checkpoint. Gemma 3 is correctly REJECTED (custom Gemma Terms + Prohibited Use Policy + viral distribution-flowdown). Gemma 4 reportedly ships Apache-2.0 but I could only confirm this via secondary/editorial sources (not a primary Google page) and it is image-multimodal, with no confirmed VIDEO capability ? treat as AMBER. Training on your OWN footage + human rally labels is clean (you own the footage; labels are facts/timestamps, not copyrightable expression; WASB code is MIT). (3) PRIVACY: Local inference is the strongest privacy lever (video never leaves device ? no cross-border issue at all). For training/fallback backends: Fireworks and Together both contractually state they do NOT train on customer data, both are SOC2 Type II (Fireworks also HIPAA, BAAs), both offer Zero Data Retention; Google Vertex AI states it 'won't use your data to train or fine-tune any models without your prior permission,' with 24h in-memory-only caching and a ZDR path. India's DPDP Act 2023 is permissive on residency (transfers allowed except to a govt 'negative list' ? currently empty as of Feb 2026), but minors are under-16 and require verifiable parental consent ? reinforcing the committed 'exclude minors from training pool' guardrail. The owned-model 'no paid-LLM outputs as training data' rule is sound: Gemini/OpenAI/Anthropic terms restrict using outputs to build competing models, so they remain eval/teacher-only.",
    "options": [

{
"name": "Path 1 ? Small TAL/segmentation head, trained LOCALLY (Option A, near-term)",
"what": "ActionFormer-class or your distilled classifier on WASB-trajectory + (later) audio/pose features ? rally [start,end]. ~4.5 GB training VRAM fits the RTX 2070 Super (8 GB) + WSL2. No VLM, no per-call fees.",
"pros": [
"Effectively $0 marginal training cost on existing hardware; full re-train on a new golden batch costs a few $ if you rent a GPU at all",
"Fastest iteration loop ? re-train per labeled batch, CI-runnable, matches nanochat-style operability goal",
"All deps permissive (ActionFormer MIT, WASB code MIT, torch BSD) ? zero licensing exposure",
"Directly attacks today's real bottleneck (golden-data volume + WASB recall), not the modeling method",
"Cheapest at inference ? best steady-state margin per COMMERCIALIZATION C7"
],
"cons": [
"Inherits WASB candidate-window recall ceiling (the structural limit Option B is meant to remove)",
"Not end-to-end multimodal ? caps eventual quality vs a frames+audio+pose model",
"Distilled learned model currently UNDERPERFORMS the threshold baseline at n=2 videos ? needs more data before it pays off"
],
"cost_note": "~$0 on the local 2070 (8 GB) for the head; if rented, a full re-train is single-digit dollars (e.g. <1-3 RunPod GPU-hrs at $0.34 RTX-4090 to $1.39 A100). Inference is near-free.",
"license": "MIT/BSD/Apache-2.0 across the stack ? clean.",
"privacy_note": "Trains and infers fully local; user video never leaves device. No cross-border/DPDP/GDPR transfer issue at all. Strongest privacy posture.",
"fit_for_us": "strong"
},
{
"name": "Path 2 ? LoRA fine-tune a 2-7B video VLM on RENTED cloud GPU (Option B bridge, self-managed)",
"what": "ms-swift/Unsloth LoRA/QLoRA on Qwen2.5-VL-7B (Apache-2.0) or Qwen3-VL-8B (Apache-2.0) over 5k-20k clip?QA pairs, rented on RunPod/Lambda/Modal. ~16-24 GB VRAM ? A100 40/80GB class.",
"pros": [
"Removes WASB recall ceiling; moves toward the owned end-to-end multimodal target",
"Per-run cost is modest and pay-as-you-go ? no standing GPU bill",
"Apache-2.0 base (at 7B/8B) keeps fine-tuned weights private, gateable, monetizable",
"RunPod is cheapest self-serve (A100 80GB ~$1.19-1.39/hr; H100 PCIe ~$1.99/hr; spot 40-50% off); you keep full control of weights + data"
],
"cons": [
"Does NOT fit the local 2070 (8 GB) ? must rent; adds a second stack to operate",
"You own MLOps: data prep, leakage guards (split by match not clip), eval gate, reproducibility",
"Self-managed rental providers offer thin/no DPA vs managed APIs ? keep training data to owned/consented footage only (which is already the plan)",
"Needs 5k-20k curated QA pairs to be worth it; current 96+32 labeled rallies are far below that band"
],
"cost_note": "Single LoRA run ? $5-40 (?4-30 A100-hrs at ~$1.2-1.4/hr; 7B LoRA over ~1k-10k examples ? 0.5-5 hrs/epoch on one A100). Full multimodal FT is ~10-15x heavier (~$70-300/epoch on 4xA100). Spot pricing cuts ~40-50%.",
"license": "Qwen2.5-VL-7B = Apache-2.0 (VERIFIED clean). Qwen3-VL-8B = Apache-2.0. AVOID Qwen2.5-VL-3B (Qwen Research License, non-commercial) and 72B (custom license, 100M-MAU gate + attribution).",
"privacy_note": "Training data = your owned/consented footage only; no user uploads, no paid-LLM outputs. Self-managed rental ? verify the provider's own DPA/retention; prefer the providers' ZDR/non-training defaults where offered. Inference can still be run locally after training.",
"fit_for_us": "medium"
},

```
      {
        "name": "Path 3 ? Managed fine-tuning backend (Together AI or Fireworks AI)",
        "what": "Token-priced LoRA SFT on a supported open base, run as a managed job. You
upload the QA dataset; they return tuned weights/an endpoint.",
        "pros": [
          "Strong, explicit privacy contracts: Fireworks states it does NOT train on customer
content and is ZDR-by-default + SOC2 Type II + HIPAA/BAA; Together states no training without
opt-in, offers ZDR, SOC2 Type II, BAAs",
          "Cheap for the small band: LoRA SFT ?16B ~$0.48-0.50/M tokens ? a 5k-20k QA-pair run
is typically single-digit to low-tens of dollars",
          "Zero MLOps for the training run itself; fits the 'agent runs a fine-tune unattended'
operability goal",
          "Together fine-tuning catalog includes Qwen variants ? keeps you on an Apache-2.0
base"
        ],
        "cons": [
          "Less control over exact base checkpoint/version and weight portability than self-
hosting",
          "Per-run economics only beat rented GPU at low/medium volume; high sustained volume
favors owning the GPU",
          "Need to confirm the specific base offered is the Apache-2.0 size (e.g. Qwen 7B/8B,
not 3B/72B) before committing",
          "Data residency: confirm region ? these are US-centric; DPDP permits the transfer
today but Significant-Data-Fiduciary localization powers exist"
        ],
        "cost_note": "Together LoRA SFT: ?16B $0.48/M, 17-69B $1.50/M, 70-100B $2.90/M tokens
(full FT ~2.5x). Fireworks fine-tuning from ~$0.50/M (?16B). A realistic 5k-20k clip?QA-pair run
lands in the ~$2-30 range.",
        "license": "Base must be an Apache-2.0/MIT open model (Qwen 7B/8B, InternVL2.5 verified-
clean variant). Provider tooling is fine; the weights' license is what governs monetization.",
        "privacy_note": "Best contractual privacy of the cloud options: Fireworks 'does not use
customer Content to train/improve models' + ZDR default; Together 'no training without opt-in' +
ZDR setting. Still: route only owned/consented footage, never user uploads or paid-LLM outputs,
and prefer in-region/ZDR config.",
        "fit_for_us": "medium"
      },
      {
        "name": "Path 4 ? Vertex AI managed tuning (the option the owner named)",
        "what": "Google Vertex AI tuning/Model Garden for an open base or Gemini tuning.
Managed, integrated with GCP storage/compute.",
        "pros": [
          "Clear data-governance statement: Google 'won't use your data to train or fine-tune
any AI/ML models without your prior permission or instruction'; 24h in-memory-only caching; ZDR
path available",
          "Mumbai (asia-south1) region available ? can keep data in-India, easing
DPDP/Significant-Data-Fiduciary concerns",
          "Mature MLOps, integrates with the existing Gemini fallback already in the provider
registry"
        ],
        "cons": [
          "GUARDRAIL TRAP: tuning Gemini itself does NOT yield an OWNED, on-device base ? it
stays Google-served; only tuning an OPEN base in Model Garden gives portable weights",
          "Tends to be pricier and more lock-in-prone than RunPod/Together for small runs",
          "Open-base catalog/licensing on Vertex must be checked per-model; don't assume
Apache-2.0",
          "Heavier setup than a single rented GPU for a solo founder's first runs"
        ],
        "cost_note": "GPU/tuning priced above the cheapest self-serve rentals; exact tuning cost
depends on the base + tokens. Use only if in-India residency or GCP integration outweighs the
premium.",
        "license": "Acceptable ONLY if you tune an Apache-2.0/MIT open base with portable
weights. Gemini-tuned models are not owned/on-device ? fails the owned-model goal (fine as a
fallback, not as 'the owned model').",
        "privacy_note": "Vertex paid tier: no training on customer data without permission;
asia-south1 (Mumbai) keeps data in-India. Good privacy + residency story; weaker on the
```

'owned/on-device' requirement.",
        "fit_for_us": "weak"
      }
    ],
    "key_sources": [
      "https://www.runpod.io/pricing",
      "https://lambda.ai/pricing",
      "https://www.spheron.network/blog/gpu-cloud-pricing-comparison-2026/",
      "https://www.together.ai/pricing",
      "https://fireworks.ai/pricing",
      "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct",
      "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/blob/main/LICENSE",
      "https://huggingface.co/Qwen/Qwen2.5-VL-72B-Instruct/blob/main/LICENSE",
      "https://github.com/QwenLM/Qwen3-VL/blob/main/LICENSE",
      "https://huggingface.co/OpenGVLab/InternVL2_5-8B-MPO",
      "https://github.com/OpenGVLab/InternVL/issues/900",
      "https://ai.google.dev/gemma/terms",
      "https://ai.google.dev/gemma/prohibited_use_policy",
      "https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-
deployment",
      "https://docs.cloud.google.com/vertex-ai/generative-ai/docs/data-governance",
      "https://docs.fireworks.ai/guides/security_compliance/data_security",
      "https://www.together.ai/blog/soc-2-compliance",
      "https://www.meity.gov.in/static/uploads/2024/06/2bf1f0e9f04e6fb4f8fef35e82c42aa5.pdf",
      "https://www.cyrilshroff.com/wp-content/uploads/2025/12/FAQs-DPDPA.pdf"
    ],
    "recommendation": "Stage it. (1) NOW: keep building the small local TAL/segmentation head
(Path 1) ? it costs ~$0 on the existing RTX 2070 Super, attacks the real bottleneck (golden-data
volume + WASB recall), and carries zero licensing/privacy exposure. Spend the budget on golden
labels (Roora/rally-annotator), not GPUs. (2) WHEN the corpus reaches the 5k-20k clip?QA-pair
band: prototype the owned VLM with a LoRA fine-tune of Qwen2.5-VL-7B or Qwen3-VL-8B (both
VERIFIED Apache-2.0) ? first on RunPod rented A100 80GB (Path 2, ~$5-40/run, cheapest + full
weight control) for the proof-of-concept; graduate to a managed backend (Path 3) if you want the
stronger DPA + zero-MLOps and the per-run economics hold. Treat Vertex (Path 4) as in-India-
residency or fallback infra, NOT as the owned model. (3) HARD licensing locks to encode in the
C14 gate: APPROVE Qwen2.5-VL-7B/32B, Qwen3-VL-4B/8B/30B, InternVL2.5 (verified-clean variant);
REJECT Qwen2.5-VL-3B (research/non-commercial), Qwen 72B (custom license), Gemma 3 (custom viral
terms). Hold Gemma 4 as AMBER until you confirm Apache-2.0 from a primary Google source AND
confirm video (not just image) capability. (4) PRIVACY: lean on local inference as the headline
privacy lever; for any cloud training, enable ZDR/non-training configs (Fireworks/Together both
contractually don't train on customer data; Vertex won't without permission), route ONLY
owned/consented footage, keep minors out of the training pool (DPDP under-16 + verifiable
parental consent), and keep paid-LLM outputs strictly eval/teacher-only.",
    "confidence": "medium",
    "caveats": [
      "GPU prices are volatile and provider-/region-/spot-dependent; the $ ranges (RunPod A100
80GB ~$1.19-1.39/hr, H100 PCIe ~$1.99/hr; Lambda H100 ~$3.29/hr) are 2026 snapshots ? re-quote
before committing budget. RunPod adds ~$0.10-0.40/instance fees on top of GPU rates.",
      "Training-cost ranges depend heavily on dataset size, epochs, sequence/frame length, and
LoRA vs full FT. The VLM LoRA figures assume the realistic 5k-20k clip?QA-pair band; full
multimodal fine-tuning is roughly 10-15x heavier per epoch. Benchmark on your real workload
before pricing the product (echoes COMMERCIALIZATION C7).",
      "Gemma 4 Apache-2.0 is the one license claim I could only confirm via secondary/editorial
sources (mindstudio.ai), not a primary Google license page; and those sources describe it as
image-multimodal with NO confirmed video capability. Verify both the license AND video support
from ai.google.dev before relying on it as a base. Gemma 3 rejection is well-grounded (custom
Gemma Terms + Prohibited Use Policy + distribution flow-down).",
      "InternVL2.5 is MIT, but a GitHub issue confirms some variants embed a Qwen-3B-Instruct
component whose license can be the non-commercial Qwen Research License ? verify the EXACT
checkpoint's bundled components before commercial use.",
      "Backend 'we don't train on your data' claims are from each vendor's own docs/blog
(Fireworks, Together, Google) ? strong but self-asserted; get the signed DPA/BAA before sending
any footage, and confirm the contracted region for India residency.",
      "India DPDP residency is permissive TODAY (empty negative list as of Feb 2026), but the
govt retains Significant-Data-Fiduciary localization powers and the DPDP Rules are still

settling ? re-check if you scale or get designated an SDF.",
      "The 'own footage + human labels = clean to train on' conclusion is a strong reading (you own the video; timestamps/reasons are non-copyrightable facts; WASB code is MIT) but is not formal legal advice ? the per-clip consent ledger + minors exclusion + separate training opt-in already planned are the right operational guardrails; confirm with counsel before scale."
    ]
  }
]

2. assistant (2026-06-08T13:21:32.498Z)

I'll analyze this study as a completeness critic. Let me first ground myself in the actual project artifacts the study claims to draw on, to verify alignment with guardrails and check claims against reality.

3. user (2026-06-08T13:21:33.285Z)

```
# Commercialization Tracker

**Status:** ? LIVING ? track this in parallel with product/quality work.
**Owner:** avidullu · **Last refreshed:** 2026-06-07

**Purpose.** Single source of truth for the **commercial** dimension of the project:
licensing/legal posture, codec & output economics, hosting unit-economics, the
golden-set vendoring spend, packaging/distribution, and the product-market-fit
risks that gate monetization. Updated as decisions land ? see the
[Changelog](#changelog) at the bottom.

**Merged from / supersedes:**
- `archives/MONETIZATION_AUDIT.md` (2026-05-30 deep-research licensing & economics
  audit) ? kept as the **research-of-record** (per-claim verification transcript,
  primary sources). This doc carries its *current* status forward.
- `archives/COMMERCIAL_READINESS_REVIEW.md` (code-health + PMF assessment) ?
  archived; its commercial findings + the quality-phase priorities live here now.

**Premise (unchanged, verified):** the product is a **closed-source hosted API** ?
users upload videos to a cloud NVIDIA GPU VM and get highlight reels back. We never
distribute code/binaries, so GPL/LGPL *distribution* copyleft is not triggered ? but
**AGPL §13 (network clause)**, **metered-API terms**, and **video-codec patent
royalties** all still apply.

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## 1. Commercialization dashboard

Legend: ? done · ? in progress / watch · ? blocker / needs action · ?? needs legal counsel · ? future
Priority: **P0** (do first) · **P1** · **P2** (lowest). Rows are grouped **Pending ? In-flight ? Completed**, each sorted by priority. (C# IDs are stable identifiers ? they don't imply order.)

| # | Workstream | Pri | Status | Where it stands | Next action / owner decision |
|---|---|---|---|---|---|
| | **? PENDING (needs action / not started)** | | | | |
| C5 | **Output codec royalties** | **P0** | ? | Highlights are still encoded **H.264 + AAC** (`libx264` in `backend/pipeline/stitcher/compiler.py`) ? encode-side AVC patent exposure remains. | Switch output to **AV1 + Opus** (royalty-free) on an **Ada-class GPU (L4/L40)** for HW encode. See §4. Decision: provision Ada-class GPU? |
| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic `setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting your approval** before code. |
| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md` (pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md` (#16). |
```

	? IN-FLIGHT (in progress / watch)				
C4	**Runtime AI strategy (A/B/C + owned model)**	**P0**	?	Live = **(C) Gemini 2.5 Flash, paid tier**. Long-term = **own a self-hosted generative video-QA model** (fine-tuned open VLM) as primary, **paid Gemini always-ready as fallback** via the provider registry.	Make "provider registry + paid fallback" a core principle; build the owned model **after** the scaffolding/flywheel are solid. See `PLATFORM_ARCHITECTURE.md` §5A.
C13	**Beachhead validation**	**P0**	?	Market research (`PMF_MARKET_RESEARCH.md`) recommends **India badminton academies** as the #1 beachhead. Unvalidated.	Cheapest tests: 10?15 academy/coach discovery interviews; rally-detection P/R benchmark on 100+ real amateur clips; fake-door pricing (B2C ~$99/yr, B2B academy ~$600/yr). See §6.
C8	**Golden-set vendoring spend**	**P1**	?	ADR-006: **Roora (Bengaluru)** selected for an initial paid pilot; licensed/owned video only (no user uploads to vendors).	Confirm quality-per-dollar from the pilot; then GS-3 `HumanVendorProvider`. See `GOLDEN_SET_VENDORING.md`.
C3	**Shuttle-tracking weights provenance**	**P1**	? ??	WASB-SBDT **code is MIT (clear)**; distributed **checkpoints are academic-data-trained** ? not covered by the MIT code grant. MonoTrack/YOLO-NAS banned (noncommercial).	Train-own weights (Phase-4 roadmap) **or** confirm per-checkpoint terms ?? before commercial deploy.
C14	**Owned-model base licensing + training-data policy**	**P1**	? ??	Owned video-QA model (PLATFORM §5A) must build on **Apache-2.0-clean, video-capable** bases ? **Qwen2.5-VL-7B/32B** (Apache; 3B/72B are *not*), **Qwen3-VL-4B/8B**, or **Gemma 4** ? to keep weights private, gate, and monetize. **Reject Gemma 3** (viral derivative clause ? permanent Google PUP), Llama Vision (naming + 700M-MAU), `gpt-oss` (text-only). **Hard rule: never train on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (competing-model clause). The moat is the **consented dataset + eval harness**, not the model.	Lock to Apache-2.0; **open-model teachers only**; separate explicit **training opt-in**; **exclude minors** from the training pool; per-clip consent ledger. See PLATFORM §5A/§7.
C6	**Inbound HEVC decode**	**P2**	? ??	Decoding users' GoPro H.264/**HEVC** uploads is unavoidable; HEVC pools are fragmented + highest-uncertainty.	Low/uncertain residual; confirm decode-side exposure with counsel before launch.
C7	**Hosting unit-economics**	**P2**	?	Per-video cost is *cheap* for Gemini (~$0.05 input for a 10-min clip); risk is volume × latency, not unit price. Self-host/offline costs unmeasured.	**Benchmark $/video & latency on the target VM** for A/B/C before pricing. See §5.
C10	**Value layer (product surface)**	**P2**	?	Minimal exports shipped (EDL/JSON/CSV + `basic_stats`, PR #51).	Deepen: per-rally tags/notes that feed golden, per-player/match stats, clip gallery. See §6.
C11	**Input-quality enforcement**	**P2**	?	Per-video observability reports surface a customer verdict + footgun flags incl. audio-present (PR #50).	Turn more `VIDEO_RECORDING_GUIDELINES` dimensions into validators/report rules.
C2	**`supervision` ByteTrack pin**	**P2**	?	Pinned `<0.30.0` (deprecation ahead).	Migrate before lifting the pin (BACKLOG). Not a legal risk, a maintenance one.
	? COMPLETED				
C1	**AGPL / copyleft removal**	?	?	ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **torchvision Faster R-CNN (BSD) + `supervision` ByteTrack (MIT)**. No AGPL code/weights in the stack.	Done. Keep new deps permissive.

---

2. Licensing & dependency posture (refreshed risk table)

Verdicts updated against the current stack (`requirements.txt` + code). Original
2026-05-30 verdicts and the 25-claim verification transcript are in
`archives/MONETIZATION_AUDIT.md`.

Dependency	License	SaaS obligation?	Current verdict	Notes
~~`ultralytics` + `yolov8n.pt`~~	AGPL-3.0	Yes (§13)	? **REMOVED**	Replaced per ADR-005 ? no longer in the stack.
torchvision detectors (Faster R-CNN/RetinaNet)	BSD-3	No	? LIVE	The chosen replacement; person = COCO class (see `yolo_hybrid.py` `_PERSON_CLASS_ID`).
`supervision` (ByteTrack)	MIT	No	? LIVE	Pinned `<0.30.0` ? migrate before bump (C2).
WASB-SBDT (shuttle)	MIT (code)	No	? code clear	Weights provenance open

(C3). |
MonoTrack / YOLO-NAS	noncommercial	?	? BANNED	Do not use.
Gemini API (paid tier)	service terms	Paid: no training/review	? LIVE WATCH	Must bill the paid tier to keep user video private (C4).
H.264 / HEVC / AAC codecs	patent pools	royalties on encode/decode	? ??	Separate from ffmpeg's license. Encode-side open (C5); decode-side residual (C6).
ffmpeg/ffprobe, google-genai, torch, opencv, numpy, fastapi/uvicorn/pydantic/jinja2/dotenv	LGPL/Apache/BSD/MIT	No	? CLEAR	Invoked as subprocess / imported; all permissive.
`obsreport` (reports)	permissive (private)	No	? soft dep	Pin the `git+ssh@vX` tag once the repo is published (BACKLOG).

Bottom line: the two hard blockers from the audit (Ultralytics; banned weights) are resolved/avoided. Remaining commercial-legal items are **codec royalties (C5/C6)** and **weights provenance (C3)**, plus the metered-API privacy posture (C4).

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3. Runtime AI strategy ? A/B/C

The semantic rally-boundary step is the one external/metered dependency. Three paths (from the audit), with current standing:

- **(A) Eliminate ? offline learned segmenter.** ActionFormer (MIT) / our own learned classifier on `candidate_segments.stats`. **Best steady-state margin** (no per-call cost, no external dep) and matches the Phase-4 roadmap (substrate = WASB shuttle-motion, decided in ADR-004). **Most engineering.** ? *target*
- **(B) Self-host VLM.** Qwen2.5-VL-7B/32B (Apache-2.0) or InternVL2.5-8B (MIT). No per-call fees; pays in GPU VRAM (~16?24 GB for 7B). ?? Qwen 3B is noncommercial.
- **(C) Keep Gemini 2.5 Flash (paid).** Fastest, cheap per video, video-native; ongoing metered cost + external dependency. **? current live path.**

Hosted alternatives for (C) comparison: Anthropic Claude (no training on inputs/outputs; customer owns outputs; **no native video** ? weaker here); OpenAI (no training by default; ZDR available); Gemini remains strongest video-native.

---

4. Codec / output economics (the live gap ? C5)

The trap: ffmpeg being free (LGPL) does **not** grant the *patent* rights to the codecs it implements. Royalties attach to the act of encode/decode.

- **Encode side (we control it):** highlights currently emit **H.264/AAC**. AVC (Via LA) is first 100k units/yr royalty-free with a permanent "free to end-users over the internet" provision ? a small SaaS likely falls under thresholds, but ?? confirm.
- **Mitigation:** emit **AV1 (SVT-AV1, BSD + AOMedia royalty-free patent grant) + Opus (royalty-free)**. NVENC AV1 HW-encode needs **Ada-class GPUs (L4/L40)**; Ampere/Turing must use **SVT-AV1 (CPU)**.
- **Decode side (unavoidable, C6):** inbound GoPro **HEVC** decode ? fragmented pools, highest uncertainty; low residual.

Recommendation: provision an **Ada-class GPU (L4/L40)** and standardize highlight output on **AV1 + Opus** ? removes the encode-side royalty question almost entirely. *Implementation lives in `backend/pipeline/stitcher/compiler.py` (codec args).*

---

5. Hosting unit-economics (C7)

Reference workload: ~30-min 1080p match on a cloud NVIDIA GPU VM.

- **(C) Gemini 2.5 Flash (paid):** input $0.30/1M tokens, output $2.50/1M; video ~263 tok/s ? 10-min clip ? 158k input tokens ? **~$0.05 input**. Cheap per unit; the economic risk is **volume × latency**. Flash-Lite is cheaper; Pro ~10×.

- **(B) Self-host Qwen2.5-VL-7B:** $0 API; cost = GPU-hours / throughput; needs ?24 GB VRAM (L4/A10). $/video vs Gemini is **open ? measure**.
- **(A) Offline ActionFormer/learned:** cheapest at inference; cost = one-time training + light GPU. **Best steady-state margin.**

> The A/B/C $/video comparison depends on GPU instance + batch throughput ?
> **benchmark on the target VM before pricing the product.**

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6. Product-market fit & the quality/data phase

Beachhead (from market research, 2026-06-07 ? see [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md))
The static-camera constraint is a **B2B filter**: the buyer must already record from a fixed angle, which points away from the median casual player and toward coaches/clubs.
Ranked beachheads:
1. **Badminton coaching academies & serious players, India** *(strongest)* ? 20M+ players, formalizing academy ecosystem (Khelo India), genuine fixed-angle recording, coaches who already monetize, badminton whitespace vs incumbents. Sell to the **academy**, not the individual; needs INR-tuned pricing.
2. **US/global pickleball players & clubs** ? fastest-growing sport (19.8M US, +45.8% YoY) but **SwingVision already owns single-camera tennis/pickleball AI** (USD 179.99/yr) ? head-to-head, not greenfield.
3. **Badminton academies/clubs in East/SE Asia + Denmark** ? highest density; China needs local hosting/partners; Denmark/EU small but high-WTP, easy to pilot.

Pricing anchors *(estimate)*: software-only BYO-camera **B2C USD 60?180/yr**, **B2B academy/club USD 300?1,500/yr** per location (undercuts hardware-bundled Veo/Pixellot/Spiideo; comparable to SwingVision software-only).

PMF risks (from the readiness review)
- **Capture quality is the gate.** Accuracy is capture-conditional; broadcast/handheld footage is a failure mode. The wedge (static, tripod, full-court) is defensible but narrows the addressable market ? enforce/guide it (C11, `VIDEO_RECORDING_GUIDELINES`).
- **Value beyond raw segments is table stakes.** Time-saved editing alone is weak; per-rally tags, stats, and a correction loop are what make it sticky (C10).
- **Golden labels are the recurring blocker** across the offline segmenter, audio fusion, and measurable quality ? the highest-leverage investment (C8).

Refined priority order (quality/data phase ? "focus on quality once new videos land")
Carried from the readiness review; commercialization items above are tracked in parallel.

1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos): ingest the new batch, expand calibration/CV/regression baselines, make label contribution + safe retraining easy and non-contaminating (eval/training separation).
2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics land.
3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/min-duration + `rally_gate`; validate WASB on real video; trial audio fusion (ADR-007).
4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ? `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ? validators/report rules; "your video is good enough" UX.
6. **Multi-sport maturation + recording best practices.**
7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
8. **Polish & distribution** (C9): installer story, first-run path, codec/output,

economics.

Recommendation: with the mid-week labeled-video batch, spend ~80% on #1 + #3
(flywheel + quality iteration on real data), then unification (#2) and deeper value (#4).

---

7. Open questions needing owner / counsel sign-off ??

1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
   insufficient (fallback). Quote-only.
2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
   confirm each (C3).
3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
   figures with counsel before relying on the "small SaaS under threshold" reading (C5/C6).
4. **Real per-video GPU cost & latency** of self-hosted VLM vs Gemini paid at expected
   throughput (C7).
5. **Provision Ada-class GPU (L4/L40)** for HW AV1 encode? (C5 ? gates the codec switch.)

Changelog

- **2026-06-07** ? Added the market-research-of-record (`PMF_MARKET_RESEARCH.md`):
  ranked beachhead (India badminton academies #1), pricing benchmarks, and a C13
  "beachhead validation" workstream. Wired the summary into §6.
- **2026-06-07** ? Created this living tracker. Merged the 2026-05-30 monetization
  audit + the commercial-readiness review. **Refreshed since the audit:** C1 AGPL
  removal now ? shipped (was the #1 blocker); added C5 as a ? (output is still H.264,
  the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
  `DEFERRED_HARDENING_PLAN.md`; reflected PR #50/#51 (input-quality reports, value-layer
  exports). Archived `COMMERCIAL_READINESS_REVIEW.md`.

See also
- `PMF_MARKET_RESEARCH.md` ? market-study-of-record (sizing, segments, competitors, beachhead; cited).
- `archives/MONETIZATION_AUDIT.md` ? licensing/economics research-of-record (sources + verification transcript).
- `archives/COMMERCIAL_READINESS_REVIEW.md` ? superseded assessment (history).
- `archives/decisions/DECISIONS.md` ? ADR-005 (permissive stack), ADR-006 (vendor process).
- `ROADMAP.md` ? Phase-4 offline segmenter (the C4 "(A) eliminate" path).
- `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
- `GOLDEN_SET_VENDORING.md` ? vendor process + Roora pilot (C8).
- `BACKLOG.md` ? `supervision` migration (C2), `obsreport` pin, codec follow-ups.

4. user (2026-06-08T13:21:34.387Z)

Platform Architecture ? Multi-User "Assistant Coach"

Status: ? Strategy / research ? for review. No code yet.
Date: 2026-06-07 · **Owner:** avidullu

Purpose. Map the path from today's **single-user, local** tool to a **multi-tenant
"assistant coach" platform**: users upload videos, get highlights **stored and
browsable by history**, and choose **where their data lives** (Google Drive / cloud
bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
execution/compute, the detection+highlight pipeline, and data-serving ? behind small

pluggable interfaces, reusing the registry pattern the codebase already uses for segmenters/providers/validators/sports/annotations.

Product framing (the north star *and* the v1 line). This is an **assistant coach**, not just a clipper ? but the strategy is **phased and unambiguous about scope**:
- **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-user history ? a coach/user correction loop that feeds the golden set** ("assistant-coach *lite*"). Essentially today's pipeline made multi-user ? this is the real v1 deliverable.
- **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the **owned video-QA model** (§5A; §6 P4?P5).
- **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete identifying templates, and deep coaching feedback under a supervising coach** (§6 P6). Inspiring, but explicitly deferred ? and the part that raises the **biometric** privacy bar (§7).

The buyer is the **academy/coach** (beachhead in [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two consequences ripple through the design: (1) the data model keeps an **optional coach ? athlete overlay over a generic core** (§4.1) ? not baked in; (2) gait/biometrics is **special-category data**, gated and deferred (§7).

> **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
> Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
> (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
> your storage, your compute"** is a genuine differentiator for privacy-sensitive
> academies ? and SwingVision proves user-side compute is commercially viable. All four
> share the same **teams ? athletes ? clips** data model, which we adopt (§4).

> Scope: this is a **direction-setting doc**. It does not start implementation; the
> first infra slice (an async job model) is the on-hold #16 in
> [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints come
> from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
> foundations) and `archives/MONETIZATION_AUDIT.md`.

1. Current state & gaps (what we're evolving from)

Concern	Today	Gap for multi-user
Identity / tenancy	none	no users, orgs, auth, or isolation
Storage	local `filepath` string; output to `output_dir`	no abstraction; can't BYO Drive/bucket; raw video must be on the box
Data model	`videos` keyed by file **MD5**, `play_segments`, `events`, `candidate_segments` (SQLite)	no owner/athlete link; **highlights aren't a first-class persisted entity** (compile writes a file and returns a path); no per-user history
Execution	pipeline runs **in-process, synchronously** in `/api/process`	blocks; can't dispatch to a user's VM/GPU; no job status
Pipeline	**already pluggable** (segmenter/provider/validator/sport/annotation registries)	reuse as-is ? this layer is the model for the rest
API	`/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth	needs auth, per-tenant scoping, async job + history endpoints

Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version` migration ladder, v1), `backend/main.py` (endpoints, sync flow), the registry pattern in `backend/pipeline/segmenters/base.py::SegmenterRegistry` and `backend/providers/base.py`.

2. Target layered architecture

Decouple into layers with a clean **control-plane vs data-plane** split ? the orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live elsewhere (even in the user's own cloud).

```
                ???????????????????????????????????????????????
   CONTROL      ?  Presentation / API  (auth, per-tenant)       ?
   PLANE        ?  Identity & Tenancy  (org?coach?athlete?user)  ?
                ?  Orchestration / Jobs (submit?status?result)   ?
                ????????????????????????????????????????????????
                                ? dispatch       ? metadata
                ?????????????????????????  ??????????????????????
   DATA         ?  Execution / Compute  ?  ?  Data / Serving     ?
   PLANE        ?  (local | BYO-VM |    ?  ?  (per-tenant DB:    ?
                ?   hosted GPU)         ?  ?   videos, segments, ?
                ?   runs the Pipeline:  ?  ?   highlights, jobs, ?
                ?   detect ? highlight  ?  ?   history)          ?
                ?????????????????????????  ??????????????????????
                            ? read input / write artifacts
                ???????????????????????????????????????????????
                ?  Storage  (local upload | Google Drive | S3 |  ?
                ?  GCS | Azure)  ? raw video + highlight outputs ?
                ????????????????????????????????????????????????
```

The two **new pluggable seams** are **Storage** and **Compute/Execution**; everything else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A **model/training layer** ? the owned generative video-QA model (§5A) ? sits atop these, fed by Storage (training corpus), Data (labels), and Compute (training + inference), and is delivered through the existing **AI-provider registry with a paid fallback**.

3. Pluggable abstractions (extend the existing registry pattern)

The codebase already proves the model: an `ABC` + a `Registry` singleton with `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with the same shape so they're plug-and-play and testable in isolation.

3.1 `StorageBackend` registry
Replace the bare `video_path: str` that flows through the pipeline with a **`StorageRef`** (a backend id + a key/URI), resolved by a backend.

```python
class StorageBackend(ABC):
    name: str
    def open_read(self, ref: StorageRef) -> BinaryIO: ...      # stream large video
    def download_to_local(self, ref: StorageRef, dst: str): ...  # for ffmpeg/cv2/WSL
    def upload(self, local_path: str, ref: StorageRef): ...     # persist highlight/artifacts
    def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
    def stat(self, ref: StorageRef) -> StorageStat: ...
    def list(self, prefix: StorageRef) -> list[StorageRef]: ...
```

- **Recommended implementation:** build the backend over **`fsspec`** (one matrix ? local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`, Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4 byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
- **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**: the tenant registers their own bucket/Drive; raw video **never leaves their account**.
- **Auth ? never persist users' long-lived keys.** For buckets, have the user create an IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned URLs** so raw video moves **browser ? the user's bucket directly, never through our servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academies, a **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
- **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while the LLM provider can take a **signed URL**. The backend exposes both `download_to_local`

(for compute) and `signed_url` (for the provider/browser), so the pipeline stops caring *where* the bytes live.
- **Migration path:** introduce `StorageRef` alongside the current local path; a `local` backend wraps today's behavior so nothing breaks; adapters land incrementally.

3.2 `ComputeBackend` / `ExecutionBackend` registry
Where the pipeline actually runs for a given job.

- **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).
- **`hosted`** ? our GPU VM pool (the SaaS default).
- **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their own VM/GPU. **Critical security property:** the worker *pulls* jobs scoped to its tenant from the control plane, pulls input from the tenant's **own** storage, runs the pipeline, and writes artifacts + results back ? so **the orchestrator never holds the user's cloud credentials** and needs **no inbound access** to their network. This mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most credential-safe v1** for BYO-compute.
- For users who want the platform to **provision GPUs in their *own* cloud account on demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot** (BYOC: everything launches in the user's accounts/VPCs; creds stay with them). **Modal** is the managed fallback for users wanting zero infra (accepting that raw video then flows through Modal).
- **?? Operational / support burden (a real adoption risk).** For the primary beachhead ? Indian badminton **academies/coaches** ? asking them to *run and maintain a worker on their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the "it stopped after our IT change" ticket). So: **default academy users to `hosted`**, and treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**, not the academy default. If/when BYO ships, plan for **auto-updating, self-healing agents + health heartbeats + clear worker status in the UI**, and budget the support cost.
- **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the **storage** stays in the user's bucket (signed URLs) while compute runs on **our** GPUs in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side). This gives the BYO *privacy* benefit (raw video never persists on our infra) **without** the user running a worker ? a pragmatic default between full-hosted and full-BYO.

The pipeline code is unchanged; only *where it's invoked* and *how it gets bytes* vary.

3.3 Reuse, don't reinvent
The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries are the proven precedent and stay as the **pipeline** layer. The `annotation` registry + `candidate_segments` table are also the natural seam for the **coach-correction / golden-set** loop in the assistant-coach phase (§6, P4).

4. Multi-tenant data model & serving layer

4.1 Entities ? generic core, with the coach model as an optional overlay
The foundational schema is deliberately **domain-neutral** so we don't overfit to the coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will iterate** (see C13 / PMF), not a load-bearing assumption. **Generic core:**
```
User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
                                  ???< PlaySegment ??< Event
Video / Highlight ??< Job (processing record: status, compute backend, timings)
```
- **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an event, a "folder") with no assumption of a coach or academy ? a power user, a coach,

or a
parent are all just `User`s with collections.
- **Highlights are first-class** (today compile only writes a file): a `highlights` table
(segment ids + output `StorageRef` + params + created_at) powers "see past videos and
highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags**
(shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings / notes.
- **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
- **Coach overlay (optional, additive):** for the academy/coach GTM, layer
`Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an `athlete_id`
tag on collections/videos) **without** baking it into the core tables ? so the schema
stays usable for solo power users and the coach model can evolve, or be dropped, as PMF
dictates.
- **History/serving API:** list collections/videos/highlights per user (and, with the
overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g. shot
type / outcome); re-stitch from stored segments without reprocessing.

4.2 Tenancy isolation ? options & recommendation

Model	Isolation	Ops cost	Best for
Shared DB + `tenant_id` (row-level, ideally Postgres **RLS**)	logical	low	**Hosted SaaS, many small tenants** ? *recommended default*
Schema-per-tenant	medium	medium	tens?hundreds of mid tenants
DB-per-tenant (incl. **SQLite-per-tenant**)	strong/physical	higher at scale	**self-hosted single-academy**, or high-isolation enterprise

Recommendation: for the hosted product, **migrate SQLite ? Postgres with shared-DB +
`tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
single-academy** deployment (it reuses today's code almost verbatim). Either way,
abstract DB access behind a repository layer now so the engine is swappable, and
graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.

5. Orchestration / async jobs

This is the serving layer that makes multi-user real. It is the **multi-user superset of
[`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ? build
that first, single-tenant, then generalize.

- **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),
status (queued|running|done|failed), progress, compute_backend, result_ref, error,
timestamps`. `/api/process` returns **202 + job_id**; `/api/jobs/{id}` polls.
- **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
content-hash discipline already in `compute_video_id`.
- **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
writes artifacts + DB rows. **Observability reports** (already per-video) become the
job artifact surfaced in history.
- **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
requirements are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
restart on a worker blip), **progress states** ? which rule out fire-and-forget Celery/RQ
for the long jobs. The *choice* should be made by prototyping in the
[`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
  - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
    Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
    async slice, with the lowest operational complexity and the easiest debugging.
  - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
    retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
    because it adds **no new datastore** once we're on Postgres, but it *is* a new
    dependency + operational/debugging surface; adopt only when long-job resumability/

  observability demands it.
- **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
  reserve for genuinely complex multi-stage pipelines at scale.

Record the decision criteria in #16: operational complexity, maturity, **debugging
experience for long GPU jobs**, and whether durability is worth a new dependency.
**Bias: start with the simplest thing that gives retries + status; graduate only on
evidence.**

5A. Owned generative model (the video-QA assistant) & the data flywheel

The north-star capability: an **owned, self-hostable generative video model** ?
fine-tuned on sports video (including **consented, user-contributed** footage) ? that
answers questions about rallies, shots, and technique. This is the conversational
brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
is introduced **behind the same abstraction, with the paid API as an always-ready
fallback**.

> **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
> assets are **(a) the consented, badminton-specific video?QA dataset** and **(b) the
> eval harness**. Build and message *those* ? not "we own an LLM."

> **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings in
> full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
> serving / fallback* only ? never as a *training* data source.** Harvesting their outputs
> to train our owned model would breach their "no competing model" terms (a terms/copyright
> violation), so it is prohibited in both engineering and legal process. **Training teachers
> are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
> distillation constraint below.

Core principle: AI is a pluggable provider with a paid fallback
The codebase already abstracts the cloud AI behind `provider_registry` /
`AIVideoProvider` (`backend/providers/`). Make that a **first-class platform principle**:
- Every generative capability routes through the **provider registry**.
- The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
  Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
  or as the quality ceiling/teacher reference).
- This lets us **ship today on the known paid stack** and **swap in the owned model when
  its quality clears a measured bar** ? with zero architectural churn and the paid path
  always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
  owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)

The model / training layer (new sub-layer, fed by the scaffolding)
```
Data curation        Training              Eval                 Serving
(consented user vid  (fine-tune an open,   (extend the existing (self-host inference
 + golden set +      commercially-usable   eval harness to QA   as a provider /
 rally/segment/event base VLM via          quality + the        ComputeBackend;
 annotations ?       LoRA/QLoRA)           regression gate)     paid fallback)
 video-QA pairs)
```
This layer **reuses every other layer**: **Storage** holds the training corpus;
Data supplies labels (the `annotation` registry + `candidate_segments` +
golden set are already the labeling seam); **Compute/Orchestration** runs training and
inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
eval harness already in `backend/eval/` extends to grade QA quality. **In other
words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
which is why scaffolding comes first (see §6).

Base model & training (Apache-2.0 only ? licensing is decisive)

Recommended near-term base: `Qwen2.5-VL-7B-Instruct` ? Apache-2.0, native long-video
(dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
Qwen3-VL-4B/8B (Apache-2.0); **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
Google-stack alternative to evaluate.
- ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
  gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
  is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
- ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
  clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
  permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch to
  Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
  model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ? Apache-2.0
  but **text-only**, so usable only as an optional text-reasoning layer, never the video
  encoder.
- **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
  avoids catastrophic forgetting of general video skills. Build the dataset by converting
  the existing rally/shot/segment annotations + golden set into templated **clip?QA
  pairs**, expand synthetically with **open-model teachers only** (see constraint below),
  and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs ?
  5k?20k curated is a realistic band for this narrow domain.
- **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is a
  comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you have
  sustained video-QA volume **+** a defensible consented dataset **+** real per-call API
  cost pain ? below that, the paid API is cheaper than a standing GPU and the second stack
  is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**

Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.

Training backend ? managed service vs self-managed (the fine-tune "forge")
Where you run the LoRA/RL fine-tune. Weigh on **video support, weight export/ownership,
smallest servable model, RL support, no-train DPA, and infra burden**.

Decisive finding (researched 2026-06): managed fine-tuning services are
image-based (JSONL + base64 images) ? for video you pre-sample frames and feed them
as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
video (+audio) training.** So for true video-QA, a self-managed framework is the
strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
serve) and keep full ownership. Managed services trade some of that for zero infra.

Option	Type	Video / VLM support	Smallest VL	RL	Export weights	Best for
ms-swift (ModelScope)	Self-managed (rented/own GPU)	**Native video+audio**; Qwen3-VL/Omni, 300+ MLLMs	any incl **7B**	? DPO/GRPO	? full ownership	**? video-QA; slots into BYO-compute (§3.2)**
LLaMA-Factory	Self-managed	Qwen3-VL + 100+ VLMs; video datasets	any incl 7B	?	?	popular, well-documented
Unsloth	Self-managed (1-GPU optimized)	Qwen3-VL LoRA (vision+lang; frames)	any incl 7B	? GRPO	?	cheapest first runs
Tinker (Thinking Machines)	Managed API (low-level)	Qwen3-VL **30B-A3B+ only**; image input (video **unverified**)	30B-A3B MoE	? PPO/GRPO, custom loss	? adapters + ckpt; no-train ToS	zero-infra RL/research at MoE scale
Together AI	Managed (tune + serve)	Qwen3-VL / Gemma-3 / Llama-4; image data	verify	partial	? adapters downloadable	managed Qwen3-VL + serving
Fireworks AI	Managed (tune + serve)	Full Qwen2.5-VL (3B?72B SFT; LoRA@32B + LoRA-upload)	**7B**	partial	?? verify export (lock-in risk)	managed *small*-VL + serving

Predibase (Rubrik)	Managed **+ VPC self-host**	VLM instruct-tune; **LoRAX** multi-adapter	verify	? (VPC)	? **RFT/GRPO** flagship	RL "creativity" + multi-adapter + privacy
~~OpenAI GPT-4o / Gemini tuning~~	Closed	?	?	?	? no export	**non-fit ? violates "own the model"**

*(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
control but are only worth it if you're already on that stack.)*

Recommendation:
- **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
  Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud): native
  video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
  ComputeBackend / BYO-compute seam (§3.2). **Likely a better fit than Tinker** for
  the video requirement *and* serving economics.
- **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
  or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support and
  adapter export.
- **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
  (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
  level RL primitives).
- **Tinker** remains a strong zero-infra RL-research option with clean weight export ? just
  not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
- **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
  backend is swappable ? same ethos as the provider/registry pattern. Don't get locked in.

Verify before committing (any backend): (1) **native video / frame-sequence**
ingestion + max frames per example; (2) **weight export** + a **commercial DPA** covering
consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the size
you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).

Training harness ? make it agent-runnable (nanochat-style operability)
The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled video
and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is the
reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
five properties are exactly what make training agent-runnable, and we copy them:
(1) one command, end-to-end; **(2)** one complexity dial (sane auto-derived defaults);
(3) an automatic **report card** ? a deterministic pass/fail vs a target (the crux);
(4) reproducible + deterministic + cheap; **(5)** minimal/readable.

Engine (don't reinvent): **`ms-swift`** already delivers this for VLMs ? CLI-driven,
native video, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
`args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ?
~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)

The harness = thin glue over components we already have ? one declarative,
agent-invokable job (call it `train_speedrun`):
1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry +
   `candidate_segments` + golden set ? the labeling seam in `backend/`).
2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an orchestration
   job (§5 / DEFERRED #16): one config, idempotent, logged.
3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness

to
   QA quality; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays) **only
   if it beats the bar**.

So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register provider**, behind one command an agent or CI runs. nanochat is the inspiration for the wrapper; ms-swift is the engine; the **golden QA eval is the report card**.

> **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA eval
> set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic and
> trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
> optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)

?? Teacher / distillation constraint (the single biggest legal landmine)
You **cannot** train the owned model on outputs from the paid frontier APIs ? a general video-QA assistant is squarely a *"competing model"*: **Gemini** ("may not use the Services to develop models that compete"), **OpenAI** ("may not use Output to develop models that compete"), **Anthropic** (bars using Outputs to train competing models). Two safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL / InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime fallback** ? serving answers is fine, *harvesting its outputs to train the model that replaces it is not*. Write a hard **"no training on competitor-API outputs"** rule into eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise: the teacher is *open models*, not the paid APIs.)

Data flywheel, personalization & consent tiers
Two complementary loops, each with its **own** opt-in (training consent is separate from processing consent ? §7):

- **Global model (shared ? raises the floor).** Train the base owned model on an
  explicitly-contributed, licensed corpus ? *not* on whatever sits in a tenant's bucket
  (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
  better shared model ? better highlights/answers ? more reason to contribute.* Higher
  consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.

- **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under exploration.**
  A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
  consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
  Together / Predibase). Why it's compelling:
  - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
    styles ? materially better detection/highlights for them.
  - **Cleanest consent story:** *"improve **your** detection using **your** data"* is far
    lower-friction and lower-risk than contributing to a global model ? the data improves
    only the user's own adapter, with **no cross-user mixing**, sidestepping much of the
    pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your video,
    your storage, your compute ? and now **your model**.* With BYO-compute the adapter can be
    trained/served on the user's own GPU and **never leave their boundary**; **deletion =
    drop the adapter** (clean erasure).
  - **Open questions (still a hunch):** cold-start (need enough of the user's labeled data
    before a personal adapter beats the base ? fall back to the shared model meanwhile); a
    per-user held-out eval to prove the adapter helps (don't ship a regression);

base +
optional global-adapter + per-user-adapter stacking; per-user training cost at scale.
Worth a spike once the harness + golden eval exist.

Two consent tiers, then: **(a)** "improve **my own** detection with **my** data"
(per-user, low friction) and **(b)** "contribute to the **shared** model" (global, higher
bar). **Default for both is off.**

6. Phased roadmap

Phase	Deliverable	Build / reuse
P0	**Async job model, single-tenant** (DEFERRED #16)	`jobs` table via `user_version` ladder; worker; 202+status endpoints
P1	**Identity + multi-tenant data model + upload + highlight persistence + history**	auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS; `highlights` table; repository layer; history API + UI
P2	**`StorageBackend` registry + BYO-storage**	new ABC+registry; `local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
P3	**`ComputeBackend` registry + BYO-compute**	`local`/`hosted`/`byo_worker`; pull-based worker agent; control/data-plane split
P4	**Rich rally/shot semantic metadata & tagging** *(the next value stage after solid rally detection + highlights)*	shot type (smash / backhand drive / net), rally **outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ? query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden set via the `annotation` registry + `candidate_segments`
P5	**Owned generative video-QA model** (§5A)	global flywheel **+ per-user adapters**; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as `owned_vlm` **provider with Gemini paid fallback**; extend the eval harness to QA quality
P6 *(visionary)*	**Gait / biomechanics under a supervising coach**	deliberately **back-burner**; technique feedback; highest Art. 9 bar ? gate behind explicit biometric + parental consent (§7)

P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the
immediate next value stage once rally detection + highlights are solid** ? e.g. *"rallies
that ended in a smash or backhand drive,"* *"rallies where the court was open but the
player made an unforced error."* **P5** owns the model; **P6 (gait) is the visionary
horizon, intentionally deferred.**
Sequencing principle (owner directive): make the scaffolding (P0?P4) **rock-solid
first**, keep the **known paid stack as an always-ready fallback**, and only then shift
focus to **owned-model quality** (P5). The provider registry keeps the paid path one
config flip away.

7. Privacy & legal (gait raises the bar ? design for it now)

- **The load-bearing nuance:** biometric data is special-category **only when used to
  uniquely identify a person** (GDPR Art. 9). **Highlight/shot detection on players a
  coach already knows is general analytics ? generally *not* Art. 9.** But
  gait/biomechanics that builds a per-athlete identifying template tips into Art. 9
  and needs **explicit, granular consent** referencing biometric data. Engineer gait
  with that boundary (consent gate, purpose limitation, no cross-account identification);
  decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB
  Guidelines 3/2019 on video devices apply.)
- **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-order,
  not an afterthought. Model consent explicitly per relationship (user?subject, coach?athlete
  where the overlay applies, parent?minor).
- **BYO-storage is the central privacy/liability lever:** if raw video stays in the

  tenant's own bucket/Drive and we process via signed URL or their compute, we minimize what we hold ? shrinking breach surface, residency, and erasure obligations.
- **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training), **not the consumer/free tier**; set short/zero retention, document it in the DPA, and surface "your footage is never used to train AI" as a trust feature for academies. For the gait phase, prefer keeping biometric inference off third-party APIs entirely.
- **Training consent is a distinct, higher bar (for the owned model, §5A).** Training on user video needs a **separate, explicit, opt-in training licence** ? not bundled with processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent issues above. Train **only on explicitly contributed/licensed video**; never train on what merely passes through a tenant's bucket. **Exclude minors' video from the training pool by default**, keep the training corpus a **separate, firewalled, opt-in sub-pool**, and log **per-clip consent/provenance** for revocation + audit. Incentives must not make consent non-free (the product stays fully usable without contributing). And the **teacher-distillation rule** (§5A): never train on paid-LLM outputs. **Two consent tiers (§5A):** *(a)* "improve **my own** detection with **my** data" ? a per-user adapter trained only on the user's own video (lowest friction; no cross-user mixing; deletion = drop the adapter); *(b)* "contribute to the **shared** model" ? pooled corpus, higher bar. Default for both is **off**.
- **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across DB + storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends here).

8. Key decisions for the owner (the forks)

1. **Hosted-first vs BYO-first.** Recommended: **hosted control-plane first** (P0?P1) with **BYO-storage soon after** (P2); **BYO-compute later** (P3) as a power-user/enterprise option ? it's the highest-complexity, highest-security item.
2. **Database now or later.** Move to **Postgres at P1** for the hosted path, or stay SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access regardless.)
3. **Auth: build vs buy** (Auth0 / Clerk / Supabase Auth / Cognito).
4. **Orchestration tech** (see research recommendation) ? bias to the simplest that gives retries + status.
5. **Gait processing location** ? local/on-tenant (privacy-max) vs cloud (capability-max).
6. **How far to push BYO-compute** given its security/ops complexity.
7. **Owned-model base + licensing (§5A).** Which commercial-OK video VLM to build on (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial derivative is permitted ? see COMMERCIALIZATION C14.
8. **Training-data sourcing & consent.** Opt-in contribution terms, incentives, and the teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
9. **When to start the owned model** ? strictly after the scaffolding + flywheel are solid (the explicit directive), with the paid provider as the standing fallback.

Risks: scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS + tests); BYO-compute security; **biometric/minor liability** (get this right before any gait feature ships); **training on user video without proper opt-in/licence**, and **base-model / teacher-distillation licence violations**; cost of hosted GPU (training *and* inference); **premature model-building** before data/eval are ready (mitigated by keeping the paid fallback and gating P5 behind the flywheel).

9. Sources

External best-practices research (2026-06-07) backing the recommendations above. Each underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and should be re-validated before committing budget.

Storage abstraction: [fsspec](https://fsspec.com/) · [cloudpathlib](https://github.com/drivendataorg/cloudpathlib) · [smart_open](https://github.com/piskvorky/smart_open) · [Apache Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) · [signed URLs across clouds](https://medium.com/@shaikhsarwan49/signed-urls-in-aws-gcp-and-azure-a-beginner-friendly-guide-e2e549425865)

Multi-tenancy: [Bytebase: multi-tenant DB patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/) · [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

BYO-compute / orchestration: [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) · [Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) · [GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank: Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-deployment) · [DevPro: Python task queues 2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) · [Hatchet](https://github.com/hatchet-dev/hatchet)

Privacy / legal: [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

Comparable products: [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

Training backends & harness (§5A): [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-

tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)
562
563   **Owned model & licensing (§5A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)
564
565   ## See also
566   - [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.
567   - [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.
568   - [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).
569   - [`CODE_MAP.md`](CODE_MAP.md) ? current architecture + the registry extension points.
570   - [`ROADMAP.md`](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).
571

5. assistant (2026-06-08T13:21:37.236Z)

Now let me verify a few specific claims against the actual code the study cites as load-bearing, particularly the label-reuse spine.

6. user (2026-06-08T13:21:37.875Z)

```
backend\eval\__init__.py
backend\eval\annotations.py
backend\eval\classifier.py
backend\eval\metrics.py
backend\eval\training.py
backend\eval\calibration.py
backend\eval\gemini_refine.py
backend\eval\calibrate_local.py
backend\eval\calibrate_wasb.py
backend\eval\distill_local.py
backend\eval\export_wasb_segments.py
backend\eval\harness.py
backend\eval\regression.py
backend\eval\batch.py
backend\eval\eval_partial_labels.py
```

7. user (2026-06-08T13:21:38.625Z)

```markdown
# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging

**Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
"config & identity consistency" tech-debt pass. Each item below becomes its own PR
only after this doc is approved.

**Why these three:** they were cataloged as out-of-scope during the consistency
pass because each is larger/design-heavy or needs the owner's machine. They are the
remaining high-value hygiene items not already done (config/structure/Result/logging/
reports) or in flight (detector abstraction, PR #49).

**Recommended sequence:** #7 Packaging (lowest risk, unblocks clean installs/CI) ?
#13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
(largest, product-facing).

---

## 1. Packaging modernization (debt #7)

### Problem
- No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done",
  but it isn't present ? reconcile with the owner.)
- The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build
  script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
- Deps live only in `requirements.txt`; `obsreport` is a commented soft dep
  (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).

### Approach
- Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring
  `requirements.txt`, with optional-dependency groups:
  - `[project.optional-dependencies] dev` ? `pytest`, coverage tooling.
  - `reporting` ? `obsreport` pinned to its git tag (`v0.1.3`, per BACKLOG).
  - `gpu` ? torch/torchvision (note: the cu124 `--extra-index-url` is **not**
    expressible in standard PEP 621 deps; document the install line or use a
    `[tool.uv]`/pip-conf approach rather than faking it).
- Console entry points: `indexer = backend.main:main` (so `indexer --server` works),
  optionally eval drivers.
- **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer
  shadows the build system; update README/`run_all_tests.py`/CI references.
- Choose a build backend (hatchling recommended) and confirm the `backend/` import
  root stays valid (keep current layout; no src/ move required).

### Risks
- `setup.py` rename touches docs and any muscle-memory invocations ? grep first.
- Torch CUDA wheels can't be pinned cleanly in PEP 621; must document the
  `--extra-index-url` install path (the current `setup.py` logic can move into docs).
- CI must switch to `pip install -e .[dev]`.

### Verification
- `pip install -e .[dev]` in a clean venv ? `run_all_tests.py` green.
- `indexer --help` resolves via the entry point.

---

## 2. Train/eval separation guardrails (debt #13)

### Problem
Separation is **by convention only**. `backend/eval/calibrate_local.py` and
`calibrate_wasb.py` do honest leave-one-video-out (LOVO), but nothing *enforces*
that a video used to train the learned segmenter
(`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
eval/regression set. Leakage would silently inflate the very metrics the product
story rests on (? in BACKLOG).
```

```
### Approach
- **Split unit = match, never clip** (per `GOLDEN_SET_PHASE1_VIDEOS.md`: split by
  match, hold ~? of each sport as eval).
- Add a `split: "train" | "eval"` field to the collector's `curate export`
  manifest (cross-repo contract, ADR-002 lineage) ? or a separate eval manifest.
- New `backend/eval/splits.py`:
  - `load_split(manifest) -> (train_ids, eval_ids)` keyed by **match id** (not just
    file MD5 ? re-encoded duplicates share a match but differ in
    `compute_video_id`, so group at match level).
  - `assert_disjoint(train_ids, eval_ids)` ? raise on any overlap.
- Wire the guard into the training entrypoint (`distill_local`/`training`) and into
  `run_regression.py` so the baseline is computed strictly on the eval split.

### Risks
- Requires a collector-side manifest change (coordinate the cross-repo contract).
- Small corpus ? tiny held-out set ? high metric variance; the guardrail is correct
  but the numbers will be noisy until the golden set grows.
- MD5-only dedup is insufficient (re-encodes evade it) ? must track match identity.

### Verification
- Unit test: overlapping train/eval ids ? `assert_disjoint` raises.
- Integration: a deliberately-leaked manifest fails the training entrypoint.

---

## 3. Async job model (debt #16)

### Problem
`POST /api/process` runs validate ? segment ? report **inline**, blocking the
request for the full duration of a long video. Unworkable as a hosted/product API.

### Approach (minimal first slice ? single self-hosted box)
- `POST /api/process` returns **202 + `job_id`**; add `GET /api/jobs/{job_id}`
  returning `{status, progress, result, reports}`.
- Run the pipeline on a background worker ? start with a `ThreadPoolExecutor`
  (segmenters already use threads internally for AI handoff, so this fits); leave a
  real queue (RQ/Celery) for later if multi-process is needed.
- Persist job state in the existing SQLite DB via the migration ladder
  (`backend/infrastructure/database.py::_init_db`, bump `PRAGMA user_version` 1 ? 2,
  add a `jobs` table). The observability report becomes the job's artifact.
- Update the static UI to poll `GET /api/jobs/{id}` instead of blocking.

### Risks
- **SQLite thread-safety:** verify `check_same_thread`/connection-per-thread;
  `_get_connection` already re-issues `PRAGMA foreign_keys=ON` per connection ?
  confirm it's safe under the executor.
- Cancellation/timeout and crash recovery (a job in RUNNING after a restart).
- In-process store is not horizontally scalable ? acceptable for the current
  single-box self-hosted target; call it out so it isn't mistaken for production-HA.

### Verification
- Submit a job ? poll ? DONE with the same segments the sync path produced.
- Server restart mid-job leaves a recoverable/clearly-FAILED job, not a hang.

---

## Out of this doc
- **Detector/WSL abstraction (#6):** in flight (PR #49).
- **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
- **ByteTrack `supervision<0.30.0` migration + real-footage validation (#14/#15):**
  need a GPU/WSL box and real video ? owner-machine work, tracked separately.

```

8. assistant (2026-06-08T13:21:42.041Z)

Let me verify the specific code claims about the label-reuse spine (`match_intervals` and `label_candidates`) and check the actual rally CSV data scale claims.

9. user (2026-06-08T13:21:43.164Z)

```
backend\eval\metrics.py:48:def match_intervals(preds: List[Interval], gts: List[Interval],
backend\eval\training.py:50:def label_candidates(candidate_intervals: List[Interval],
gt_intervals: List[Interval],
```

10. user (2026-06-08T13:21:43.835Z)

```
# Roadmap ? from annotated data to a self-improving, offline rally segmenter

This document is the **narrative thread**: how the indexer evolves from an
AI-API-dependent prototype into a self-contained tool that learns rally detection
from its own commercially-licensed, annotated video corpus. It explains *how we
got here*, *why each step matters*, and *how it shapes the final tool*. Decisions
are recorded as ADRs in [DECISIONS.md](DECISIONS.md); the scoring mechanics live
in [EVALUATION.md](EVALUATION.md).

## The thesis

The indexer detects badminton rallies in long recordings. Its strongest mode
(`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
works, but it costs money, needs the network, and never improves on its own.

> *Naming note:* the `yolo_hybrid` registry key is retained for back-compat, but
> AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
> ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
> player-motion gating stage, not the Ultralytics library.

A sibling project, **sports-data-collector**, curates sports videos with
per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
this roadmap: **use that annotated, commercially-usable data to (1) measure the
pipeline objectively and (2) train an offline replacement for the paid Gemini
stage** ? turning the tool into one that improves as more data is curated, runs
air-gapped, and generalizes across sports.

Why it matters: it removes the per-run API cost and network dependency, gives us
a defensible quality metric instead of eyeballing, and creates a flywheel ?
more curated matches ? a better offline segmenter ? better highlights.

## The four phases

### Phase 1 ? Bridge the repos (DONE)
A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
the `is_commercial` flag. ? *Why:* one clean producer/consumer contract that works
for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.
See **ADR-002**.

### Phase 2 ? Acquire full-resolution video (DONE)
The corpus was 256×144 (the original crawl used `worstvideo`). The `fetch` command
re-acquires every video at **best available resolution (1080p here)**, **in place**
(preserving annotations), via **authenticated Firefox Premium cookies** and yt-dlp's
`default` client. ? *Why:* YOLO/shuttle detection needs real pixels; and the
journey surfaced non-obvious traps (SABR streaming ? 144p; Chrome DPAPI; rate
limiting) now encoded in the tooling. **22/22 fetchable matches are now 1080p.**
See **ADR-003**.

### Phase 3 ? Evaluate + tune (IN PROGRESS)
`run_phase3_eval.py` walks the manifest, processes each video, and scores detected
rallies against the ground truth ? per video and corpus-aggregate (precision/recall/
```

F1, mean IoU, boundary error), for both the `candidates` (raw detector) and `final` (refined) stages. ? *Why:* the candidates-vs-final split tells us *where* error lives ? a **detection** problem (candidates miss rallies) vs. a **segmentation** problem (candidates catch them, refinement is loose). That diagnosis decides where effort goes, and establishes the baseline any trained model must beat.

Phase 4 ? Learn an offline, API-free segmenter (NEXT)
Replace the paid Gemini refinement with a **lightweight classifier trained on the rally annotations**, consuming features the pipeline already computes and stores (`candidate_segments.stats`). No external API, no new labeling. ? *Why:* this is the payoff ? a free, offline, self-improving rally mode. The **substrate was chosen from data**: the ADR-004 update (2026-06-02) measured the YOLO/player-motion candidate stage at **recall ~0.05 @ IoU 0.5** on `shuttleset_8` (median best coverage of a rally only ~20%) ? too low a ceiling to build on. So the offline segmenter's substrate is **WASB shuttle-motion** (ADR-001), not player-motion. See **ADR-004**.

Complementary local cues (beyond the segmenter swap)

Once the substrate is WASB shuttle-motion, the next lever is **fusing additional fully-local cues** to raise rally recall/precision without reintroducing a paid API. The first one explored is **audio "thwack" hit-detection** (**ADR-007**, accepted as a direction 2026-06-07): a pure-numpy spectral-flux onset detector in `backend/pipeline/audio/`. The **E1 GO/NO-GO probe** has run on real GoPro audio ? verdict **AMBER / leaning-negative**: recall signal is real (hit-clusters recover WASB-missed rallies) but the classical-onset precision ceiling (~2.2×) is too weak to fuse cleanly, and a band-pass doesn't rescue it. The only path past that is a *learned timbre classifier* ? which, like Phase 4, is **gated on golden labels**. See `AUDIO_E1_FINDINGS.md`. Pose/court serve-gating is parked as the #2 cue.

The recurring blocker across Phases 3?4 and the audio work is the same: **more golden labels** is the single highest-leverage move (see the cross-cut note below).

How this shapes the final tool

- **Offline & free by default.** The production rally mode becomes a local model; Gemini is kept only as an upper-bound reference, not a dependency.
- **Self-improving.** Every newly curated, annotated match is more training signal; re-training is a local, deterministic step.
- **Measurable.** Every change is scored against the golden set, so "better" is a number, not a vibe.
- **Sport-generic.** The label ? feature ? classifier machinery and the export bridge already span sports; adding a sport is data, not new architecture.
- **Commercially clean.** Only `is_commercial` data flows through the bridge, so the trained model and any derived product rest on a licensable foundation.

Status at a glance

Phase	What	Status
1	Export bridge (`curate export` ? manifest + CSVs)	? Done
2	1080p acquisition (`fetch`, in-place, cookies)	? Done (22/22)
3	Eval + tuning loop (`run_phase3_eval.py`)	? In progress
4	Offline learned segmenter (WASB shuttle-motion substrate)	? Next (substrate decided: WASB, ADR-004)
+	Audio hit-detection cue (`backend/pipeline/audio/`, ADR-007)	? E1 probe run ? AMBER, gated on golden labels

> **Cross-cutting:** golden-set ground truth (which powers Phases 3?4 and on-demand regression testing)
> is produced via a vendor process + pluggable annotation provider ? see
> GOLDEN_SET_VENDORING.md and **ADR-006**. As of 2026-06-02 the annotation
> guideline is authored (collector `docs/annotation/`) and **Roora** (Bengaluru) is selected for an
> initial **3-sport pilot** (badminton, pickleball, table tennis). Input-capture quality

requirements
> that bound achievable precision/recall are tracked in
VIDEO_RECORDING_GUIDELINES.md.
>
> **Observability:** every processed video now emits an always-on per-video diagnostics
> report (spec-driven warnings/errors with FAQ refs, soft-dependency on `obsreport`), so
> pipeline regressions and footguns surface without re-running eval ? see
> CODE_MAP.md §2 (reporting/observability).

11. assistant (2026-06-08T13:21:51.228Z)

The code anchors check out (`match_intervals` at metrics.py:48, `label_candidates` at training.py:50, an `eval_partial_labels.py` exists matching the recent PR #60). Let me verify two more things the study leans on heavily: the human-GT partial-label eval result (the F1?0.73 / "loses at n=2" claim) and whether the project has stated anything about its actual data corpus scale (128 rallies, 26GB).

12. user (2026-06-08T13:21:51.864Z)

docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\CODE_MAP.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
docs\ROADMAP.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\decisions\DECISIONS.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\COMMERCIALIZATION.md
docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\README.md

13. user (2026-06-08T13:21:52.933Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-505 of 825 total (34710 tokens, cap 25000). Call Read with offset=506 limit=505 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

---
name: current-state
description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not only on request). What just landed, what's next, open decisions."
metadata:

  node_type: memory
  type: project
  originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301

Current state ? updated continuously (see [[working-agreement-checkpoints]])

Checkpoint: 2026-06-08 (later+2) · ? **Multi-video eval corpus + owned-model research underway.** User fully tagged Adarsh-and-Avi (40?**96 rallies**) + provided **Kushagra** (32 rallies, 4K) for the TRAINING mix, and **gBaaddy** (new, 1080p, untagged) + TestLargeVideo as held-out TEST videos. Generated full-video WASB trajectories (single 8GB GPU, serialized via wait-loops): `~/clips/{adarsh_avi_full,kushagra,gbaaddy}_traj.csv` (+ %TEMP% copies). Scripts: `~/clips/run_wasb_{avi_full,kushagra,gbaaddy}.sh`.
- **Adarsh-full eval (96 rallies, IoU?0.5, window [0,1405.8s]):** baseline F1 0.632, **tuned+gate2 0.727** (P0.80/R0.67, best out-of-box), sweep **ceiling 0.750** (vel0.02/merge1.0/min_dur2.0/gate off, fit-on-self). vs the opening-10-min (40 rallies): best 0.667 / ceiling 0.742. **Best-config REGION is identical across the slice and full match** (merge_gap 1.0, min_dur 2.0; velocity irrelevant; gate-off=recall, gate-on=precision) ? early sign tuning may transfer; LOVO is the real test. Per-rally: `output/adarsh_avi_full_vs_human.csv`.
- ? **Kushagra trajectory inferring on GPU** (~15 min) ? then **first cross-video LOVO** (`calibrate_local`, manifest of Adarsh-full + Kushagra, human CSVs as labels) + per-video sweeps. gBaaddy queued behind it (race-free, gated on kushagra_traj.csv). ? LOVO honesty: 1 entry per DISTINCT MATCH (group Adarsh-main + Adarsh-last-game as one unit; calibrate_local has no grouping ? keep matches as separate entries only).
- ? **OWNED-MODEL TRAINING RESEARCH launched** (background workflow) ? Vertex AI vs alternatives + model architectures for rally temporal localization + Option-B end-to-end multimodal + label reuse + cost/licensing, aligned to guardrails (Apache-2.0 base, no paid-LLM training data, privacy/local). Reuse all labeled videos for training.
- ?? **NEXT:** Kushagra lands ? LOVO + sweeps + report; research lands ? roadmap-aligned study doc; then **doc checkpoint** (update quality-iterations iteration doc with full-match + multi-video + research ? PR). After user tags gBaaddy/TestLarge ? held-out test eval.

Checkpoint: 2026-06-08 (later+1) · ? **3 work items shipped (user said "go ahead with all three").**
1. **Tuning sweep** ? added `--sweep` to `backend/eval/eval_partial_labels.py` (+test); indexer branch `feat/eval-human-labels` (commits 2c950f6 harness, 58d10cd sweep; NOT pushed/PR'd ? offer to PR). On Adarsh-and-Avi: **F1 0.667?0.742** (velocity 0.03 / merge_gap 1.0 / **min_dur 2.0** / gate OFF; P0.67/R0.83). Dominant lever = min_window_duration; gate over-prunes this camera; velocity ~irrelevant. ? fit-on-self (one video, optimistic) ? real number = LOVO.
2. **Collector P0 normalizer + positioning doc ? PR #13** (https://github.com/avidullu/sports-data-collector/pull/13, branch `feat/manual-ingestion-normalizer`, pushed). `src/cli/normalize_annotations.py` (+test) = standalone pre-import guard: drops blank/?start/**negative-start** rows (else pydantic validator/safe_float_required abort the WHOLE import) + renumbers fractional/dup rally_number 1..N by start (else int(float()) PK-collides + DELETE-INSERT destroys one). Also mm:ss?sec, pipe ending_reason?primary. `docs/MANUAL_INGESTION_POSITIONING.md` = positioning + schema map + P1/P2 roadmap. **Validated full chain on the REAL file vs a TEMP db**:

normalize 40?40 ? import-local
40 ? export 40-interval indexer CSV. Collector suite **124 green** (was 116).
3. **Adversarial review** (1 agent) caught a REAL hole pre-merge: negative start_time was undefended (pydantic
`start_time must be non-negative` ? import abort) ? fixed (drop branch + test, commit 1fe1c24) + corrected 2 doc
inaccuracies (abort mechanism = pydantic validator not safe_float_required; ending_reason is free-text not enum).
4. **Iteration doc archived + indexer PR'd:** `docs/archives/quality-iterations/2026-06-human-gt-tuning/` (STATUS
COMPLETED; results table, sweep, failure modes, bugs, infra gotchas, reproduce) + updated the quality-iterations
index README + 2026-06-initial seed forward-pointer. **Indexer PR #60 OPEN**
(https://github.com/avidullu/badminton-highlight-indexer/pull/60; branch feat/eval-human-labels =
harness 2c950f6 + sweep 58d10cd + docs; 5 files, +446/-1, full suite 251 green).
- **GOTCHA learned:** import-local does NOT copy the video (records local_path); writes obs report NEXT TO THE VIDEO
(clean those up after validation). DB path from `config/config.yaml` `database_path` ? override via temp config for
side-effect-free validation (no `--config`/`--db` CLI flag exists).
- ? **BOTH PRs MERGED ? both repos CLEAN on master, no open PRs, branches deleted.** Indexer **#60** (squash
`90fb8b3`: eval_partial_labels harness + sweep + iteration doc). Collector **#13** (squash `099cf67`: normalize_annotations
guard + MANUAL_INGESTION_POSITIONING.md). NOTE: user said "review is merged" while live API showed both OPEN ? I
completed both squash merges + synced.
- ?? **NEXT:** after user's 60-min tagging ? full-video re-run (`backend/eval/eval_partial_labels.py`, window auto =
last labeled end) + (recommended) tag BREADTH across 3?4 videos for LOVO. P1 collector follow-ups (labeled-window in
manifest, maturity ladder needs_review?confirmed?gold) when prioritized.

Checkpoint: 2026-06-08 (later) · ? **FIRST human-GT eval landed + collector manual-ingestion positioning done.**
User did ~10 min of VLC rally-annotator tagging ? partial human CSV `Annotation Setup/Collect/Adarsh and Avi.rallies.csv`
(40 rallies, covers 0?605s of a 23m29s video; schema `rally_number,start_time,end_time,ending_reason,sport`).
- **Built `backend/eval/eval_partial_labels.py` (+test, 3 green) ? committed on branch `feat/eval-human-labels` (commit
2c950f6, NOT pushed/PR'd).** Scores WASB rally segments vs a PARTIAL human CSV, **restricted to the labeled window
[0, last_end]** so correct detections in the un-labeled tail aren't counted as FP (THE key methodology for partial
labels). Reuses calibrate_wasb.make_predict_fn/load_golden_gts + rally_gate + metrics + export_wasb_segments.build_comparison.
- **Generated WASB trajectory for the labeled window** (first 610s ? 36,564 frames ? `~/clips/adarsh_avi_traj.csv` +
Windows `%TEMP%\adarsh_avi_traj.csv`; WSL script `~/clips/run_wasb_avi.sh`; ~18 min inference). ? GOTCHA: passing a
long multi-statement WSL command via the Bash tool with `run_in_background` MANGLES `$VAR`/`$()` (empty expansion) ?
put the job in a .sh FILE (tr-strip CRLF into WSL) and background `bash file.sh`.
- **FIRST trustworthy human-GT numbers (IoU?0.5, 40 rallies in-window):** baseline F1 **0.650** (P/R 0.65/0.65);
tuned 0.623 (P0.50/R0.825, over-segments); **tuned+gate2 best F1 0.667** (P0.79/R0.575). Boundary err ~1s.
Per-rally CSV `output/adarsh_avi_vs_human.csv`. **Failure modes:** detector catches LONG rallies well (IoU 0.7?0.95,
tight boundaries) but MISSES short 3?5s rallies (most of the 17 misses) and FRAGMENTS long ones (golden#1's 34.8s
split into 2 sub-fires; most "FPs" are real-rally fragments + one possibly-untagged early rally @23.5s, NOT

hallucinations ? precision is recoverable via merge_gap). gate2 alone hurt recall hard here (0.65?0.475) ? net-axis est may be off for this camera. Levers: min_window_duration? (short rallies), merge_gap? (fragmentation).
- **Collector manual-ingestion positioning (ultracode 10-agent workflow `wf_d251ad6c-472`, verdict confidence HIGH):** position sports-data-collector as the **single canonical system-of-record**; indexer consumes ONLY intervals-only export (manifest.json + per-video start,end CSV). **Architecture ~80% there ? the lean .rallies.csv imports via `import-local` TODAY with ZERO code changes** (header matches DictReader, curate.py:232-298). **2 P0 landmines** ? thin pre-import normalizer: (a) blank/?start end_time RAISES and aborts the WHOLE import (parsing.py:22-28) ? drop bad rows; (b) fractional rally_number (7.5/16.5) truncates via int(float()) and PK-COLLIDES, DELETE-then-INSERT destroys one (db.py:217) ? renumber 1..N by start_time, keep orig in notes. DON'T widen PK to float (indexer ignores rally_number). End-state P1/P2: persist `labeled_window_start/end` in manifest (so indexer auto-restricts instead of inferring max-end ? WRONG when a video is tagged past its last rally); maturity ladder needs_review?confirmed?gold (? CurationStatus); provenance/annotation_source+timestamp; unify adapters + run Issue detection on manual imports; freeze intervals-only export contract. Rich `rally_review.csv` (mm:ss, `rally` col, pipe reasons) NOT directly ingestible ? offline converter (P2, only if it's source of truth ? OPEN question). Enrichment rides in MANIFEST, never the eval CSV.
- ?? **NEXT:** after the 60-min tagging ? re-run full-video (extract full frames + WASB + eval, window auto = max end of full labels). For LOVO (learned>baseline, n=2 blocker), recommend tagging BREADTH (3?4 videos partially) over finishing one. Offer: P0 collector normalizer as a small PR; `docs/MANUAL_INGESTION_POSITIONING.md`.

Checkpoint: 2026-06-08 · ? **rally-annotator v1.3 ? PR #1 MERGED to `main`** (merge `d4c682a`, commit `b058262`, 5 files +711/?127; local `main` fast-forwarded). Goal: in-plugin Help text + a documented ending-reason taxonomy + UX fixes the user asked for. Files changed in `C:\Users\avidu\Projects\rally-annotator`: `vlc/rally_annotator.lua` (full v1.2?v1.3 rewrite), `README.md`, `docs/CSV_FORMAT.md`, `CHANGELOG.md`, + NEW `docs/ENDING_REASONS.md`. **Installed to `%APPDATA%\vlc\lua\extensions\rally_annotator.lua` for the user's live smoke test** (Reload extensions / restart VLC).
- **VLC "About" finding (verified, high-confidence):** the empty "About ?lua" box the user screenshotted is VLC's Add-ons `AddonInfoDialog`; its "Lua script" body is a hardcoded constant from the addon scanner (`modules/misc/addons/fsstorage.c`) ? an extension CANNOT set it. descriptor `shortdesc`/`description` DO show, but only in the *Active Extensions* tab "More information" dialog. So real, author-controlled help = IN-DIALOG (Help button + `add_html`). Enriched the descriptor too.
- **v1.3 plugin changes:** ending reason now REQUIRED + non-sticky (placeholder `-- choose reason --`; reset after save via dropdown `clear()`+rebuild ? VLC 3.0.x dropdowns have NO `set_value`, first value after clear auto-selects); two-step commit (Mark END *arms* ? Save Rally writes; button shows the # it will write); editable Start/End text fields; Recent-rallies `add_list` with Edit/Delete (uses `get_selection()`); Undo last (cancel edit / clear mark / pop last). Switched to an in-memory rows model + atomic CSV rewrite (tmp+rename, Windows `.bak` rollback). CSV schema UNCHANGED (`rally_number,start_time,end_time,ending_reason,sport`); forgiving parser preserves extra columns.

115   - **Taxonomy (verified via a 3-agent workflow + adversarial check):** OUT-of-bounds = the HITTER's error, forced ONLY
116   if pressured else unforced (NEVER auto-forced, never the opponent's winner); into-net same; serve-into-net =
117   service_fault; rally net-cord-in = live (often a winner); serve net-cord diverges by sport; default-to-unforced
118   tie-breaker. One fix the verifier caught and I applied: a badminton serve CAUGHT on the net = service_fault (BWF Law
119   13.2), NOT a let. Full guide in `docs/ENDING_REASONS.md`; summary table in README.
120   - **Verification:** no Lua interpreter on PATH; adversarial static review (subagent) ? state machine + VLC 3.0.x API
121   usage clean; fixed the 2 cross-platform bugs it found (POSIX `file://` path lost leading `/`; `save_all` deleted the
122   live CSV before rename). ? STILL NEEDS the user's live VLC click-test (I cannot drive VLC).
123   - **Follow-up (same session, also installed):** (a) RESUME ? on activate the plugin reloads the video's existing
124   `.rallies.csv` (Recent list + continued numbering); `Refresh` now re-points to whatever video is playing now and
125   loads its rallies (guarded against reload-during-edit so `edit_index` can't go stale; playback *position* is NOT
126   restored). (b) UNDO clarity ? `Undo last` button shows the row it will remove (`Undo last (#N)`) + a "Last row
127   (what Undo removes)" line in the status panel. README/CHANGELOG updated.
128   - **Reinstall gotcha (FIXED):** after merging PR #1 the user hit "plugin not reloading" ? the installed copy was
129   STALE (28251B, an earlier file-copy) vs merged source (28905B). Deleted + reinstalled the merged file to
130   `%APPDATA%\vlc\lua\extensions\rally_annotator.lua`, hash-verified byte-for-byte (28905B). **VLC 3.x "Reload
131   extensions" is unreliable for Lua code changes ? a full VLC restart is the reliable path** (VLC was not running).
132   Lesson: always reinstall to the VLC dir AFTER the final edit, and verify by SHA-256, not just byte count.
133   - ? **REAL root cause found (plugin never showed under View) ? fix PR #2 OPEN, now grown to v1.4**
134   (github.com/avidullu/rally-annotator/pull/2, branch `fix/extension-scan-ipairs`, commits `8d42a71` load-fix +
135   `424eb3c` v1.4). The
136   stale file was a red herring; v1.3 NEVER loaded. **VLC scans an extension's `descriptor()` in a RESTRICTED Lua
137   sandbox that does NOT expose base globals (`ipairs`, etc.).** v1.3 built `REASON_ID`/`SPORT_ID` with **top-level
138   `ipairs` loops** ? scan-time error `attempt to call global 'ipairs' (a nil value)` ? VLC silently skipped it. Fix:
139   move the table-building into `activate()` (full env); **NEVER call a global function at the top level of a VLC Lua
140   extension.** (v1.2 worked only because its `ipairs` calls were all INSIDE functions = activate-time.)
141   - ? **Reusable diagnosis method:** installed `lua5.1` in WSL ? `luac5.1 -p <file>` (syntax) + load chunk in an EMPTY
142   env via `setfenv` (simulates the scan sandbox); and launched VLC with `-vv --file-logging --logfile` + `--no-one-instance`
143   to read the actual `lua debug: Scanning Lua script ?` / `Error loading ?` lines. After fix the log shows the scan with
144   `capability flags: 0x0` and NO error. WSL has no lua by default; `sudo apt install -y lua5.1` adds `lua5.1`+`luac5.1`.
145   Also built a reusable **headless `lua5.1` harness** (`%TEMP%\ra_harness.lua`) that STUBS the vlc dialog API + loads
146   the extension ? drives `activate()`/`show_help()`/`save_rally()` and asserts state ? catches dialog-logic bugs
147   without VLC.
148   - ? **v1.4 (commit `424eb3c`, in PR #2):** (a) **Help toggle fixed** ? was "stuck"; VLC didn't repaint the in-place
149   `set_text` on the shared status widget ? now a DEDICATED `add_html` panel removed via

`del_widget` (same trick). (b) **Reason-stickiness fixed** ? selecting `winner` blocked the next rally also being `winner`; VLC didn't repaint the `clear()`-based dropdown reset ? now the reason RESETS to a savable **`unknown`** default by RECREATING the dropdown (del_widget+add_dropdown at the same cell; VLC has no `set_value`). Reason no longer hard-required. (c) **"Next rally #" field** ? resume/insert numbering anywhere; auto-advances to next FREE number; Undo/Delete re-sync it (no gaps). `unknown` added to README/ENDING_REASONS/CSV_FORMAT. **Adversarial-review workflow (3 lenses) caught 2 real gap bugs (undo/delete didn't re-sync Next#) ? fixed before commit.** ? **VLC dialog lesson:** in-place widget updates (`set_text`, dropdown `clear()`+re-add) DON'T reliably repaint ? use `del_widget`+re-add.
- ?? **NEXT:** merge PR #2 (load fix + v1.4), then user fully restarts VLC ? extension under View ? smoke-test (mark/save/edit/delete/undo/resume/help/reason/next# × 5 sports); iterate on feedback.

Checkpoint: 2026-06-07 (FINAL, session wrap) · ? **ALL MERGED ? observability feature fully shipped; both repos CLEAN.** Zero open PRs across all 3 repos (indexer/collector/obsreport); both repos on `master`, 0 uncommitted, no stray branches. Merged this session: indexer **#41/#42/#43/#44**, collector **#10/#11/#12**, obsreport on GitHub (private, v0.1.3).
- **#44 (last one)** = docs+tooling handoff: **`docs/CODE_MAP.md` REGENERATED** for the #40 layered layout + config/ reporting subsystems (build-code-map workflow, 0 broken anchors, 325 lines) · **`run_all_tests.py`** at repo root (one-command full suite; README points to it) · Grok coordination notes in `docs/BACKLOG.md` + `LOW_REGRET_MOVES_PLAN.md`.
- ?? **TWO config-migration bugs flagged for Grok** (in BACKLOG + CODE_MAP §2.0 gotchas, found by the regen, NOT fixed ? Grok's config domain): (1) `/api/compile` ignores `--output-dir` (`OutputConfig` has no `output_dir` field ? always `"output"`); (2) `--output-dir` lives only on dynamic `global_config._runtime_overrides`. Fix: add `output_dir` to a config/runtime model + read consistently.
- **Suites:** indexer 231, collector 116, obsreport 74 ? all green. Run via `python run_all_tests.py`.
- ?? **NOTHING PENDING for the observability work.** Optional future (not started): HTML customer renderer; real AI-provider timing (via Grok item #7); pin `obsreport` git+ssh tag once SSH/CI access exists; extend report with audio-readiness (Grok item #8). Original rally-detection modeling still PARKED pending golden labels.

Checkpoint: 2026-06-07 (latest+2) · ? **FEATURE COMPLETE ? observability in both tools, merged; Phase D done.** MERGED to master: obsreport (GitHub private, v0.1.3) · indexer **#42** (config fix) + **#41** (observability) · collector **#10** (parser fix) + **#11** (observability). **Debuggability eval done** (15-agent Workflow: all 7 broken-run scenarios self-debuggable; fresh readers given only README/FAQ landed on the root cause + cited the FAQ). Refinements applied as follow-up PRs (OPEN, green, awaiting owner merge): indexer **#43** (README Q7 lists wasb_hybrid/tracknet_hybrid; new fps Q14 + repoint fps_fallback off Q5; numeric fps fallback; motion synthetic at data level; silent_provider_noop segmenter?handoff + .env path; blur remediation; **faq-resolution golden test** ? 231 green) and collector **#12** (frame-level evidence on delivery flags; review-only

framing for possible_missed_rally/shot_density; **faq-resolution golden test** ? 116 green). Eyeballed reports via rendered HTML screenshot + customer summaries (clean, audience-appropriate). Eval artifacts: `%TEMP%\obseval\`.
?? **NEXT (when owner is back):** merge #43 + #12 (both green, disjoint). Then the per-video observability feature is fully shipped across both tools + the shared obsreport lib. Optional future: HTML customer report renderer (obsreport); thread real AI-provider timing through the segmenters (currently 'not_captured'); pin obsreport via git+ssh once SSH/CI access is set up.

Checkpoint: 2026-06-07 (latest+1) · ? **INDEXER OBSERVABILITY MERGED.** Both indexer PRs merged to master: **#42** (config fix `AppConfig.get` nested?dict; review nits addressed: dynamic section scan + shim comment + tests) and **#41** (per-video observability reports) ? master `98bb1d9`, **229 tests green**, feat branch deleted. Full pipeline now sound end-to-end (validators run via #42; every run emits report via #41). obsreport on GitHub (private, tags v0.1.0?v0.1.3). **Collector observability PR #11 STILL OPEN** (rebased on #10, 113 green).
?? **NEXT: Phase D refinements (follow-up PRs) + golden tests.** Eval verdict: all 7 broken-run scenarios self-debuggable. Apply: (HIGH) repoint `fps_fallback_used` faq_ref off README#Q5 (keyframes) to a NEW fps FAQ; add `wasb_hybrid`/`tracknet_hybrid` to README Q7 segmenter list. (MINOR) numeric fps fallback `30 (fallback)`; motion "synthetic" surfaced at data level; self-remediating blur line (name `validation.min_blur_threshold`); collector overlap frame-level detail; `possible_missed_rally` review-only framing. Add golden tests: faq-ref-resolution (every flag's cited anchor exists in its doc) + expected flag-set per fixture, in BOTH repos. Indexer refinements ? new branch off master; collector ? push to #11. Eval artifacts in `%TEMP%\obseval\` (manifest.json + reports/ + gen_*.py).

Checkpoint: 2026-06-07 (latest) · ? **RE-SYNCED onto parallel work + found/fixed a config runtime bug.** Indexer master advanced: **#39 (typed config backend/config/)** + **#40 (layered directory restructure: backend/pipeline/{validators,segmenters,stitcher,detectors,audio}, backend/infrastructure/database.py, backend/api/static/)**. Rebased **observability PR #41 onto #40** (git rename-detection merged my main.py reporting onto the new imports; only repointed 2 test imports) ? **228 tests green**; #40 also fixed the 2 config tests I'd flagged. Collector observability **PR #11 rebased onto #10** (parser fix merged) ? **113 green**. obsreport **74 green**.
- ? **FOUND + FIXED a real config runtime bug ? indexer PR #42 (`fix/appconfig-nested-get`).** The typed AppConfig is passed to validators/segmenters which use legacy chained `config.get("validation",{}).get("k")`; AppConfig.get returned the nested *model* (no `.get`) ? **all 4 validators threw `'ValidationConfig' object has no attribute 'get'` on every real `python -m backend.main` run** (tests passed because they inject dicts). Fix: AppConfig.get returns nested sections via `model_dump` (deep dict-compat); +regression test; **215 green**; CLI smoke now runs all validators. **Surfaced BY the observability report** (nice dogfood). Owner approved fixing it (its own small PR #42).
- **Open PRs now:** indexer **#41** (observability, green), indexer **#42** (config fix, green), collector **#11** (observability, green). obsreport on GitHub (private, tags v0.1.0?v0.1.3). Suggested merge order: **#42 ? #41**

(so #41's runtime path is unbroken), and **#11** any time.
- ?? **STILL PENDING: Phase D** ? debuggability eval Workflow was running (fresh-reader agents over 7 broken-run
reports); apply synthesized wording refinements + render a report to an image to eyeball + add golden faq-resolution
tests, then push to #41/#11.

Checkpoint: 2026-06-07 (later) · ? **NEW FEATURE IN PROGRESS ? per-video OBSERVABILITY REPORTS** across
BOTH tools (indexer + collector) + a NEW shared lib. Approved plan:
`C:\Users\avidu\.claude\plans\drifting-snuggling-squirrel.md` (READ IT to resume). Decisions: always-on,
reports written NEXT TO THE VIDEO by default (override via `report.output_dir`); BOTH tools emit a technical
report (debuggable by a README+FAQ reader) AND a customer-friendly summary (strip internals); shared schema via
a **new private repo `sports-obsreport`** (pkg `obsreport`) ? a reusable "report config language."
Research: Soda/SodaCL = ELv2 (disqualified, not OSS); GE/Evidently/ydata/Frictionless license-safe but
wrong shape ? built a small lib on permissive primitives (pydantic/Jinja2/PyYAML).
- ? **PHASE C DONE ? `sports-obsreport` lib built + LOCAL git tag `v0.1.0`** at `C:\Users\avidu\Projects\sports-obsreport\`
(NOT on GitHub yet ? local only; editable-installed into Python 3.13 via `pip install -e . --no-build-isolation`).
71 tests green. Modules: `envelope.py` (typed ReportEnvelope = single source of truth), `spec.py` (YAML config
language: sections + rules), `rules.py` (sub-Turing clause engine, no eval), `render/{json_render,markdown}.py`
(technical + customer views from same envelope), `io.py` (next-to-video paths, atomic, NEVER raises),
`recorder.py` (RunRecorder: in-flight stage capture, finalize?rules?verdict, write), `diff.py` (stable_json for
cross-run diffs), `schema/report_envelope.schema.json` (contract + drift test). specs/example.yaml ships
silent_provider_noop + fps_fallback_used. **Footgun caught:** correct build-backend is `setuptools.build_meta`
(underscore) ? collector's pyproject has the hyphen typo `setuptools.build-meta` = latent `pip install -e` break (flag/fix).
- ? **obsreport bumped to v0.1.1** (local tag) ? added optional `customer_message` on flags/rules: customer report
shows ONLY curated flags (no technical leakage); technical always uses `message`. Schema still 1.0 (additive).
- ? **PHASE A DONE (indexer)** on branch `feat/observability-reports` (NOT pushed; commit 9dbc764). 222 tests green
(209 baseline + 13). Added `run_all_with_context` (backward-compat; run_all unchanged); `backend/reporting/`
(soft-import wrapper ? no-op if obsreport absent + never-raises; pure `harvest.py`; `spec.yaml` footgun rules);
wired `main.py` CLI + API in finally blocks; README FAQ Q10-Q13; requirements.txt commented git+ssh dep. Verified
end-to-end via CLI smoke (report files appear next to video; customer summary clean; technical findings cite FAQ).
- ? **PHASE B DONE (collector)** on branch `feat/observability-reports` (NOT pushed). 113 tests green. `src/reporting/`
(soft-import wrapper ? never raises; `adapter.py` Issue?Flag + delivery/import recorders reusing rally_cvat;
`spec.yaml` sections + customer_verdict + item_noun=rally + license_noncommercial/missing_fps rules). Wired
curate.py validate-delivery / convert-cvat / import-local; import report falls back to annotations dir when video
absent (not cwd). docs/INGESTION_PIPELINE.md FAQ added. Fixed build-backend typo. .gitignore ignores
*.report.json/.report.md/.summary.md. Verified via validate-delivery CLI smoke.
- ?? **obsreport bumped to v0.1.3** during B (local tags v0.1.0/.1/.2/.3): added

customer_message (curated customer
flags), customer_verdict (per-tool verdict phrasing), item_noun (rally/point/highlight). 74 lib tests green.
- ? **DUP RISK RESOLVED (2026-06-07):** the collector `load_canonical_csv` falsy-zero bug (drops a rally starting
at 0.0s) landed **exactly ONCE** via the dedicated branch `fix/canonical-csv-zero-start` ? **PR #10 MERGED**
(merge commit `99dfa22`; fix `f7cab11` + regression test `test_load_canonical_csv_keeps_rally_starting_at_zero`).
master shows the parser fix once ? **no double-merge**. The observability branch's duplicate `fix(parser)` commit
was dropped, so collector **PR #11 (observability) carries NO parser fix** and merges cleanly (doesn't touch
rally_cvat.py). Local fix branch deleted; remote auto-deleted on merge. (Originated from spawned chip task_7a3d8081 +
this session's branch/PR ? same branch name, one PR.)
- ? **SYNCED TO REMOTE (2026-06-07):** obsreport ? github.com/avidullu/sports-obsreport (PRIVATE, main + tags
v0.1.0?v0.1.3). Indexer ? **PR #41** (rebased onto #39 typed-config `backend/config/`; main.py conflict resolved
= typed attr access for segmenter + my reporting; reporting normalizes AppConfig?dict). Collector ? **PR #11**
(observability-ONLY, still OPEN; the dup parser-fix commit was DROPPED ? the dedicated `fix/canonical-csv-zero-start`
branch owned it as **PR #10, now MERGED to master** `99dfa22`).
- ?? **Indexer PR #41 has 2 PRE-EXISTING red tests from #39** (NOT mine, left untouched to avoid clashing with Grok's
config work): test_config.py::test_load_default_config (config.json default_segmenter=gemini vs model default
yolo_hybrid) + test_save_default_config_roundtrip (Path==str on Windows). My reporting: 226 pass. Collector: 112 pass.
- ?? **NEXT: finish PHASE D debuggability eval** (ultracode Workflow: fresh-reader agents over broken-run reports ?
refine wording; render a report to an image + eyeball) + committed golden tests (faq_ref-resolution + flag-set) in both repos.
- **TODO (HELD for owner OK):** create private GitHub repo `sports-obsreport` + push (tags v0.1.0/v0.1.1); then
pin obsreport via git+ssh in both repos' deps; then push indexer/collector branches + open PRs.
- Phasing tasks tracked in the harness task list (C1-C4 + A done; B in progress; D pending).

--- (prior checkpoint) ---
Checkpoint: 2026-06-07 · ? **SESSION SUNSET.** master clean (PRs #25?#35 merged; 209 tests green; only
feat/khelacharya-shot-intelligence retained; no open PRs). Audio E1 done ? classical onset = ~2.2× ceiling, NOT
adopted (detector kept, PR #35); quantitative findings + lessons in [[audio-e1-findings]] (+ plan §10). VLC
annotator = standalone repo github.com/avidullu/rally-annotator (MIT, multi-sport).
?? RESUME NEXT SESSION on EITHER: (1) **GOLDEN EVALS** ? owner self-rates clips in the VLC annotator ?
`import-local`/`--labels` ? unblocks distilled-model LOVO (n?3) + human-gold E1 re-confirm = the highest-leverage
unblock; OR (2) **AUDIO E2** ? learned timbre classifier (needs labeled hits). Also queued: pose/stance #2, owner's
"other hunch", Option B (owned end-to-end model). Read [[audio-e1-findings]], [[code-map-artifact]] (docs/CODE_MAP.md),
docs/RALLY_DETECTION_RESEARCH.md, ADR-007 to ramp.
? **PR #36 (docs/AUDIO_E1_FINDINGS.md) MERGED to master 2026-06-07** (owner restored the branch after an accidental
close; reopened + squash-merged). Record now in-repo + [[audio-e1-findings]] (memory) + plan §10. master fully clean, no open PRs.

--- (prior checkpoint) CLEAN SLATE. Indexer PRs #25?#32 MERGED; master 202 tests green,

tree clean, branches pruned
(kept feat/khelacharya-shot-intelligence). Audio decided (ADR-007). VLC annotator ? standalone public repo
github.com/avidullu/rally-annotator (MIT, multi-sport). **ALL PRs #25?#34 MERGED; slate FULLY clean** (master only + retained feat/khelacharya-shot-intelligence; no open
PRs; tree clean). #33 removed the stale annotator copy; #34 expanded ADR-007 (fusion + Option B) + repointed the
ref ? both merged safely. **PARKED: modeling waits on golden labels (user self-rates via annotator); next code
lever = AUDIO (E1 go/no-go probe).**
ARCHITECTURE DECISION captured (ADR-007 §Integration architecture, PR #34): audio fuses in OUR rally layer
(WASB stays frozen black box ? feature-level into distill_local + decision-level recall-recovery); **Option B = the
intended end-state** ? once a healthy golden corpus exists, train an OWNED end-to-end multimodal model (frames+audio+
pose+??rallies) fine-tuned on our data, dropping WASB-weight dependence + recall ceiling. Modular cues now (audio?
stance?other) are how we earn Option B (each generates the labels it needs). See resumption runbook ?.

DISTILLED MODEL BUILT (2026-06-06) ? user wants this approach; honest result = needs more data
- **ALL PRs #27-#31 MERGED to master** (0c6f2c1). master now has: code-map, gate-A/rally_gate, gemini-refine
  (+provider hardening + extract_video_chunk scale_width), calibrate_local, distill_local. This session's 5
  branches PRUNED (local+remote). Retained: feat/khelacharya-shot-intelligence (PR #11). Pre-existing merged
  stragglers still on origin (left, optional prune): docs/ending-reason-forced-unforced(#19), feat/rally-slicer(#20),
  feat/wasb-substrate(#21).
- **User will RATE the clips ? cached artifacts LEFT IN PLACE** (do NOT delete): Temp\{adarsh,testlarge}_frames +
  trajectories (~/clips + Temp), Temp\verylarge_frames (partial 12246/46080). Adarsh+TestLargeVideo trajectories
  ready ? when user import-locals golden labels, calibrate_local/distill_local use them with zero re-extraction.
  (#30 stranded calibrate_local on gate branch ? RECOVERED in PR #31.)
- `backend/eval/distill_local.py` (PR #31, OPEN): learned model ? WASB recall-oriented candidates (0.005,1.0,0.5)
  ? features [duration, peak_velocity, mean_velocity, active_ratio, net_crossings] (resolution/fps-INVARIANT)
  ? classifier.LogisticRegression trained on Gemini labels (training.label_candidates IoU). Honest LOVO + saves
  output/local_rally_model.json. Reuses classifier/training/rally_gate/metrics/gemini_refine.
- **HONEST RESULT (2 videos): LEARNED UNDERPERFORMS baseline** ? mean held-out F1 0.359 (learned) vs 0.438
  (threshold baseline). Causes: (1) n=2 too few ? no transferable model; (2) WASB proposal recall CEILING 0.43
  on TestLargeVideo (its windows don't align with Gemini rallies; Adarsh ceiling 0.81). Framework READY; needs
  more teacher videos + better WASB recall to beat baseline. Final weights: mean_velocity + duration dominate,
  net_crossings barely used (unreliable at n=2).
- ? **VeryLargeTestVideo (4K, 46080 frames, 12m49s, 11.5GB) = held-out FINAL test for USER MANUAL EVAL.**
  Extraction running (task bzt8hc4ks, frames?1920). Then WASB ? baseline + learned rally lists ? user manually
  evals (manual = GT; no Gemini needed on it). User's manual eval becomes a 3rd-video human GT ? strengthens model.
- PR #31 (feat/distill-local) OPEN ? merge to get distill_local + calibrate_local onto master.

? AUDIO SCAFFOLDING + PRELIMINARY E1 DONE (2026-06-07) ? PR #35 (feat/audio-hit-detector-e1, OPEN). Verdict = AMBER.
- Built `backend/audio/hit_detector.py` (ffmpeg?numpy spectral-flux onset?adaptive peak; NO scipy; 5 tests) +
  `backend/eval/audio_e1_probe.py` (E1 diagnostic). 207 tests green. WASB untouched (Option A).
- Ran preliminary E1 on EXISTING GoPro audio vs **Gemini SILVER** labels (no wait for human labels): Adarsh (36
  rallies) + TestLargeVideo (7).
- **RESULT (honest, AMBER):** ? audio fires in **100% of rallies** and **100% of WASB-missed rallies have
  hit-clusters** ? real RECALL potential (the bottleneck). ?? but **discrimination weak**: 1.3x @k=2.5, peaks
  ~2.8x @k=4, then a CLIFF (k=6 ? ~0 hits); hit rate high (56/min @k=2.5 = many non-shuttle transients:
  footwork/voices/ambient); **audio-only F1 0.13-0.27 < WASB-only (0.33-0.54).** So audio is NOT clean enough
  to stand alone ? it's a **fusion/recall cue**, not a standalone segmenter.
- **OPTION A DONE (band-pass) ? NEGATIVE RESULT (2026-06-07, committed to PR #35):** added a `band` knob to the
  onset detector + swept low/mid/high bands × k at 32kHz on both videos. **Band-pass does NOT help** ? HF
  "shuttle-click" bands gave LOWER discrimination, not higher. **Ceiling stays ~2.2x** (in-rally vs MID-MATCH
  dead-time). **Not a warm-up/label artifact** (mid-match dead alone ~0.6 hits/s vs ~1.3 in-rally on both).
  Root cause: between-point activity (shuttle tosses=our false-fire, footwork, talk) = transients a classical
  onset detector can't separate from rally hits. `k` only trades coverage for discrimination. Kept the band knob
  (cheap/optional) but it's NOT the fix. Full writeup: AUDIO_RALLY_DETECTION_PLAN.md §10.
- **HONEST VERDICT: classical-onset audio is too weak to be a clean cue (~2.2x); band-pass won't rescue it.**
  Remaining audio path = **E2 learned TIMBRE classifier** (MFCC ? shuttle-thwack vs toss/footstep/voice; needs
  labeled hits ? real work, the only plausible way past 2.2x). Cheaper: re-confirm on human-gold (unlikely to
  move much per the warm-up check). **Recommend re-prioritizing:** owner's "other hunch" / pose-stance #2 /
  more golden data for WASB+distilled, rather than over-investing in classical audio. **User to decide when back.**
- **PR #35 review addressed (2026-06-07):** owner left 2 refactor comments (PR otherwise LGTM). (1) made
  `backend/eval/audio/` package + moved probe ? `backend/eval/audio/e1_probe.py` (folded INTO #35 since it's a
  new unmerged file ? cleaner than a stacked PR; import/doc refs updated). (2) broader tests/ de-fragmentation
  filed as a TODO in `docs/BACKLOG.md` (Code & test organization) ? grouping metric is owner's design call,
  dedicated PR later. 209 tests green; replied on the PR. **#35 still OPEN, ready to merge as the single PR.**

? DECISION DOCUMENTED (2026-06-07): AUDIO next, STANCE parked ? PR #32 (docs/audio-plan-adr007)
- **ADR-007** (docs/DECISIONS.md): adopt fully-local AUDIO "thwack" hit detection FUSED with WASB trajectory as
  the next rally cue (attacks recall bottleneck); park pose/stance as #2 (harder); Gemini = offline-teacher only;
  gate experiments on golden labels. **Sequence: AUDIO now ? [TBD owner hunch] ? POSE/STANCE.**
- **docs/AUDIO_RALLY_DETECTION_PLAN.md** ? detailed plan: extract?onset/peak?confidence?hit-cluster?fuse-with-WASB;
  experiments E1(GO/NO-GO recall probe on real GoPro audio) / E2(LR+MFCC FP suppression) / E3(audio features into

distill_local) / E4(boundary tighten); deps numpy+scipy (permissive); top risk = shuttle SNR (quiet vs tennis).
New module planned: `backend/audio/hit_detector.py`.
- **docs/VIDEO_RECORDING_GUIDELINES.md** ? AUDIO now a first-class capture REQUIREMENT (record w/ audio on, mic
unobstructed). User will add "enable audio always" to camera-setup docs.
- **Research checked in** (was uncommitted): docs/RALLY_DETECTION_RESEARCH.md. **VLC annotator checked in**:
tools/vlc_rally_annotator/ (source+README). All in PR #32.
- Owner has "another hunch" to slot BEFORE stance (TBD).

? RESEARCH DONE (2026-06-06) ? rally-detection methods ? `docs/RALLY_DETECTION_RESEARCH.md` (full cited report)
Verdict: teacher?student is SOUND but NOT uniquely best ? augment with a COMPLEMENTARY LOCAL multi-cue stack.
**#1 cheapest lever = AUDIO hit detection (shuttle "thwack") fused with the trajectory ? directly attacks the
WASB recall bottleneck with inputs we already have.** #2 = pose/court serve-ready START + flight-gradient END
(our "gate B", validated). #3 = compact HitNet GRU (MonoTrack ~16K params). Audio+visual fusion lifts rally
RECALL (tennis HMM: 83% fused vs 54% visual/63% audio). Keep Gemini OFFLINE-teacher only. ? caveats: most lit
numbers are BROADCAST + per-frame (not IoU boundary F1, not amateur single-cam) ? won't transfer cleanly; ?
LICENSE TBD for ShuttleSet/MonoTrack/WASB-weights (commercial). Cheap next experiments E1-E4 in the doc (E1:
audio energy-peaks overlay on WASB windows ? recover misses; E3: add audio features to the distill student).

? VLC RALLY ANNOTATOR ? STANDALONE PUBLIC REPO (2026-06-07): github.com/avidullu/rally-annotator
MOVED OUT of the indexer into its own **MIT, public, multi-sport** repo (default branch `main`) per user ? for
open-source release + iteration across **net-separated racquet sports** (badminton/tennis/table_tennis/
pickleball/padel). Local clone: `C:\Users\avidu\Projects\rally-annotator\` (git identity = Avi Dullu
<avi.dullu@gmail.com>, mirrored from indexer). Layout: `vlc/rally_annotator.lua` (v1.2 ? added Sport dropdown +
`sport` CSV column; generic naming) + README + LICENSE(MIT) + docs/CSV_FORMAT.md + CHANGELOG. CSV now
`rally_number,start_time,end_time,ending_reason,sport`. **v1.2 INSTALLED** to `%APPDATA%\vlc\lua\extensions\
rally_annotator.lua` (replaced v1.1). ? v1.2 multi-sport build NEEDS user's live VLC smoke test (I can't click
VLC). Removed from indexer (PR #32 now docs-only; ADR-007 ref repointed to the new repo).
Roadmap (in the new repo README): live test all 5 sports; per-sport hotkeys; python-vlc/HTML5 front-ends.
--- historical (v1.1, badminton-only, was in indexer tools/) ---
Button-dialog (not pause-hook;
pause callback flaky macOS/broken VLC4): View menu ? "Badminton Rally Annotator" ? Mark START / ending_reason
dropdown (winner/forced_error/unforced_error/service_fault/let/other) / Mark END / Undo + live status. Reads
vlc.var.get(input,"time")/1e6 ? decimal sec. Appends `rally_number,start_time,end_time,ending_reason` to
`<video-stem>.rallies.csv` next to the video (continues numbering across re-enable ? no dup rally_number).
CSV ingests UNCHANGED by backend/eval/calibrate_wasb.py --labels, backend/tools/rally_slicer.py --labels, AND
collector import-local. Verify=high confidence (fixed dup-rally_number, HTML status label, install-doc repo
mixup) but NEEDS user's live VLC test (I can't click VLC). Alternatives if cramped:

python-vlc+Tkinter (global hotkeys) or HTML5 page (shareable w/ remote raters). This is the golden-data generation tool ? feeds the model.

? GOLDEN DATA via SELF-RATING (confirmed 2026-06-06) ? preferred over Gemini silver
User can be a RATER: make a CSV `rally_number,start_time,end_time` (sec; JSON ok; optional shots_count/ending_reason)
? `python -m src.cli.curate import-local --sport badminton --video-path <vid> --annotations-path <csv> --license Proprietary`
(validates rallies via validate_badminton_rallies, registers CURATED) ? `curate export` ? indexer eval format ?
add to calibrate_local/distill_local manifest. HUMAN golden > Gemini silver ? directly fixes the 2 caps (too few
videos + label quality). Collector `import_local` @src/cli/curate.py:52; CSV parse @_parse_local_annotations:144.
Pause decision 2026-06-06: modeling PAUSED until more golden data (user self-rating + next week's vendor videos).
VeryLargeTestVideo 4K extraction STOPPED at 12246/46080 (partial frames Temp\verylarge_frames deletable). Research
workflow wxzp6h7xk still running ? post report then fully paused.

?? NEXT-WEEK RESUMPTION RUNBOOK (more teacher videos ? re-run calibrate_local + distill_local; learned beats baseline once n grows)
Goal: more teacher-labeled videos ? stable leave-one-video-out ? then build the LEARNED distilled model
(classifier.py on Gemini labels) ? the real local-precision lever (threshold-tuning ceiling'd at ~0.44 P/F1, overfits on 2 videos).
FIRST: merge the open PR stack so master has the tools: #27 (code-map), #28 (gate-A/rally_gate), #29
(gemini-refine + provider hardening + extract_video_chunk scale_width), #30 (calibrate_local, stacked on #28 ? retarget to master after #28).
Per NEW video (repeat for each):
1. Probe: `ffprobe ... <video>` ? note fps, width. (TestLargeVideo was 5.3K ? extract frames downscaled to 1920.)
2. Extract frames: Windows `ffmpeg -i <video> [-vf scale=1920:-2] -q:v 2 <Temp>/<name>_frames/%08d.jpg`.
3. WASB trajectory (WSL, checkpointed): re-stage wasb_infer.py to `~/models/WASB-SBDT/src`, then
   `conda activate wasb && cd ~/models/WASB-SBDT/src && python wasb_infer.py --frames_dir /mnt/c/.../<name>_frames
   --weights ../pretrained_weights/wasb_badminton_best.pth.tar --out ~/clips/<name>_traj.csv --batch-size 8 --num-workers 4 --flush-every 500`
   then copy CSV to a Windows path.
4. Gemini teacher labels (paid key in env GEMINI_API_KEY; NEVER store key): `python -m backend.eval.gemini_refine
   --video <video> --full-video --out output/<name>_gemini_fullvideo.csv` (model gemini-flash-latest; clips auto-downscaled <500MB).
5. Add `{name,trajectory,fps,frame_width,labels}` to the calibration manifest (see output/local_calib_manifest.json).
Then re-run calibration + build the learned model:
6. `python -m backend.eval.calibrate_local --manifest <manifest> --metric f1` (and --metric precision) ? honest LOVO across N videos.
7. **NEW (the real lever):** train `backend/eval/classifier.py` (LogisticRegression) on pooled Gemini labels: label each WASB
   candidate rally-vs-not by IoU-match to Gemini labels (training.py::label_candidates), features = candidate stats
   (peak/mean velocity, active_frame_count) + net-crossings (rally_gate). Score held-out (LOVO). Compare to threshold-calib.
Artifacts that persist (don't redo): trajectories `~/clips/{adarsh,testlarge,sample_long}_traj.csv` (+ Windows Temp copies);
Gemini labels output/{testlarge,adarsh}_gemini_fullvideo.csv; manifest output/local_calib_manifest.json.
Deletable (big, regenerable from trajectories): Temp/{adarsh_frames(7.9GB), testlarge_frames, sample_long_frames(4.1GB)}.

OLD checkpoint context below ? (this session's journey)

DEMO RUN ? 2nd video "Adarsh and Vasanth.MP4" (2026-06-06, stress + checkpoint test)
- Video: `C:\Users\avidu\Projects\Annotation Setup\Collect\Adarsh and Vasanth.MP4`, 1920x1080, 59.94fps,
  40,536 frames, 11m16s, 5.5GB. Stress-tested infra (1.9x Sample Long).
- Frames: ffmpeg -> `C:\?\Temp\adarsh_frames` (40,536 JPGs, 7.9GB, deletable). Inference 40,534 windows
  in 1272s @ 31.9 win/s; 13,224/40,536 visible (32.6%). Checkpoint cache kept pace (no crash, so resume
  not exercised, but cache fully built). Trajectory: WSL `~/clips/adarsh_traj.csv` + `C:\?\Temp\adarsh_traj.csv`.
- Tuned export (cfg 0.05/1.0/1.0, no golden labels for this video): **65 rally segments, 251.4s play** ->
  `output/adarsh_tuned_segments.csv`. User eyeballing; baseline NOT yet generated (held until user anchors).
  NOTE tuned cfg was fit on Sample Long (different video) -> transfer is the open question.

BASELINE vs TUNED on Adarsh (2026-06-06) ? tuned config does NOT transfer + over-fire pattern
- Baseline (0.02/2.0/2.0): 45 segs, 303.8s, mean 6.8s. Tuned (0.05/1.0/1.0): 65 segs, 251.4s, mean 3.9s.
- Tuned = +44% segments but -17% play: 30 tuned-only segs (almost all 1-2s = false fires); 10 baseline-only
  segs that tuned LOST are the LONG rallies (17.7/10.9/9.6/9.2/8.9/7.6s) ? merge_gap=1.0 fragments them.
- **Honest verdict: no "% improvement" (no GT for this video); tuned REGRESSES here. Baseline is the better
  operating point on Adarsh.** Concrete proof the Sample-Long-tuned cfg overfits -> need leave_one_video_out
  (2nd labelled video). `output/adarsh_baseline_segments.csv` written.
- **OVER-FIRE PATTERN (user-documented, data-confirmed):** loser tossing/flicking shuttle back for service =
  single short trajectory -> WASB detects as rally start. Real rally = shuttle crosses net MULTIPLE times.
- **FIX SPECTRUM (design, grounded in current substrate):** (A) shuttle-only: gate candidates on net/midline
  crossings >=K (?shot count) / direction-reversals ? cheap, no new model, kills single-toss fires. (B) user's
  START-STANCE STATE MACHINE: IDLE->await-stance(players settled+diagonal in service zones)->ARMED->shuttle->
  RALLY->end->IDLE; reuse the existing person detector (yolo_hybrid torchvision FRCNN+ByteTrack) as a VERIFIER
  run only ~2s around each candidate start (not full-video, which was ~50min/video). (C) pose-based serve
  detection ? richest, new model + license review, likely overkill. Rec: A now, B robust target; B needs research.
- ? **GATE A SHIPPED ? PR #28** (`feat/rally-start-gate`): `backend/detectors/rally_gate.py` (pure,
  camera-agnostic, no new model) ? estimate play axis (larger shuttle spread) + net (median coord),
  count (coord-net) sign-changes per window, keep crossings>=K. Wired into export via `--min-crossings`
  (0=off, 2 rec). 6 tests, suite 193 green. **Validated on Adarsh (side camera auto-detected, net x?716):
  real rallies 5-13 crossings, junk 0-1; K>=2 kills 18 of ~19 short tuned false fires, keeps all long
  rallies.** `output/adarsh_baseline_gated.csv` = BASELINE+gate(K=2) = 38 segs (45->38) = the "stronger
  baseline now". 1 yellow flag to eyeball: 4:39.5-4:44.7 (5.2s,1 crossing) removed ? real rally or one-sided?
- **STRATEGIC (user, 2026-06-06): paid-Gemini as a cost-controlled QUALITY fallback.** User OK slicing
  videos <1GB + paid Gemini to build a stronger baseline NOW + as a product "calculated-

bit-expensive,
quality-significantly-improving" fallback. KEY: the hybrid Phase-3 ALREADY does this (WASB candidates ->
Gemini verifies each PADDED CANDIDATE CLIP, not whole video -> cost bounded by #candidates×cliplen, not
duration); it died only on FREE-tier quota. So paid key re-enables it. Recommended stack: WASB propose ->
gate A (free, cuts junk -> fewer Gemini calls) -> Gemini verify survivors (paid, precision+boundaries).
? PRIVACY TRADE-OFF: sending video to Gemini breaks the "vendors/AI never see user uploads" privacy lever
([[monetization-plan]]) ? fine for OWNED/test/golden video; for end users make it an explicit opt-in
premium tier. More annotated videos arrive ~next week (-> leave_one_video_out for honest cross-video).

GEMINI REFINE (2026-06-06) ? WASB candidates -> Gemini verify/refine (in progress)

- `backend/eval/gemini_refine.py` (NEW, uncommitted on master working tree; + provider retry edits in
  `backend/providers/gemini.py`). Reuses GeminiProvider + badminton prompt + extract_video_chunk +
  make_predict_fn + write_segments_csv (no reinvent). Drives Gemini over SHORT WASB candidate clips
  (~baseline 45, padded 3s, reencoded ? tiny uploads ~450MB total, NOT GB chunks). Gemini returns the
  true rally per clip or [] (rejects between-point feeds it can see). **Envelope per candidate** (1 WASB
  candidate ? 1 rally) so Gemini fragments don't shatter a long rally below min_dur. User has a PAID key
  (env GEMINI_API_KEY only ? NEVER store the key in memory/disk/commits).
- **MODEL GOTCHA (verified 2026-06-06):** `gemini-2.5-flash` works but hits frequent **503 "high demand"**
  spikes; `gemini-2.0-flash` is **404 (deprecated for generateContent)** despite appearing in models.list.
  ? use **`gemini-flash-latest`** (clean, no 503, current stable flash). Account also has gemini-3.x /
  gemini-3.5-flash (newer than my training). Added upload+generate retries (3x backoff) to the provider;
  per-call client per worker (genai client not thread-safe ? socket errors when shared).
- Smoke (6 cands) clean on flash-latest: tightens boundaries (e.g. WASB 21.3-27.5 ? Gemini 23.8-25.8),
  keeps long rally whole. Full 45-candidate run launched (task bkap9jri2) -> output/adarsh_gemini_refined.csv.
- NEXT after run: compare Gemini-refined vs WASB baseline/baseline+gate (Gemini as SILVER-GT to finally
  quantify Adarsh); commit gemini_refine + provider retries as a PR. Recall ceiling = WASB candidates
  (Gemini only sees proposed windows; rallies WASB missed entirely won't appear).

PRECISION FINDING (2026-06-06): full-context Gemini > candidate-refine (envelope over-merges)

- Added `full_video_rallies()` to gemini_refine + `--full-video` (whole clip, downscaled 1280, chunked
  <=120s to stay under 500MB) + clip downscaling in extract_video_chunk (scale_width). Reuses provider/prompt.
- **TestLargeVideo (5.3K, 100s) head-to-head:** candidate-refine = 5 rallies/49s; **full-context = 7 rallies/
  34.5s**. Refine collapsed the 0:23-0:43 region into ONE 22s blob; full-context split it into 3 real rallies
  (23.5-26.5, 31.5-33.5, 39-42.5) with dead time between. Root cause: WASB merge_gap=2.0 bundled 3 rallies
  into 1 candidate, then my **envelope-per-candidate** collapsed Gemini's fragments back into 1 ? swept ~13s
  dead time into a "rally" = PRECISION LOSS. Full-context is a strict improvement (same rallies, tighter, +

correct split). Files: output/testlarge_gemini_rallies.csv (refine), output/testlarge_gemini_fullvideo.csv (full).
- ? **USER CONFIRMED (2026-06-06): full-context is correct** ? 0:23-0:43 IS three rallies (24-26, 31-34,

14. assistant (2026-06-08T13:21:58.895Z)

I now have strong grounding. The current-state confirms the actual numbers (96+32 rallies, F1 0.727 tuned on Adarsh-full, the n=2 LOVO result, WASB recall ceiling 0.43?0.81), and the study's data claims are honest. Let me do a couple of targeted checks to surface what the study omitted: the codec/AV1 output economics (a live P0 in COMMERCIALIZATION the study never connects), the actual `eval_partial_labels.py` methodology, and whether the study addresses the C3 WASB-weights provenance issue.

15. user (2026-06-08T13:22:00.979Z)

4:**Owner decisions captured:** licensed/owned video only · rally `[start,end)` depth · tiered turnaround (fast automated + slow human) · vendor bake-off · **Bangalore/India-local vendor preference**
30:- **Golden sets are built only from footage we license or own** ? starting with the founder's **constantly-growing personal collection** of match videos, plus ShuttleSet (ADR-003) and any purpose-recorded clips. **No end-user uploads ever flow to a vendor.**
34:> **Honest tradeoff:** a licensed/owned corpus may not perfectly mirror real users' cameras, courts, and lighting. A green regression on our corpus is therefore *strong evidence*, not a *guarantee*, for production traffic. Revisit if field behavior diverges from corpus behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by steering users toward capture conditions our corpus represents.
51:One trusted expert (initially: the founder) hand-annotates **5?10 licensed clips** to near-perfect. This is the answer key vendors are graded against ? kept private during the bake-off.
188:2. **Corpus representativeness** ? how well does the owned/licensed collection mirror eventual user capture? (Ties directly to VIDEO_RECORDING_GUIDELINES.md.)

16. user (2026-06-08T13:22:01.474Z)

```
"""Evaluate the rally pipeline against PARTIAL human golden labels.

The VLC rally-annotator produces a human-rated CSV (rally_number,start_time,end_time,
ending_reason,sport). While a video is only partially tagged, the labels cover a
PREFIX of the video ? every rally up to some point is marked, and everything after is
simply un-watched (not "no rally there"). Scoring naively against such labels is wrong:
the detector's *correct* rallies in the un-labeled tail would be counted as false
positives and crush precision.

This harness fixes that by restricting the evaluation to the LABELED WINDOW
[window_start, window_end] (window_end defaults to the last labeled rally's end) and
dropping predicted segments that fall outside it. Within that window the human labels
ARE complete, so precision/recall/F1 are honest.

It reuses everything from the existing eval stack (no new windowing/metrics/gate logic):
  - calibrate_wasb.make_predict_fn / load_golden_gts / BASELINE
  - pipeline.detectors.rally_gate.apply_gate   (net-crossing "gate A")
  - eval.metrics.match_intervals / score
  - export_wasb_segments.build_comparison / write_comparison_csv  (per-rally detail CSV)

Run:
    python -m backend.eval.eval_partial_labels \
        --trajectory C:/Users/avidu/AppData/Local/Temp/adarsh_avi_traj.csv \
        --labels "C:/Users/avidu/Projects/Annotation Setup/Collect/Adarsh and Avi.rallies.csv" \
        --fps 59.94 --frame-width 1920
"""
from __future__ import annotations

import argparse
```

```
import os.path as osp
from typing import List, Optional, Tuple

from backend.eval.calibrate_wasb import make_predict_fn, load_golden_gts, BASELINE
from backend.eval.metrics import score
from backend.eval.export_wasb_segments import build_comparison, write_comparison_csv
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.pipeline.detectors import rally_gate

Interval = Tuple[float, float]
Config = Tuple[float, float, float]

# Operating points to compare. The tuned config wins on its fit-video but over-segments
# and does not transfer; baseline (+ net-crossing gate) is the stronger general point.
TUNED: Config = (0.05, 1.0, 1.0)


def restrict_to_window(intervals: List[Interval], w0: float, w1: float) -> List[Interval]:
    """Keep intervals that overlap the labeled window [w0, w1].

    A predicted segment is evaluated iff it overlaps the labeled region; segments wholly
    in the un-labeled tail (start >= w1) or before the window (end <= w0) are dropped so
    they are neither rewarded nor penalized. GTs are assumed already inside the window.
    """
    return [(s, e) for (s, e) in intervals if e > w0 and s < w1]


def _eval_point(predict_fn, cfg: Config, points, fps: float, gts: List[Interval],
                window: Tuple[float, float], gate_crossings: int, iou: float) -> dict:
    """Score one (config, gate) operating point inside the labeled window."""
    preds = predict_fn(cfg)
```

17. assistant (2026-06-08T13:22:11.312Z)

I have everything I need. The study's code anchors are real and accurate, the data-scale claims match the project's actual state, and the guardrail framing is faithful to CLAUDE.md/PLATFORM/COMMERCIALIZATION. Now I can identify the genuine gaps, omissions, and weak claims. Let me compile the structured critique.

The key findings from my grounding:
- The C3 WASB-weights provenance issue (academic-data-trained checkpoints, NOT covered by MIT code grant ? `wasb_badminton_best.pth.tar`) is a flagged ?? commercial blocker that the study treats as fully resolved ("frozen MIT detector," "WASB code is MIT"). This is a real misalignment ? the study conflates code license with weights provenance.
- The study never connects to C5 (AV1/Opus codec output economics, a live P0) or C6 (HEVC decode) ? out of scope but the owned-VLM serving stack interacts with GPU/codec provisioning.
- The MAU-cap rejections ("100M-MAU cap") are dismissive without noting the founder is nowhere near that ? a possibly over-conservative rejection that forecloses 72B/78B options.
- ending_reason as a trajectory-only task is correctly flagged as near-impossible, but the 5k-20k QA band assumes ending_reason multiplexing as a "free" amplification lever ? internal tension.

18. user (2026-06-08T13:23:48.544Z)

Structured output provided successfully

19. assistant (2026-06-08T13:23:59.691Z)

I've completed the completeness critique and returned it via StructuredOutput. Summary of what I grounded against and found:

Verification performed: I confirmed the study's code anchors are real (`match_intervals` at `backend/eval/metrics.py:48`, `label_candidates` at `backend/eval/training.py:50`, plus the `eval_partial_labels.py` harness from PR #60), and cross-checked its data/guardrail claims against `CLAUDE.md`, `COMMERCIALIZATION.md`, `PLATFORM_ARCHITECTURE.md`, `ROADMAP.md`, `DEFERRED_HARDENING_PLAN.md`, and the live `current-state.md` memory. The study is honest and well-sourced ? it does not fabricate provenance and correctly identifies data starvation as the bottleneck.

Most material findings:
- **Guardrail blind spot:** the study treats WASB as a clean "frozen MIT detector," but `COMMERCIALIZATION.md` C3 is an open ?? blocker ? the actual checkpoint (`wasb_badminton_best.pth.tar`) is academic-data-trained, NOT covered by the MIT code grant. This taint propagates through every generation (features + pseudo-labels).
- **The ~0.73 ship-gate** is a fit-on-self number the project itself flagged as "LOVO is the real test" (not yet landed when the study was written).
- **Gemma-4 "confirmed-clean" flip** rests on post-cutoff secondary sources; should stay AMBER until a raw HF LICENSE revision is read, given the committed Gemma-3 rejection.
- **Dropped option:** InternVideo2 encoder + grounding head (the only data-realistic owned-video experiment NOW) was cut from the phased plan.
- **Two unratified policy items:** the guardrail literally says "Apache-2.0 only" but the approved list includes MIT bases; and the existing distilled model was trained on Gemini silver labels ? which the study's own eval-only policy now forbids, implying a required purge it never calls out.